

Derived categories of sheaves

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About this edition

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Sheaves of modules on a ringed space

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The first task in derived sheaf theory is to decide what an exact sequence means. A map can lift every germ and still fail to lift a global section. A cokernel must therefore be constructed locally, whereas a kernel can be constructed directly on sections. Once this distinction is clear, the basic operations on sheaves acquire precise exactness properties. These properties will determine which operations need resolutions in later lessons.

The prerequisites are the elementary definitions of a topological space, a ring, a left module, a functor, a limit and an adjunction. Section 1 recalls presheaves and proves the filtered-colimit, module-category, sheafification and stalk facts used below. In particular, [Lemma 1.3](#) supplies the germ construction, its universal property and the isomorphism test without assuming an abelian category of sheaves. Definitions and the germ construction are also freely readable in the Stacks project authors' [Sheaves on Spaces](#), in the AI Integrated Stacks Project edition.

After this lesson, *Injective modules and bounded-below derived functors* will construct resolutions using skyscrapers. *Flat modules and K-flat resolutions* will use extensions by zero to construct flat resolutions. The distinction between inverse image and module pullback is the starting point for *Derived pullback and pushforward*.

The main construction source is the Stacks project authors' *Sheaves of Modules*, in the [AI Integrated Stacks Project edition](#). The kernels, cokernels, colimits and sheaf operations developed below supply the foundations for the resolution lessons. Every proof prerequisite for this lesson is supplied below. Source attribution and the licence for adapted passages appear in the [course notice](#).

1. Coefficients, germs and finite relations

Let X be any topological space and let $\mathcal{O}$ be a sheaf of associative rings with identity. Commutativity is not assumed. A sheaf of left $\mathcal{O}$ -modules assigns to each open V an $\mathcal{O}(V)$ -module, with restriction maps compatible with restriction of scalars. Write $\text{Mod}(\mathcal{O})$ for their category. The zero ring and the empty space are allowed. All indexing sets and diagrams lie in one fixed universe.

A stalk F_x consists of germs of sections near x . To define its action by $\mathcal{O}_x$, represent the scalar and section on a common neighbourhood and multiply there. Changing representatives does not change the resulting germ. The ring identities and module identities follow on such a common neighbourhood. In particular the action is meaningful for noncommutative rings, with the order of multiplication preserved.

Lemma 1.1 (filtered colimits of modules). For a fixed ring A , filtered colimits in $\text{Mod}(A)$ preserve short exact sequences.

Proof. A category is filtered if every finite collection of objects and arrows has a compatible map to a common object. Equivalently, it is nonempty, any pair of objects maps to a common object, and any two parallel arrows become equal after a further arrow. Elements of a filtered colimit can be represented at one object. A finite list of elements can be moved to one common object. Equality of two representatives can be checked after maps to one further object: the relations defining a colimit involve finitely many arrows, and filteredness moves all of them to a common object and equalizes parallel arrows. Thus a representative is zero precisely when it becomes zero after some further arrow.

Given a filtered diagram of exact sequences $0 \rightarrow A_i \rightarrow B_i \rightarrow C_i \rightarrow 0$, suppose an element of $\varinjlim A_i$ maps to zero in $\varinjlim B_i$. Represent it at i , and move to an object where its image in B is zero. Injectivity of $A \rightarrow B$ at that object makes the representative zero there. This proves injectivity on colimits. If a representative in B_i maps to zero in the colimit of C_i , move to an object where its image in C is zero and use exactness there to lift it from A . Finally every representative in C_i lifts from B_i . These give exactness at the middle and surjectivity at the right. The constructions respect addition and the A -action. $\square$

In a neighbourhood diagram used to construct a stalk, the rings $\mathcal{O}(V)$ vary. We apply Lemma 1.1 to the underlying abelian groups, which all have the fixed coefficient ring $\mathbb{Z}$. The induced equalities of kernels and cokernels respect the $\mathcal{O}_x$ -action, since every scalar germ and each finite list of section germs can be represented on a common neighbourhood. Thus there is no assumption that the neighbourhood diagram has one fixed section ring.

The word *filtered* cannot be removed. For a pushout diagram written $L \leftarrow M \rightarrow R$, consider the following componentwise short exact sequence of diagrams:

$$(\mathbb{Z} \leftarrow 0 \rightarrow 0) \longrightarrow (\mathbb{Z} \xleftarrow{1} \mathbb{Z} \rightarrow 0) \longrightarrow (0 \leftarrow \mathbb{Z} \rightarrow 0).$$

The first map is the identity on the left object and the zero inclusion on the other objects; the second is the identity on the middle object and the quotient elsewhere. The maps commute with both arrows. The three pushouts are respectively $\mathbb{Z}, 0, 0$. The first resulting map is not injective. This example already occurs on a one-point space.

Lemma 1.2 (the elementary module category). Left modules over any ring form an abelian category. Kernels are submodules, cokernels are quotient modules, and a sequence is exact if its ordinary image equals its ordinary kernel. Small limits exist as compatible tuples in products; small colimits exist as a direct sum modulo the relations from the arrows.

Proof. The kernel of $u : M \rightarrow N$ is $\{m : u(m) = 0\}$; its universal property follows because a map killed by u lands there. The cokernel is $N/u(M)$; a map vanishing on $u(M)$ factors uniquely through that quotient. The map $M/\ker u \rightarrow u(M)$, $[m] \mapsto u(m)$, is linear, onto and one-to-one. Thus the canonical coimage-to-image map is an isomorphism. Morphisms add, composition is bilinear, and finite direct sums are simultaneously products and coproducts. These facts establish the abelian-category axioms and the stated exactness criterion. For a diagram, compatible tuples in the product have the universal property of its limit. Quotienting the direct sum by all elements $m_i - u(m_i)$ associated with arrows $i \rightarrow j$ gives the colimit's universal property. These constructions are sets in the chosen universe. $\square$

Sheafification will be denoted P^+ .

Sheafification and the stalk test

Lemma 1.3 (sheafification and the stalk test). Let X be any topological space.

1. Every presheaf of sets P has a sheaf P^+ and a map $\eta : P \rightarrow P^+$ that induces a bijection on every stalk. Every section of P^+ is locally the image of a section of P . For each sheaf H , composition with η gives a bijection $\text{Hom}(P^+, H) \cong \text{Hom}(P, H)$.
2. If P is separated, η_U is injective for each open U .
3. For a presheaf of rings, the same construction preserves addition, multiplication and the identity. For a presheaf of left modules over a sheaf of associative unital rings $\mathcal{O}$, it produces a sheaf of left $\mathcal{O}$ -modules with the same universal property for linear maps. Commutativity is unnecessary.
4. A map of sheaves of sets whose stalk maps are all bijective is an isomorphism. This also holds for sheaves of rings and of modules.

Proof. Recall the elementary definitions. A presheaf gives sections on opens and restriction maps compatible with composition. A sheaf glues every family of sections agreeing on overlaps to a unique section on the union. A separated presheaf has the uniqueness part of this condition. Empty covers are included, so a sheaf of sets has exactly one section on the empty open.

For an open U , define $P^+(U)$ to be the families $(a_x)_{x \in U}$, with $a_x \in P_x$, having the following property: each x has a neighbourhood $V \subset U$ and $p \in P(V)$ such that $a_y = p_y$ for every $y \in V$. Restrictions discard coordinates, and $\eta_U(p) = (p_x)_{x \in U}$. Compatible families on a cover glue coordinate by coordinate. At a point, any local representative from a covering member represents the glued family on that same neighbourhood. This proves existence and uniqueness of gluing in P^+ , including its one empty family on the empty open.

A germ of a section of P^+ at x has a local representative $\eta(p)$; hence η_x is surjective. If two germs from P_x have the same image, their images as families agree on a neighbourhood of x . Their coordinates at x are therefore equal in P_x , proving injectivity. If P is separated and $\eta_U(p) = \eta_U(q)$, equality of germs gives a cover of U on which p and q agree. Separatedness then gives $p = q$, proving (2).

Sections of any sheaf H with the same germs agree: germ equality gives equality on a neighbourhood of each point, and the uniqueness part of the sheaf condition gives equality on the union. Given $b : P \rightarrow H$, choose local representatives p_i of a family in $P^+(U)$. On an overlap, $b(p_i)$ and $b(p_j)$ have the same germs, so they agree. Glue them in H . Changing representatives produces sections with the same germs and hence the same glued section. The construction commutes with restrictions and extends b . Any extension must agree with it locally on all representatives, hence globally by uniqueness of gluing. This proves the universal property in (1).

For (3), form the algebraic operations coordinatewise on germs. To check local representability of a sum or product, restrict the finitely many representatives to one common neighbourhood and perform the corresponding operation there. Zero and the identity are locally represented by their section values. All ring identities therefore hold on germs and on their families, with the given order of multiplication. The extension of a ring map preserves these operations, as is checked on those local representatives. For a module presheaf, define $(c \cdot a)_x = c_x a_x$ for $c \in \mathcal{O}(U)$. Near a point, represent a by p and use $c|_V p$; thus the action preserves local representability. The module identities and restriction compatibility follow from the section identities. The universal extension of a linear map is linear because that identity can be checked on local representatives

and then glued. On the empty open the ring and module have their unique zero element. None of these arguments uses an abelian category of sheaves.

Finally let $u : F \rightarrow G$ have bijective stalk maps. Injectivity on stalks implies injectivity on sections, by the germ-equality argument above. Given $g \in G(U)$, lift its germ at each point, represent the lift on a neighbourhood, and shrink until its image equals the restriction of g . The local lifts agree on overlaps by sectionwise injectivity. They glue to an $f \in F(U)$ with $u(f) = g$. Thus every u_U is bijective. Its inverse commutes with restrictions, since restricting a lift is the unique lift on the smaller open. For rings and modules the inverses preserve the operations because the original bijections do. This proves (4). $\square$

The germ construction follows the Stacks project authors' [Sheaves on Spaces, sheafification](#), in the AI Integrated Stacks Project edition. The proof above includes the stalk-isomorphism argument and the associative-ring module operations in full. Adapted material retains GFDL-1.2-or-later; the expansion and checks are by GPT-6.1 Sol (OpenAI), at Ultra.

2. The abelian category of module sheaves

Theorem 2.1. Let $\mathcal{O}$ be any sheaf of rings on X .

1. $\text{Mod}(\mathcal{O})$ is additive. A morphism $\varphi : F \rightarrow G$ has a kernel, computed on sections: $(\text{Ker } \varphi)(U) = \ker \varphi_U$. It has a cokernel: the sheafification of $U \mapsto \text{coker } \varphi_U$.
2. For each x , the stalk functor $F \mapsto F_x$, from $\text{Mod}(\mathcal{O})$ to $\text{Mod}(\mathcal{O}_x)$, is additive, and $(\text{Ker } \varphi)_x = \ker \varphi_x$, $(\text{Coker } \varphi)_x = \text{coker } \varphi_x$.
3. $\text{Mod}(\mathcal{O})$ is abelian, and each stalk functor is exact.
4. A sequence $F \rightarrow G \rightarrow H$ with zero composite is exact at G if and only if $F_x \rightarrow G_x \rightarrow H_x$ is exact at G_x for every x . A morphism φ is a monomorphism (epimorphism, isomorphism) if and only if every φ_x is one.

Proof. (1) Morphisms add on sections, and composition is bilinear. The zero sheaf is a zero object. Finite direct sums are formed on sections: $U \mapsto F(U) \oplus G(U)$ is a sheaf, because sections glue coordinatewise. Let $\varphi : F \rightarrow G$. The presheaf $U \mapsto \ker \varphi_U$ is a sheaf. Indeed, let a section s of F lie in the kernel on each set of an open cover. Then $\varphi(s)$ vanishes on that cover, so $\varphi(s) = 0$, since G is a sheaf. Sections of the kernel that agree on overlaps glue in F , and by this remark the glued section lies in the kernel. Each φ_U is $\mathcal{O}(U)$ -linear, so the kernel is a subpresheaf of left modules. A morphism $E \rightarrow F$ whose composite with φ is zero lands in it on every open set. So it is the kernel. For the cokernel, let $P(U) = \text{coker } \varphi_U$, a presheaf of left $\mathcal{O}$ -modules. By [Lemma 1.3](#), morphisms $P^+ \rightarrow H$ into a sheaf H correspond to morphisms of presheaves $P \rightarrow H$. These are the morphisms $G \rightarrow H$ that vanish after φ . So P^+ is the cokernel.

(2) The stalk $F_x = \lim_{\rightarrow U \ni x} F(U)$ is a left module over $\mathcal{O}_x = \lim_{\rightarrow U \ni x} \mathcal{O}(U)$, and the functor is additive. The open neighbourhoods of x form a filtered system, and filtered colimits are exact ([Lemma 1.1](#)). Hence $(\text{Ker } \varphi)_x = \lim_{\rightarrow U} \ker \varphi_U = \ker \varphi_x$. By [Lemma 1.3](#), η is bijective on stalks, so $(P^+)_x = P_x$; and by ([Lemma 1.1](#)) again, $P_x = \lim_{\rightarrow U} \text{coker } \varphi_U = \text{coker } \varphi_x$.

(3) The coimage of φ is the cokernel of $\text{Ker } \varphi \rightarrow F$. The image is the kernel of $G \rightarrow \text{Coker } \varphi$. By (2), at each x the canonical map $\text{Coim } \varphi \rightarrow \text{Im } \varphi$ becomes the canonical map $\text{Coim } \varphi_x \rightarrow \text{Im } \varphi_x$ of $\mathcal{O}_x$ -modules. That map is an isomorphism, because modules over a ring form an abelian category. By [Lemma 1.3\(4\)](#) the map of sheaves is an isomorphism. Its inverse is $\mathcal{O}$ -linear, as the map is. So $\text{Mod}(\mathcal{O})$ is abelian. The stalk functor preserves kernels and cokernels by (2), so it is exact.

(4) Exactness at G means that $\text{Im}(F \rightarrow G) \rightarrow \text{Ker}(G \rightarrow H)$ is an isomorphism. Both sides are built from kernels and cokernels. By (2), the map at x is the same map for the stalk sequence. If every stalk sequence is exact, [Lemma 1.3\(4\)](#) gives exactness. Conversely, exact functors preserve exactness. A sheaf whose stalks all vanish is zero, since each section then vanishes on an open cover. Hence φ is a monomorphism iff $\text{Ker } \varphi = 0$ iff every $\ker \varphi_x = 0$. The same holds with cokernels. An isomorphism is a morphism that is both. $\square$

The proof never uses commutativity of $\mathcal{O}$. In particular, [Theorem 2.1](#) holds for the constant sheaf k_X of any unital ring k , commutative or not.

3. Limits, colimits, stalks, and sections of sums

Theorem 3.1. Let $\mathcal{O}$ be any sheaf of rings on X .

1. Every small diagram (F_i) in $\text{Mod}(\mathcal{O})$ has a limit, computed on sections. So $\Gamma(U, -)$ commutes with limits.
2. Every small diagram has a colimit: the sheafification of the presheaf colimit. Stalks commute with colimits: $(\varinjlim F_i)_x = \varinjlim (F_i)_x$.
3. Filtered colimits are exact.
4. Finite direct sums are computed on sections.
5. If U is a quasi-compact open set, the canonical map $\bigoplus_i \Gamma(U, F_i) \rightarrow \Gamma(U, \bigoplus_i F_i)$ is bijective for every family. Without quasi-compactness it can fail to be onto (Exercise 5).
6. Products exist and are computed on sections, but they need not be exact (Exercise 6).

Proof. (1) Let $L(U) = \varprojlim_i F_i(U)$, with coordinatewise restriction and module structure. It is a sheaf. A local section on a cover is a compatible family on each piece. Glue each coordinate in F_i . The glued coordinates again form a compatible family, because the transition maps are morphisms of sheaves and sections of a sheaf are determined locally. A morphism $E \rightarrow L$ is the same as a compatible family of morphisms $E \rightarrow F_i$, open set by open set. So L is the limit, and $\Gamma(U, L) = \varprojlim_i \Gamma(U, F_i)$.

(2) Let $P(U) = \varinjlim_i F_i(U)$. By [Lemma 1.3](#), morphisms $P^+ \rightarrow H$ into a sheaf correspond to compatible families of morphisms $F_i \rightarrow H$. So P^+ is the colimit. At x ,

$$(P^+)_x = P_x = \varinjlim_{U \ni x} \varinjlim_i F_i(U) = \varinjlim_i \varinjlim_{U \ni x} F_i(U) = \varinjlim_i (F_i)_x,$$

since colimits commute with colimits.

(3) Let $0 \rightarrow A_i \rightarrow B_i \rightarrow C_i \rightarrow 0$ be exact, naturally in i over a filtered category. At each x , by (2), the stalk sequence of the colimits is the filtered colimit of the exact sequences $0 \rightarrow (A_i)_x \rightarrow (B_i)_x \rightarrow (C_i)_x \rightarrow 0$. It is exact by (Lemma 1.1). Since exactness is checked on stalks ([Theorem 2.1\(4\)](#)), the colimit sequence is exact.

(4) This was shown in the proof of [Theorem 2.1\(1\)](#).

(5) Let $S = \bigoplus_i F_i$, the sheafification of $P(V) = \bigoplus_i F_i(V)$. The germ at x of $p = (p_i) \in P(V)$ is $(p_{i,x})_i \in \bigoplus_i (F_i)_x$. If all germs of p vanish, each p_i has zero germs and is zero. So P is separated, and $\theta : P(V) \rightarrow S(V)$ is injective for every V . For $V = U$ this is the injectivity of the canonical map. For surjectivity, let $t \in S(U)$. By [Lemma 1.3](#), every point of U has an open neighbourhood $V \subset U$ and some $p \in P(V)$ with $\theta(p) = t|_V$. As U is quasi-compact, finitely many such pairs $(V_1, p_1), \dots, (V_m, p_m)$ cover U . Each p_k has finitely many nonzero coordinates. Let J be the finite set of indices that occur, and $S_J = \bigoplus_{i \in J} F_i$. By (4), S_J is a sheaf. It is also a subpresheaf of P that contains all p_k . On $V_k \cap V_l$, the restrictions of p_k and p_l have the same image under the injective map θ , so they are equal. Glue them to $s \in S_J(U) = \bigoplus_{i \in J} F_i(U)$. Then $\theta(s)$ and t agree on each V_k , so $\theta(s) = t$.

(6) Products are limits, so (1) applies. Exercise 6 proves that a product of epimorphisms need not be an epimorphism. $\square$

4. Exact inverse image

Theorem 4.1. Let $f : X \rightarrow Y$ be continuous and $\mathcal{O}_Y$ a sheaf of rings on Y .

1. On X , $f^{-1}\mathcal{O}_Y$ is a sheaf of rings, with stalks $(f^{-1}\mathcal{O}_Y)_x = \mathcal{O}_{Y,f(x)}$. For $G \in \text{Mod}(\mathcal{O}_Y)$, $f^{-1}G$ is a sheaf of left $f^{-1}\mathcal{O}_Y$ -modules, and the canonical map $G_{f(x)} \rightarrow (f^{-1}G)_x$ is an isomorphism of $\mathcal{O}_{Y,f(x)}$ -modules.
2. $f^{-1} : \text{Mod}(\mathcal{O}_Y) \rightarrow \text{Mod}(f^{-1}\mathcal{O}_Y)$ is exact and commutes with colimits.
3. For a unital ring k , $f^{-1}k_Y = k_X$ as sheaves of rings. So $f^{-1} : \text{Mod}(k_Y) \rightarrow \text{Mod}(k_X)$ is exact, and $(f^{-1}G)_x = G_{f(x)}$ as k -modules. This holds for every unital ring k , commutative or not.
4. f^{-1} is left adjoint to $f_* : \text{Mod}(f^{-1}\mathcal{O}_Y) \rightarrow \text{Mod}(\mathcal{O}_Y)$.

All four assertions are proved below. *References:* [Kashiwara–Schapira 1990, Chapter II]; [Stacks, Tag 01AJ] for exactness on sheaves of abelian groups.

Proof of (1)–(3). Recall that $f^{-1}G$ is the sheafification of the presheaf $V \mapsto \lim_{\substack{\longrightarrow \\ W \supset f(V)}} G(W)$, where W runs over the open sets containing $f(V)$. We use its description by germs. For $V \subset X$ open, $f^{-1}G(V)$ is the set of maps s that give each $x \in V$ an element $s(x) \in G_{f(x)}$, such that each point of V has an open neighbourhood $V' \subset V$, an open $W \supset f(V')$ and a section $g \in G(W)$ with $s(x') = g_{f(x')}$ for all $x' \in V'$. The condition is local, so this is a sheaf of sets. Take $G = \mathcal{O}_Y$ to get $f^{-1}\mathcal{O}_Y$. Sum, product and the action are defined pointwise, through the ring $\mathcal{O}_{Y,f(x)}$ and its module $G_{f(x)}$. They preserve the local condition: if near a point s is represented by $g \in G(W)$ and a by $h \in \mathcal{O}_Y(W')$, then near that point as is represented by $h|_{W \cap W'} g|_{W \cap W'}$. This gives the ring and module structures. Evaluation at x is a ring map, and a module map over it.

Evaluation $(f^{-1}G)_x \rightarrow G_{f(x)}$ is onto: $g \in G(W)$ with $W \ni f(x)$ defines a section over $f^{-1}(W)$. It is one-to-one: if $s(x) = 0$, represent s near x by $g \in G(W)$. Then $g_{f(x)} = 0$, so g vanishes on an open $W_0 \ni f(x)$, and s vanishes on $V' \cap f^{-1}(W_0)$. The canonical map $G_{f(x)} \rightarrow (f^{-1}G)_x$ sends the germ of g to the germ of the section defined by g . It is inverse to evaluation. This proves (1).

(2) Exactness can be checked on stalks (Theorem 2.1(4)). By (1), the stalk of $f^{-1}G$ at x is $G_{f(x)}$, so on stalks f^{-1} acts as the identity, and f^{-1} is exact. For a colimit, the canonical map $\lim_{\substack{\longrightarrow \\ i}} f^{-1}G_i \rightarrow f^{-1} \lim_{\substack{\longrightarrow \\ i}} G_i$ is, at x , the map $\lim_{\substack{\longrightarrow \\ i}} (G_i)_{f(x)} \rightarrow (\lim_{\substack{\longrightarrow \\ i}} G_i)_{f(x)}$. This map is bijective, because stalks commute with colimits (Theorem 3.1(2)). So the canonical map is an isomorphism by Lemma 1.3(4).

(3) A germ of a section of k_Y is the germ of a locally constant function. So the sections of $f^{-1}k_Y$ over V are the maps $V \rightarrow k$ that are constant near each point, that is, the sections of k_X . The ring operations agree pointwise. The argument never uses commutativity of k . $\square$

Proof of (4). Let G be an $\mathcal{O}_Y$ -module and F an $f^{-1}\mathcal{O}_Y$ -module. A map $\alpha : f^{-1}G \rightarrow F$ sends a section $g \in G(W)$ to the image under α of its pullback section on $f^{-1}(W)$. This defines

$$\beta_W : G(W) \longrightarrow F(f^{-1}(W)) = (f_*F)(W).$$

The maps commute with restriction and are $\mathcal{O}_Y(W)$ -linear through the canonical map $\mathcal{O}_Y(W) \rightarrow (f^{-1}\mathcal{O}_Y)(f^{-1}(W))$.

Conversely, suppose $\beta : G \rightarrow f_*F$ is $\mathcal{O}_Y$ -linear. Locally a section s of $f^{-1}G$ on V is represented by a section $g \in G(W)$, with $f(V) \subset W$. Send it to $\beta_W(g)|_V$. If two representatives have the same germ at x , their germs in $G_{f(x)}$ agree, so they agree on some smaller neighbourhood W' of $f(x)$. Their proposed images therefore agree near x . This proves independence of representatives; the proposed local images glue uniquely. Multiplication is checked after locally representing the scalar by a section of $\mathcal{O}_Y$, so the glued map is $f^{-1}\mathcal{O}_Y$ -linear. The two constructions undo one another on local representatives, and hence everywhere. They commute with maps of G and F ; this is the adjunction. $\square$

Module pullback and extension of scalars

The inverse image changes the space. A morphism of ringed spaces also changes the coefficient ring. Let

$$f : (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

consist of a continuous map and a specified homomorphism $f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$. Write $A = f^{-1}\mathcal{O}_Y$ and $B = \mathcal{O}_X$. Then B is a (B, A) -bimodule: its right A -action is multiplication by the image of A . For an A -module N , the sheaf $B \otimes_A N$ is the sheafification of $V \mapsto B(V) \otimes_{A(V)} N(V)$. It is a sheaf of left B -modules.

Lemma 4.3 (tensor and stalks). There is a natural isomorphism

$$(B \otimes_A N)_x \cong B_x \otimes_{A_x} N_x.$$

Proof. A tensor on either side is a finite sum of elementary tensors. Choose representatives for its finitely many factors on one neighbourhood of x ; this constructs the inverse of the canonical map from the left side to the right. The tensor relations are finite sums of additivity and balancing relations $ba \otimes n = b \otimes an$. Any finite collection of germs witnessing those relations can again be represented on a common neighbourhood. Thus two representing expressions give the same germ exactly when the corresponding tensors agree on the right. Sheafification preserves stalks, so this verifies both injectivity and surjectivity for the displayed sheaf tensor product. $\square$

Theorem 4.4 (pullback adjunction and exactness). Define

$$f^*G = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}G.$$

For sheaves of left modules over arbitrary unital coefficient rings, f^* is left adjoint to f_* . It is right exact; f_* is left exact. At a point,

$$(f^*G)_x = \mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,f(x)}} G_{f(x)}.$$

Proof. First, extension of scalars is left adjoint to forgetting the B -action. Given a B -linear map $t : B \otimes_A N \rightarrow F$, define $a : N \rightarrow F$ by $n \mapsto t(1 \otimes n)$. It is A -linear by the balancing relation. Conversely an A -linear map a defines $b \otimes n \mapsto ba(n)$ on elementary tensors. This is balanced, hence gives a map on the presheaf tensor product and uniquely on its sheafification. The two formulas undo one another, commute with restriction, and are natural in both modules. Combining them with Theorem 4.1(4) gives

$$\mathrm{Hom}_{\mathcal{O}_X}(f^*G, F) \cong \mathrm{Hom}_{f^{-1}\mathcal{O}_Y}(f^{-1}G, F) \cong \mathrm{Hom}_{\mathcal{O}_Y}(G, f_*F).$$

Here the middle term regards F as an $f^{-1}\mathcal{O}_Y$ -module through the specified ring map. Thus the adjunction has the required coefficient structures. The stalk formula follows from Lemma 4.3 and Theorem 4.1(1).

For completeness, tensor over a ring is right exact. If $N \rightarrow N' \rightarrow N'' \rightarrow 0$ is exact, present N'' as the quotient of N' by the image of N . A balanced map on $B \times N''$ is exactly a balanced map on $B \times N'$ vanishing on $B \times \mathrm{im} N$. The defining universal property of tensor therefore identifies $B \otimes N''$ with the cokernel of $B \otimes N \rightarrow B \otimes N'$. This also proves surjectivity of the map to that cokernel. Apply this fact to the stalk formula and use Theorem 2.1 to obtain right exactness of f^* .

Finally $(f_*F)(V) = F(f^{-1}V)$. Kernels of module-sheaf maps are computed on sections, so f_* preserves kernels. It preserves finite products as well, again by their sectionwise description. These are the left-exactness assertions. No commutativity was used. $\square$

The tensor of two *left* $\mathcal{O}$ -modules used in later lessons requires commutative $\mathcal{O}$. The tensor in Theorem 4.4 has a specified right action on its first factor and therefore does not need that hypothesis.

Example 4.5 (an exact inverse image with a nonexact pullback). Take both spaces to be a point and use the coefficient map $\mathbb{Z} \rightarrow \mathbb{Z}/2$. Inverse image on abelian groups is the identity. Module pullback is $\mathbb{Z}/2 \otimes_{\mathbb{Z}} -$. The injection $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ becomes the zero map $\mathbb{Z}/2 \rightarrow \mathbb{Z}/2$, so pullback is not left exact. If the specified structure map $f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$ is an isomorphism, the formula instead identifies f^* with f^{-1} , and pullback is exact.

5. Extension by zero and skyscrapers

Lemma 5.1 (extension by zero from an open set). Let $j : U \hookrightarrow X$ be open and B a sheaf of left $\mathcal{O}|_U$ -modules. For $V \subset X$ open put

$$j_!B(V) = \{ s \in B(U \cap V) : \mathrm{supp} s \text{ is closed in } V \}.$$

1. $j_!B$ is a sheaf of left $\mathcal{O}$ -modules, with $(j_!B)_x = B_x$ for $x \in U$ and $(j_!B)_x = 0$ for $x \notin U$. So $j_!$ is exact.
2. $\mathrm{Hom}_{\mathcal{O}}(j_!B, G) \cong \mathrm{Hom}_{\mathcal{O}|_U}(B, G|_U)$, naturally in B and G .

3. $\text{Hom}_{\mathcal{O}}(j_! \mathcal{O}_U, G) \cong G(U)$ by $\varphi \mapsto \varphi(1)$. For opens $V \subset W$, the morphism $j_{V!} \mathcal{O}_V \rightarrow j_{W!} \mathcal{O}_W$ that corresponds to $1 \in \mathcal{O}(V) = (j_{W!} \mathcal{O}_W)(V)$ is a monomorphism, and under these identifications it induces restriction $G(W) \rightarrow G(V)$.

Proof. (1) Supports of sections are closed in their domains. Closedness of a support in V is a local condition on V , so sections of B glue within $j_! B$, and $j_! B$ is a sheaf. Module operations do not enlarge supports. If $x \in U$, the opens inside U are cofinal among neighbourhoods of x , and there $j_! B = B$; so $(j_! B)_x = B_x$. If $x \notin U$ and $s \in j_! B(V)$ with $x \in V$, then $x \notin \text{supp } s$, and this support is closed in V ; so s vanishes near x and the stalk is 0. Exactness is checked on stalks (**Theorem 2.1(4)**), and on each stalk $j_!$ acts as the identity or as zero; so $j_!$ is exact.

(2) A morphism $j_! B \rightarrow G$ restricts over U to a morphism $B \rightarrow G|_U$. Conversely, let $\alpha : B \rightarrow G|_U$ and $s \in j_! B(V)$. The section $\alpha(s)$ on $U \cap V$ and the section 0 on $V \setminus \text{supp } s$ agree on the overlap. The two open sets cover V , because $\text{supp } s \subset U \cap V$. Glue them to a section of G over V . This defines the inverse bijection; linearity and naturality are checked on the two pieces.

(3) By (2), a morphism $j_! \mathcal{O}_U \rightarrow G$ is a left-linear map $\mathcal{O}_U \rightarrow G|_U$, and such a map is $a \mapsto as$ with s the image of $1 \in \mathcal{O}(U)$. For $V \subset W$, $(j_{W!} \mathcal{O}_W)(V) = \mathcal{O}(V)$. At $x \in V$ the morphism is the identity of $\mathcal{O}_x$; at other points its source stalk is 0. So it is injective on every stalk, hence a monomorphism by **Theorem 2.1(4)**. Composing with it sends the morphism with $\varphi(1) = s \in G(W)$ to the one with value $s|_V$ at 1. $\square$

Lemma 5.2 (skyscrapers). Let $x \in X$ and let J be a left $\mathcal{O}_x$ -module. Put $x_* J(V) = J$ if $x \in V$ and 0 otherwise, with identity restrictions between opens containing x , and let $\mathcal{O}(V)$ act through $\mathcal{O}(V) \rightarrow \mathcal{O}_x$. Then $x_* J$ is a sheaf of left $\mathcal{O}$ -modules, and

$$\text{Hom}_{\mathcal{O}}(F, x_* J) \cong \text{Hom}_{\mathcal{O}_x}(F_x, J), \quad (5.2)$$

naturally in F and J .

Proof. Let $x \in V$ and let (V_i) cover V . Some V_i contains x . Compatible sections agree on all V_i that contain x , since their pairwise overlaps contain x ; this common value is the glued section, and it is unique. If $x \notin V$, all sections are 0. A morphism $\varphi : F \rightarrow x_* J$ has components φ_V , for $x \in V$, that are compatible with restriction. So they define an $\mathcal{O}_x$ -linear map $F_x \rightarrow J$. Conversely, given $g : F_x \rightarrow J$, put $\varphi_V(s) = g(s_x)$ if $x \in V$ and 0 otherwise. These constructions are inverse. $\square$

6. Three local and global computations

Example 6.1 (the two-point space). Let $S = \{o, c\}$, with opens $\emptyset, \{o\}, S$. Fix any ring A , and give S the constant coefficient sheaf A_S . A module sheaf is exactly a homomorphism $r : M \rightarrow N$ of left A -modules: its sections on S are M , on $\{o\}$ are N , and the only nontrivial restriction is r . Every covering of S contains S , and the sheaf condition imposes no further restriction. The stalk at c is M , since its only neighbourhood is S ; the stalk at o is N . Kernels, cokernels and exactness are therefore computed componentwise.

For $j : \{o\} \rightarrow S$, restriction sends $(M \rightarrow N)$ to N , extension by zero sends Q to $(0 \rightarrow Q)$, and direct image sends Q to $(Q \xrightarrow{1} Q)$. A section of $j_! Q$ on S has support contained in $\{o\}$; that subset is not closed in S , so only the zero section is allowed. For the closed point, $c_* Q = (Q \rightarrow 0)$. For the open point, the skyscraper is $o_* Q = (Q \xrightarrow{1} Q)$. Thus a skyscraper at a point need not have zero stalks at all other points. No closed-point assumption appeared in Lemma 5.2.

As a concrete cokernel, map $(A \xrightarrow{1} A)$ to $(A \rightarrow 0)$ by the identity at c and zero at o . Its kernel is $(0 \rightarrow A)$, which is precisely $j_! A$. This gives the short exact sequence

$$0 \rightarrow j_! A \rightarrow o_* A \rightarrow c_* A \rightarrow 0.$$

It records the open and closed pieces of this space without any separation hypothesis. In the lesson on supports the same space will make a localization triangle completely explicit.

Example 6.2 (a surjection with no global lift). On $X = \mathbb{C}^\times$, let $\mathcal{C}$ be the sheaf of continuous complex-valued functions, written additively, and let $\mathcal{C}^\times$ be the sheaf of continuous nowhere-zero functions, written multiplicatively. They are sheaves of abelian groups, hence left $\mathbb{Z}_X$ -modules. They are not being regarded as modules over the sheaf of complex-valued functions. Exponentiation gives

$$0 \longrightarrow (2\pi i\mathbb{Z})_X \longrightarrow \mathcal{C} \xrightarrow{\exp} \mathcal{C}^\times \longrightarrow 0.$$

The kernel consists of continuous functions with values in the discrete set $2\pi i\mathbb{Z}$, so it is the indicated locally constant sheaf. Every nonzero complex number has a neighbourhood on which a continuous logarithm is defined: choose a sufficiently small disk avoiding zero and a continuous branch of the argument there. For a nowhere-zero function, continuity places a neighbourhood of each point in such a disk. Composition gives a local logarithm, proving stalkwise surjectivity.

The global section $z \mapsto z$ has no continuous logarithm on X . Suppose one existed, and restrict it to $z = e^{2\pi it}$, $0 \leq t \leq 1$. The function

$$t \mapsto \frac{g(e^{2\pi it}) - 2\pi it}{2\pi i}$$

takes values in $\mathbb{Z}$ and is continuous, so it is constant. This forces $g(e^{2\pi i}) - g(1) = 2\pi i$, although the two arguments are the same point. The contradiction proves failure of global surjectivity. The cokernel sheaf is zero even though the cokernel on global sections is nonzero.

Example 6.3 (extension by zero at an endpoint). Let $X = [0, 1]$, $U = (0, 1)$, and let $\mathcal{O}_X$ be the sheaf of continuous real-valued functions. A global section of $j_!\mathcal{O}_U$ is a continuous function on U whose support is closed in X and contained in U . Its support is compact and misses both endpoints. Consequently the section vanishes on a neighbourhood of each endpoint. Conversely any such function has the required support and extends by zero to a continuous function on X . The constant function one on U is a section of $j_*\mathcal{O}_U$, but not of $j_!\mathcal{O}_U$ on X . At either endpoint $j_!\mathcal{O}_U$ has zero stalk; at an interior point it has stalk $\mathcal{O}_x$. These stalks, rather than the global section module, give the later flatness calculation.

7. Exercises with complete solutions

The first two exercises test the exactness criteria. The next two compare constructions. The last two concern finiteness and topology.

Exercise 1 (easy: an isomorphism criterion). Show directly that a morphism of module sheaves with isomorphisms on all stalks has an inverse sheaf morphism. Explain why the inverse is module-linear.

Solution. If two sections have the same image, their germs have the same image at every point. Stalkwise injectivity makes their germs equal, hence the sections equal. Given a section t in the target, lift each of its germs, choose local representatives, and shrink until the image of each representative is t on that neighbourhood. On overlaps the representatives have the same image, so the just-proved injectivity makes them agree. They glue to a unique section in the source. These inverses commute with restriction by uniqueness. The inverse sends at to as , because the forward map sends as to at , again with a unique preimage. It preserves sums for the same reason. Thus it is a linear inverse sheaf morphism.

Exercise 2 (medium: filtered and unfiltered colimits). Prove exactness of filtered colimits of module sheaves. Then verify every arrow in the pushout counterexample in Section 1 and compute its three pushouts.

Solution. Stalks of a colimit are the colimits of the stalks by Theorem 3.1(2). For a diagram of short exact sequences, the stalk diagrams are exact by Theorem 2.1. Lemma 1.1 gives exactness after taking their filtered colimits. Theorem 2.1 then gives exactness of the sheaf sequence. In the counterexample, at the left object the sequence is $0 \rightarrow \mathbb{Z} \xrightarrow{1} \mathbb{Z} \rightarrow 0 \rightarrow 0$, at the middle object it is $0 \rightarrow 0 \rightarrow \mathbb{Z} \xrightarrow{1} \mathbb{Z} \rightarrow 0$, and at the right all terms vanish. The square for the arrow middle-to-left in the first morphism is $0 \rightarrow \mathbb{Z}$ followed by the identity on the left, and on the other side the zero inclusion followed by the identity middle-to-left; both composites

from the zero group are zero. In the second morphism, both composites from the middle $\mathbb{Z}$ to the last left object zero are zero. The middle-to-right squares are zero throughout. A module pushout of $L \leftarrow M \rightarrow R$ is $(L \oplus R)/\{(a(m), -b(m)) : m \in M\}$, as its universal property shows. This gives $\mathbb{Z}$, 0 and 0. Thus the pushout functor does not preserve the displayed injection of diagrams.

Exercise 3 (medium: extension by zero). Given a map $B \rightarrow G|_U$, construct its adjoint $j_!B \rightarrow G$ on every open set. Show this construction gives the identity in both adjunction compositions, and deduce exactness of $j_!$.

Solution. For $s \in j_!B(V)$, glue its image on $V \cap U$ with zero on $V \setminus \text{supp } s$. These sets cover V , and the images agree on the intersection because s is zero there. Restriction commutes with gluing, as do sums and scalar multiplication. Restricting the resulting map to U recovers the original map, since $j_!B|_U = B$. Starting with a map $j_!B \rightarrow G$, the reconstructed section has the same restriction on $V \cap U$ and is zero off the support; these are also the restrictions of the original image. Uniqueness of gluing recovers it. At a point of U , the stalk functor of $j_!$ is the identity; elsewhere it is zero. Both preserve exact sequences, so stalkwise exactness proves exactness of $j_!$.

Exercise 4 (medium: scalar extension). For a homomorphism of possibly noncommutative rings $A \rightarrow B$, write the unit and counit of $B \otimes_A - \dashv \text{Res}_A^B$. Check both triangle identities. Apply the formulas to the pullback adjunction in Theorem 4.4.

Solution. The unit is $M \rightarrow \text{Res}(B \otimes_A M)$, $m \mapsto 1 \otimes m$. It is A -linear because $1 \otimes am = a \otimes m$, with a acting through $A \rightarrow B$. The counit is $B \otimes_A \text{Res } N \rightarrow N$, $b \otimes n \mapsto bn$, which is balanced and B -linear. The first composite sends $b \otimes m$ to $b \otimes (1 \otimes m)$ and then to $b \otimes m$. The second sends n to $1 \otimes n$ and then to n . These verify the triangle identities. For sheaves the formulas hold on local tensor representatives and pass through sheafification. Compose with the inverse-image/direct-image adjunction: the pullback unit sends a section g over $V \subset Y$ to $1 \otimes f^{-1}g$ over $f^{-1}V$. The pullback counit sends a local tensor $b \otimes s$, where s is locally represented by a section of f_*F , to b times its restriction to the relevant open subset of X . These are precisely the formulas in the proof of Theorem 4.4.

Exercise 5 (hard: sections of an infinite direct sum). Let $X = \mathbb{N}$ be discrete, let $A \neq 0$ be a ring, and let F_n be the skyscraper A at n . Compute $\Gamma(X, \bigoplus_n F_n)$ and compare it with $\bigoplus_n \Gamma(X, F_n)$.

Solution. A sheaf on a discrete space has sections on an open subset V equal to the product of its stalks at the points of V . Indeed, the singleton opens cover V , their intersections are empty, and the sheaf condition says that all choices of singleton sections glue uniquely. The direct sum sheaf has stalk A at each n , since at that stalk only F_n is nonzero. Its sections on X are therefore $\prod_n A$. In contrast $\Gamma(X, F_n) = A$, so the sum of the global sections is $\bigoplus_n A$. The canonical map is the inclusion of finitely supported sequences into all sequences, and is not onto: the sequence with every entry one is absent. The open X is not quasi-compact, as the singleton cover has no finite subcover. This does not contradict Theorem 3.1(5).

Exercise 6 (hard: products need not preserve epimorphisms). Let $X = \mathbb{R}$, let k be a field, and put $U_n = (-1/n, 1/n)$. For each n , extend the identity germ at zero to a morphism $j_{n!}k_{U_n} \rightarrow 0_*k$. Prove it is an epimorphism and show that the product of these maps is not an epimorphism.

Solution. By Lemma 5.2 a map to 0_*k is determined by the map of stalks at zero. Choose the identity $k \rightarrow k$. At zero it is onto; at every other point the target stalk is zero because zero is closed in $\mathbb{R}$. Thus Theorem 2.1 makes it an epimorphism. Products are computed on sections. A germ at zero of a section of $\prod_n j_{n!}k_{U_n}$ is represented on one common neighbourhood V of zero. Choose a connected interval W containing zero inside V . Each factor restricts to a locally constant function on $W \cap U_n$, whose support is closed in W . For every sufficiently large n , the interval U_n is contained in W . A nonzero constant on U_n would have support U_n , which is not closed in W ; hence that factor is zero. The images of the germ at zero thus have only finitely many nonzero coordinates. The target product sheaf has stalk $\prod_n k$ at zero: on every neighbourhood of zero its sections are that product and all restrictions are identities. The germ with every coordinate one is not in the image. The product map is not onto on the stalk, hence is not an epimorphism. This argument uses one common neighbourhood for all factors; commuting a stalk with an infinite product would lose exactly that condition.

References

- **[Stacks]** The Stacks project authors, *The Stacks project*, chapter *Sheaves of Modules*, [kernels and cokernels](#), [the abelian category](#), and [limits and colimits](#), in the AI Integrated Stacks Project edition. Stacks-derived material retains GFDL-1.2-or-later; the edition is an unofficial AI-integrated derivative.
- **[Kashiwara–Schapira 1990]** Masaki Kashiwara and Pierre Schapira, *Sheaves on Manifolds*, Grundlehren der mathematischen Wissenschaften 292, Springer, 1990. For module sheaves and sheaf operations.
- **[Kashiwara–Schapira 2006]** Masaki Kashiwara and Pierre Schapira, *Categories and Sheaves*, Grundlehren der mathematischen Wissenschaften 332, Springer, 2006. Scholarly reference for categorical constructions. The statements used in this lesson have complete proofs above.

Complexes, cones and localization

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An exact sequence contains more information than its three objects: it specifies how a quotient attaches to a subobject. A complex contains still more attachment information, in each degree. Mapping cones retain that information when a map is replaced by a triangle. Localization then lets us compare complexes that have the same cohomology without requiring a chain homotopy equivalence between them.

This common reading precedes the resolution lessons. Its algebra prerequisite is Wen-Wei Li’s *Methods of Algebra*, Volume 2: “[Definition of an Abelian Category](#)”, and “[Some Diagram Lemmas](#)”, especially the exactness criterion by epimorphic lifting, the Snake Lemma and the Five Lemma. “[Complexes on an Abelian Category](#)” proves the long exact cohomology sequence using that Snake Lemma. We recall its construction here and supply the cone and localization arguments in full.

The cone and triangulation arguments follow the Stacks project authors’ *Derived Categories*, in the [AI Integrated Stacks Project edition](#). We develop localization by roofs and common refinements, then prove that cone triangles descend to the derived category. Source attribution and the licence for adapted passages appear in the [course notice](#).

1. Maps of complexes and homotopies

Let $\mathcal{A}$ be an abelian category. A **cochain complex** K consists of objects K^n and differentials $d_K^n : K^n \rightarrow K^{n+1}$ satisfying $d_K^{n+1}d_K^n = 0$. A chain map $f : K \rightarrow L$ satisfies $d_L f = f d_K$. Its cohomology is

$$H^n(K) = \ker(d_K^n) / \operatorname{im}(d_K^{n-1}).$$

We set $K[r]^n = K^{n+r}$, with differential $(-1)^r d_K$. A homotopy from g to f is a family $s^n : K^n \rightarrow L^{n-1}$ with $f - g = d_L s + s d_K$. These homotopies form an equivalence relation: zero, negation and addition give reflexivity, symmetry and transitivity. Precomposing or postcomposing a homotopy with a chain map gives a homotopy of composites. Thus homotopy classes form an additive category $K(\mathcal{A})$, with the same objects as the category of complexes. Its finite sums and zero object are those of complexes. A **homotopy equivalence** is an isomorphism in this category. A complex is **contractible** when its identity is null-homotopic.

The group-valued Hom complex, as distinct from an internal sheaf Hom, is

$$\operatorname{Hom}^r(K, L) = \prod_{n \in \mathbb{Z}} \operatorname{Hom}_{\mathcal{A}}(K^n, L^{n+r}), \quad D(h) = d_L h - (-1)^r h d_K. \quad (1.1)$$

Lemma 1.1. Formula (1.1) defines a complex of abelian groups, and

$$H^r \operatorname{Hom}^\bullet(K, L) = \operatorname{Hom}_{K(\mathcal{A})}(K, L[r]). \quad (1.2)$$

Proof. Applying the differential twice cancels the two terms involving $d_L h d_K$; the other terms vanish because $d_L^2 = d_K^2 = 0$. A degree- r family is a cocycle precisely when $d_L h = (-1)^r h d_K$, which is the chain-map equation into $L[r]$. If $h = D(v)$ with v of degree $r - 1$, then

$$h = d_L v + (-1)^r v d_K = d_{L[r]}((-1)^r v) + ((-1)^r v) d_K.$$

So boundaries are exactly the null-homotopic maps into $L[r]$. Conversely a homotopy s gives the boundary of $(-1)^r s$. This proves (1.2). Composition of homogeneous families obeys

$$D(gh) = D(g)h + (-1)^{\deg g} g D(h) :$$

expanding the right side cancels its two middle terms and leaves $d_M gh - (-1)^{\deg g + \deg h} gh d_K$. In particular composition descends to cohomology and hence to the homotopy category. $\square$

Lemma 1.2 (cohomology of a short exact sequence). A degreewise short exact sequence $0 \rightarrow A \xrightarrow{a} B \xrightarrow{b} C \rightarrow 0$ has a natural long exact cohomology sequence. Homotopic maps induce the same cohomology map.

Proof. For the second assertion, on $\ker d_K^n$ the summand sd_K vanishes and $d_L s$ has image in $\operatorname{im} d_L^{n-1}$. It therefore induces zero on cohomology.

For the first assertion, apply the Snake Lemma to the exact rows

$$\begin{aligned} \operatorname{coker} d_A^{n-2} &\longrightarrow \operatorname{coker} d_B^{n-2} \longrightarrow \operatorname{coker} d_C^{n-2} \longrightarrow 0, \\ 0 &\longrightarrow \ker d_A^n \longrightarrow \ker d_B^n \longrightarrow \ker d_C^n, \end{aligned}$$

whose vertical maps are induced by d^{n-1} . The kernels of these vertical maps are H^{n-1} ; their cokernels are H^n . The first row is right exact by the cokernel part of the Snake Lemma applied to the degreewise exact diagram with vertical maps d^{n-2} ; the second is left exact by its kernel part for d^n . Thus the new Snake Lemma sequence joins $H^{n-1}(C)$ to $H^n(A)$ and gives exactly the six consecutive terms of the cohomology sequence. Varying n joins the sequences. Naturality follows from the naturality of the Snake Lemma construction by kernels, cokernels and pullbacks.

In modules, its boundary has the familiar precise description: lift a cocycle c to b , write $d_B b = a(a')$, and send $[c]$ to $[a']$. A changed lift adds a boundary, and a changed cocycle by a boundary also changes a' by a boundary. In an arbitrary abelian category these lifts are performed after pullback along an epimorphism, as in the exactness criterion in the prerequisite; the kernel and cokernel description just given supplies the resulting canonical morphism. $\square$

A **quasi-isomorphism** is a map inducing an isomorphism on every cohomology object. A complex is **acyclic** when all its cohomology objects vanish. A homotopy equivalence is a quasi-isomorphism by Lemma 1.2, but an acyclic complex need not be contractible.

2. Cones and triangles from split sequences

For $f : K \rightarrow L$, define

$$C(f)^n = L^n \oplus K^{n+1}, \quad d(l, k) = (d_L l + f k, -d_K k). \quad (2.1)$$

Squaring this differential gives zero because f is a chain map. Let $i : L \rightarrow C(f)$ be inclusion and $p : C(f) \rightarrow K[1]$ projection. Our **cone triangle** is

$$K \xrightarrow{f} L \xrightarrow{i} C(f) \xrightarrow{-p} K[1]. \quad (2.2)$$

The minus sign is intentional. With this convention, a triangle attached to a short exact sequence has the Snake Lemma's positive boundary.

Lemma 2.1 (comparison of cones). If $bf - f'a = dh + hd$, the maps

$$c^n = \begin{pmatrix} b^n & h^{n+1} \\ 0 & a^{n+1} \end{pmatrix} : C(f)^n \longrightarrow C(f')^n$$

give a morphism of cone triangles in $K(\mathcal{A})$.

Proof. The diagonal entries of $dc = cd$ are the chain-map equations for a, b . Equality of the upper-right entries is $dh + f'a = bf - hd$, exactly the assumed homotopy equation. The lower-left entries are zero. Also $ci = i'b$ and $p'c = a[1]p$ hold already as chain maps. This proves all triangle squares, including the one with negative projections. The choice of h need not be unique; the third triangle map is not canonically determined by a square in the homotopy category. $\square$

Lemma 2.2 (split sequences). Suppose $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is split in each degree. Write

$$B^n = A^n \oplus C^n, \quad d_B = \begin{pmatrix} d_A & \delta \\ 0 & d_C \end{pmatrix}. \quad (2.3)$$

Then $\delta : C \rightarrow A[1]$ is a chain map, and the triangle $A \rightarrow B \rightarrow C \xrightarrow{\delta} A[1]$ is isomorphic to the cone triangle of $A \rightarrow B$.

Proof. The upper-right entry of $d_B^2 = 0$ is $d_A\delta + \delta d_C = 0$, the asserted chain-map equation. In degree n , the cone of inclusion is $A^n \oplus C^n \oplus A^{n+1}$. Define

$$r(a, c, a') = c, \quad t(c) = (0, c, -\delta c).$$

Both are chain maps by (2.3), and $rt = 1$. The homotopy $H(a, c, a') = (0, 0, a)$, landing in degree $n - 1$, satisfies

$$(dH + Hd)(a, c, a') = (a, 0, a' + \delta c) = (1 - tr)(a, c, a').$$

Thus r, t are inverse in K . Inclusion of B is carried to its quotient map into C , and $(-p)t = \delta$. These give the claimed triangle isomorphism. Formula (2.3) also shows that $H^n(\delta)$ is exactly the boundary constructed in Lemma 1.2. $\square$

Lemma 2.3 (replacement by a split injection). Every map $f : K \rightarrow L$ factors as a degreewise split injection $K \rightarrow \tilde{L}$ followed by a homotopy equivalence $\tilde{L} \rightarrow L$. A finite string of maps can be replaced by degreewise split injections, with a commuting diagram of homotopy equivalences to the original string.

Proof. Put $\tilde{L} = L \oplus C(1_K)$, with degree- n coordinates (l, k, k') , differential $(d_L l, d_K k + k', -d_K k')$, and injection $k \mapsto (fk, k, 0)$. Projection π to L has chain-map section $s(l) = (l, 0, 0)$. The homotopy $H(l, k, k') = (0, 0, k)$ gives

$$(dH + Hd)(l, k, k') = (0, k, -d_K k) + (0, 0, d_K k + k') = (0, k, k') = 1 - s\pi.$$

The injection is split by projection onto K^n . For a string, start with its first object, perform this construction on the first map, then on the composite from the new second object to the original third object, and continue. Every resulting square commutes as a square of chain maps. Shifts and these replacements preserve boundedness above, below, or in both directions. $\square$

Lemma 2.4 (representable exactness and cone uniqueness). For a cone triangle, applying $\text{Hom}_K(T, -)$ or $\text{Hom}_K(-, T)$ gives a long exact sequence after shifts. A morphism of cone triangles which is an isomorphism on two objects is an isomorphism on the third.

Proof. There is a chain isomorphism

$$\text{Hom}^\bullet(T, C(f)) = C(\text{Hom}^\bullet(T, f)).$$

It follows by distributing products over the two summands in (2.1); its differential has exactly the matrix (2.1). For the contravariant version put $U = \text{Hom}^\bullet(L, T)$, $V = \text{Hom}^\bullet(K, T)$, and $u = f^* : U \rightarrow V$. A degree- r map from the cone has coordinates $(a, b) \in U^r \oplus V^{r-1}$ and differential

$$(a, b) \mapsto (Da, Db + (-1)^{r+1}u(a)).$$

The map $(a, b) \mapsto ((-1)^{r-1}b, a)$ identifies this complex with $C(u)[-1]$, whose differential is $(v, a) \mapsto (-Dv - u(a), Da)$. Lemma 1.2 for these group complexes and Lemma 1.1 give both exact sequences. Signs of individual arrows do not change exactness.

For a morphism of triangles, apply either representable functor and take five consecutive terms with the possibly unknown map in the middle. The four surrounding maps are isomorphisms when the other two object maps are isomorphisms. The Five Lemma makes the middle map an isomorphism for every T . If $c : Z \rightarrow Z'$ induces bijections $\text{Hom}(T, Z) \rightarrow \text{Hom}(T, Z')$ for every T , surjectivity for $T = Z'$ gives a right inverse v , and injectivity for $T = Z$ gives $vc = 1$. Thus c is an isomorphism. Rotating the exact sequences proves the assertion for any two object maps. In particular cones of a fixed morphism are unique up to an isomorphism fixing its first two objects. Lemma 2.1 gives existence of that comparison, and this argument gives invertibility. $\square$

3. The triangulated structure of the homotopy category

A triangle means $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$. It is **distinguished** in $K(\mathcal{A})$ when it is isomorphic to (2.2). A triangulated category is an additive category with an additive shift equivalence and distinguished triangles satisfying four requirements: identity and completion triangles exist and are closed under isomorphism; rotation $(f, g, h) \mapsto (g, h, -f[1])$ preserves distinguished triangles in both directions; a commuting square between their first maps extends to a morphism of triangles; and composable maps have compatible cone triangles forming an octahedron. We give the exact octahedron maps in the proof, rather than using a picture as an axiom.

Theorem 3.1. These definitions make $K(\mathcal{A})$ a triangulated category. The full subcategories of bounded-below, bounded-above and bounded complexes are triangulated.

Proof. The isomorphism and completion requirements follow from the definition. The cone $C(1_X)$ is contractible by Lemma 2.3, so its cone triangle is isomorphic to $X \xrightarrow{1} X \rightarrow 0 \rightarrow X[1]$.

For rotation, $0 \rightarrow L \rightarrow C(f) \rightarrow K[1] \rightarrow 0$ is split in each degree, with off-diagonal map $f[1]$. By Lemma 2.2 its triangle $(i, p, f[1])$ is distinguished. Changing the sign on its third object identifies it with $(i, -p, -f[1])$, the rotation of (2.2). Conversely, suppose a rotated triangle is represented by a split sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ with boundary δ . The inverse rotation is $C[-1] \xrightarrow{-\delta[-1]} A \rightarrow B \rightarrow C$. Formula (2.3) identifies B with $C(-\delta[-1])$ by the map $(a, c) \mapsto (a, -c)$; its projection into C becomes the negative cone projection. Hence this inverse rotation is distinguished. Every cone triangle has a split-sequence representative: use Lemma 2.3 on its first map, Lemma 2.1 and Lemma 2.4 to compare cones, and Lemma 2.2 on the new split injection. Thus the converse applies to all distinguished triangles.

The square-extension requirement is precisely Lemma 2.1 after replacing the rows by cone triangles.

For the octahedron take composable maps $A \xrightarrow{\alpha} B \xrightarrow{\beta} C$. Lemma 2.3 replaces this string by degreewise split injections. Lemmas 2.1, 2.2 and 2.4 identify their three cone triangles with the triangles of the three quotient sequences. Choose degreewise decompositions $B = A \oplus Q_1$ and $C = A \oplus Q_1 \oplus Q_3$. Their differentials have the forms

$$d_B = \begin{pmatrix} d_A & u \\ 0 & d_1 \end{pmatrix}, \quad d_C = \begin{pmatrix} d_A & u & v \\ 0 & d_1 & w \\ 0 & 0 & d_3 \end{pmatrix}. \quad (3.1)$$

Set $Q_2 = C/A = Q_1 \oplus Q_3$, with differential $\begin{pmatrix} d_1 & w \\ 0 & d_3 \end{pmatrix}$. The three boundary maps are u , (u, v) , and (v, w) . The split sequence $0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow Q_3 \rightarrow 0$ has boundary w , so its triangle is distinguished. Its first two maps are inclusion $a : Q_1 \rightarrow Q_2$ and projection $b : Q_2 \rightarrow Q_3$. The triples $(1_A, \beta, a)$ and $(\alpha, 1_C, b)$ are morphisms of the three original triangles. All their squares commute strictly except possibly the second boundary square. The difference

$$(v, w)b - \alpha[1](u, v) = \begin{pmatrix} -u & 0 \\ 0 & w \end{pmatrix}$$

is the homotopy boundary for $H : Q_2^n \rightarrow B[1]^{n-1} = B^n$, $H(q_1, q_3) = (0, q_1)$: substituting (3.1) into $d_{B[1]}H + Hd_{Q_2}$ gives $(-uq_1, wq_3)$. Finally $w = p_1[1](v, w)$, where $p_1 : B \rightarrow Q_1$. Thus

$$Q_1 \xrightarrow{a} Q_2 \xrightarrow{b} Q_3 \xrightarrow{p_1[1]\delta_3} Q_1[1]$$

is the required fourth triangle, with exactly the required comparison maps. Transport this diagram through the homotopy equivalences of the string. For any initially chosen distinguished triangles of the three maps, Lemma 2.4 compares them by isomorphisms fixing their first two objects. Transport through those too. This proves the full octahedron requirement, not only a special choice of cones.

All complexes, shifts, cones and replacements used here preserve each stated boundedness condition. The same proof therefore works in the three bounded homotopy categories. $\square$

Corollary 3.2. An additive functor $\Phi : \mathcal{A} \rightarrow \mathcal{B}$ induces a triangulated functor on homotopy categories. A quasi-isomorphism f is characterized by acyclicity of $C(f)$.

Proof. Applying Φ degreewise preserves differentials, homotopies, shifts, finite sums and every entry of (2.1), hence cone triangles. For the second assertion apply Lemma 1.2 to $0 \rightarrow L \rightarrow C(f) \rightarrow K[1] \rightarrow 0$; its connecting map is $H(f[1])$. Exactness proves that the middle complex is acyclic exactly when each $H^n(f)$ is invertible. $\square$

4. Fractions at quasi-isomorphisms

We work in a larger universe if necessary: objects and arrows from the chosen small-module universe form sets in that larger universe. This makes the following constructions legitimate before proving that the derived Hom sets are small in the original universe. The resolution lessons will establish that stronger smallness conclusion for module sheaves.

Let S be the quasi-isomorphisms in $K(\mathcal{A})$. It contains identities, is closed under composition, and has the two-out-of-three property, because these assertions hold for isomorphisms after every H^n .

Lemma 4.1 (fraction squares and cancellation). Given $f : M \rightarrow Y$ and $t : N \rightarrow Y$ with $t \in S$, there are $r : P \rightarrow M$ in S and $a : P \rightarrow N$ with $fr = ta$ in K . If $tf = tg$ for $t : Y \rightarrow Z$ in S , there is $r : P \rightarrow X$ in S with $fr = gr$. The dual assertions, with all arrows reversed, also hold.

Proof. For the square take $P = C((f, -t) : M \oplus N \rightarrow Y)[-1]$. Its coordinates in degree n are (y, m, n') , with differential $(-d_Y y - fm + tn', d_M m, d_N n')$. Let r, a be projections to m, n' . Their required equality

holds up to the homotopy $(y, m, n') \mapsto -y$. The kernel of r is $C(-t)[-1]$, which is acyclic by Corollary 3.2. Since r is degreewise split onto, Lemma 1.2 makes r a quasi-isomorphism. This proves the square assertion.

For cancellation put $v = f - g$. Choose a homotopy $tv = d_Z h + h d_X$. The map $(-h, v) : X \rightarrow C(t)[-1]$ is a chain map: its first component's chain-map equation is $d_Z h - tv = -h d_X$, exactly the chosen homotopy equation. Its second component projects to v . Write this map as $e : X \rightarrow E$, where $E = C(t)[-1]$ is acyclic. Projection $r : C(e)[-1] \rightarrow X$ is a quasi-isomorphism, since its kernel $E[-1]$ is acyclic. Moreover er is null-homotopic by projection onto the first cone coordinate with a minus sign, exactly as in the square computation. Therefore $vr = 0$.

For the dual square take the cone of $(t, -f) : M \rightarrow N \oplus Y$; the inclusion of Y has acyclic quotient $C(t)$, and the two composites agree up to the cone's homotopy. For the dual cancellation, representable exactness in Lemma 2.4 factors a map v satisfying $vt = 0$ through the acyclic cone $C(t)$. If this factor is $e : C(t) \rightarrow Z$, the inclusion $Z \rightarrow C(e)$ is a quasi-isomorphism and kills e up to homotopy. It consequently kills v . These are the asserted reversed constructions. $\square$

A **roof** from X to Y is $X \xleftarrow{s} M \xrightarrow{f} Y$, with $s \in S$. Declare two roofs equivalent if they have a common refinement: maps $u : P \rightarrow M$, $v : P \rightarrow M'$ such that $su = s'v \in S$ and $fu = f'v$. Two-out-of-three then implies $u, v \in S$.

Theorem 4.2 (localization). Roof classes form a category $D(\mathcal{A})$. The map $Q : K(\mathcal{A}) \rightarrow D(\mathcal{A})$, $f \mapsto (1, f)$, inverts quasi-isomorphisms and is their localization. Its morphism sets are abelian groups, and Q is additive. Moreover $Qf = Qg$ if and only if $fr = gr$ for some quasi-isomorphism r into their common source.

Proof. Equivalence is reflexive and symmetric. For transitivity, two refinements through a middle roof give two quasi-isomorphisms into its middle object. Lemma 4.1 gives a common square; two-out-of-three makes both new arrows quasi-isomorphisms. Composing the refinements gives the third required refinement. Thus the relation is transitive.

Compose roofs $(s, f) : X \rightarrow Y$ and $(t, g) : Y \rightarrow Z$ by using Lemma 4.1 to choose $r : P \rightarrow M$ in S and $a : P \rightarrow N$ with $fr = ta$, and set the composite to (sr, ga) . Two choices $(r, a), (r', a')$ have a common refinement of r, r' . On it, $ta = ta'$; cancellation in Lemma 4.1 refines once more to make $a = a'$. Their resulting roofs are therefore equivalent. Replacing the first roof by a refinement is handled by a common square between its refinement arrow and r ; replacing the second by a refinement is handled by a square between a and that refinement arrow. Each produces precisely a refinement of the original composite. Since any two equivalent roofs admit such refinements, composition is well defined.

For associativity take three roofs $(s, f), (t, g), (u, h)$. First form a square $gr = ub$, with $r : P \rightarrow N$ in S . Next form $fr' = trc$, with $r' : P' \rightarrow M$ in S . For the other order of composition, choose (r', rc) as the square for the first two roofs. Their numerator is $grc = ubc$, so choose the identity on P' for the last square. Both parenthesizations give (sr', hbc) . Independence of choices, already proved, gives associativity for every choice. The roof $(1, 1)$ is an identity by choosing identity squares.

For $s : M \rightarrow X$ in S , the roof $(s, 1_M)$ is inverse to $Q(s)$; either composite refines the identity roof. A functor F inverting S must send (s, f) to $F(f)F(s)^{-1}$. Common refinements leave this expression unchanged, and the composition square proves that it preserves composition. It defines the unique factorization through Q . Natural transformations also descend: a naturality square for s remains natural after inverting $F(s)$ and $G(s)$, and every roof is a composite of such an inverse and an ordinary arrow.

For $Qf = Qg$, a common refinement of their identity roofs is exactly an $r \in S$ with $fr = gr$; the converse is the same refinement. This is the stated equality criterion.

To add roofs, first give them a common denominator s by Lemma 4.1 and set $(s, f) + (s, g) = (s, f + g)$. If two choices or representatives are made, refine their denominators to a common one and use cancellation to equalize each pair of numerators. Adding the equal numerators proves independence. Negation and zero are $(s, -f)$ and $(s, 0)$; the group laws follow after common refinement. Composition is bilinear. For example, for a fixed second denominator t , construct squares for two numerators f, g , then refine their source arrows together. The equalities $fr = ta, gr = tb$ imply $(f + g)r = t(a + b)$, so the composite distributes over their

sum. Distributivity in the other variable follows directly after giving its two roofs a common denominator. Thus Q is additive. Its original finite-sum inclusions and projections retain their equations, including $i_1 p_1 + i_2 p_2 = 1$, and so remain biproducts. The zero object's identity remains zero, making it a zero object. This proves additivity of the localized category. $\square$

5. Triangles after localization

Theorem 5.1. A triangle in $D(\mathcal{A})$ is distinguished when it is isomorphic to the image of a cone triangle. With this definition $D(\mathcal{A})$ is triangulated, Q is triangulated, and each H^n descends to a functor that takes distinguished triangles to long exact cohomology sequences.

Proof. Shift acts on roofs by shifting both arrows; its inverse is the negative shift. Identity triangles descend. A roof $(s, f) : X \rightarrow Y$ is, through the isomorphism $Q(s) : M \rightarrow X$, the first map of the image of the cone triangle of f . Thus every localized map has a completion. Rotation in both directions follows from Theorem 3.1.

We give the full square-lifting argument. Suppose $bf = f'a$ in D , for chain-homotopy classes $f : X \rightarrow Y$, $f' : X' \rightarrow Y'$, and localized maps $a : X \rightarrow X'$, $b : Y \rightarrow Y'$. Write a as $X \xleftarrow{s} U \xrightarrow{a_0} X'$ and b as $Y \xleftarrow{t} V \xrightarrow{b_0} Y'$. Form a square $tv = fsu$, with $u : W \rightarrow U$ a quasi-isomorphism. In D , the maps $b_0 v$ and $f' a_0 u$ are equal. The equality criterion of Theorem 4.2 gives a quasi-isomorphism $k : W' \rightarrow W$ equalizing them in K . We now have a map of first arrows $W' \xrightarrow{vk} V$ to $X \xrightarrow{f} Y$, whose vertical maps suk, t are quasi-isomorphisms, and a map to $X' \xrightarrow{f'} Y'$, whose vertical maps are $a_0 uk, b_0$. Lemma 2.1 extends each square to cone triangles. The first cone comparison is a quasi-isomorphism by their cohomology sequences and the Five Lemma. Invert it in D and compose with the second comparison. This gives precisely the missing third map for the original localized square. It proves the square-extension axiom.

We also need cone uniqueness to handle initially chosen octahedra. For fixed T , the roofs construction writes $\text{Hom}_D(T, -)$ as the filtered colimit of $\text{Hom}_K(M, -)$ over denominators $M \rightarrow T$, ordered by refinements. This index category is filtered: common squares give common refinements, and cancellation equalizes parallel refinements. Filtered colimits of abelian groups are exact by Lemma 1.1 of *Sheaves of modules on a ringed space*. Apply that result in the larger universe when required. Lemma 2.4 therefore gives long exact representable sequences in D . The dual fraction description, furnished by the dual statements in Lemma 4.1 and the same proof as Theorem 4.2, gives the contravariant sequences as well. The Five Lemma and the inverse construction in Lemma 2.4 show that two isomorphisms in a morphism of localized triangles imply the third. Combined with square extension, this gives uniqueness of completion triangles up to an isomorphism fixing their first two objects.

Finally take two composable localized maps represented by $X \xleftarrow{s} U \xrightarrow{a} Y$ and $Y \xleftarrow{t} V \xrightarrow{b} Z$. Form $th = ar$, with $r : P \rightarrow U$ a quasi-isomorphism. The string $P \xrightarrow{h} V \xrightarrow{b} Z$ maps to the original string through the isomorphisms $Q(sr), Q(t), 1_Z$. Theorem 3.1 supplies its octahedron, whose image supplies an octahedron for the original maps. Cone uniqueness just proved compares its three chosen triangles with any three initially specified triangles, fixing their first two objects. Transport gives the full localized octahedron axiom.

Every H^n inverts quasi-isomorphisms and therefore descends by Theorem 4.2. Its long exact sequence on an image triangle is Lemma 1.2 applied to the cone, with the signs in (2.2); an isomorphic triangle has the transported sequence. This finishes the proof. $\square$

Proposition 5.2 (short exact sequences and boundedness). Every short exact sequence of complexes gives a distinguished triangle in $D(\mathcal{A})$. A complex whose cohomology vanishes below a is quasi-isomorphic to a complex whose terms vanish below a .

Proof. For $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$, the map $r : C(f) \rightarrow C$, $(b, a') \mapsto g(b)$, is a chain map. Its kernel is $C(1_A)$, under the identification $\ker g = A$, and is contractible. Lemma 1.2 makes r a quasi-isomorphism. Thus the cone triangle identifies with $A \rightarrow B \rightarrow C \rightarrow A[1]$ in D , with boundary $(-p)r^{-1}$. When the original sequence splits degreewise, Lemma 2.2 identifies this boundary with its positive Snake Lemma boundary. In

general the same agreement follows from the lift formula in Lemma 1.2, or from its naturality applied to $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ and the cone construction.

For the last assertion use the good truncation $\tau_{\geq a} K$: it is zero below a , equals $K^a / \text{im } d_K^{a-1}$ in degree a , and equals K^n above a . The outgoing differential factors through this quotient because $d^2 = 0$. The natural map from K is an isomorphism on cohomology at and above a , by kernels and images, and both cohomologies vanish below a . It is the required quasi-isomorphism. The dual good truncation uses $\ker d_K^b$ at the top degree b . $\square$

We write $D^+(\mathcal{A})$ for the full subcategory whose cohomology vanishes sufficiently far below zero, and similarly D^- and D^b . They are triangulated: shifts preserve the relevant bounds, and the long exact sequence bounds the cohomology of a cone whenever its first two objects have those bounds. Proposition 5.2 provides bounded representatives where needed. It is cohomological boundedness, rather than the terms of a particular representative, that defines these subcategories.

6. Exercises with complete solutions

Exercise 1 (easy). For a module M , show that the complex $M \xrightarrow{1} M$ in degrees $n, n+1$ is contractible. Explain why this proves that adding it to any complex does not change its object in K .

Solution. Take the only nonzero homotopy component to be the identity from degree $n+1$ to degree n . Then $dH + Hd$ is the identity in both nonzero degrees. Projection from the direct sum to the original complex has its inclusion as inverse up to the homotopy that is zero on the original summand and the indicated contraction on the new summand. Thus it is an isomorphism in K .

Exercise 2 (medium). The complex $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}/2$ in degrees $-1, 0, 1$, with the quotient as second differential, is acyclic. Prove that it is not contractible.

Solution. Multiplication by two is injective, its image is the quotient map's kernel, and that quotient map is onto. Thus all cohomology groups vanish. A contraction at degree 1 would give a homomorphism $s : \mathbb{Z}/2 \rightarrow \mathbb{Z}$ such that the quotient composed with s is the identity. Every such homomorphism is zero, since $\mathbb{Z}$ has no element of order two. No contraction exists. Localization makes this complex zero, because its map from the zero complex is a quasi-isomorphism, whereas the homotopy category retains it as a nonzero object.

Exercise 3 (medium). A map $f : M[0] \rightarrow N[0]$ of modules has a cone supported in degrees $-1, 0$. Compute its cohomology and the signs in its cone triangle.

Solution. The cone is $M \xrightarrow{f} N$, in those degrees. Its cohomology is $\ker f$ in degree -1 , $\text{coker } f$ in degree 0 , and zero otherwise. The triangle is $M[0] \xrightarrow{f} N[0] \rightarrow C(f) \xrightarrow{-p} M[1]$, so the degree -1 component of the last map is -1_M . This sign is consistent with the split-sequence calculation of Lemma 2.2; replacing it silently by a positive projection would change the adopted triangle convention.

Exercise 4 (hard). Let I be a complex such that $\text{Hom}^\bullet(A, I)$ is acyclic for every acyclic A . Without using roofs, prove that $\text{Hom}_K(L, I) \rightarrow \text{Hom}_D(L, I)$ is bijective.

Solution. If $s : M \rightarrow N$ is a quasi-isomorphism, its cone is acyclic. The contravariant Hom-cone identity in Lemma 2.4 implies that $s^* : \text{Hom}^\bullet(N, I) \rightarrow \text{Hom}^\bullet(M, I)$ is a quasi-isomorphism. The induced degree-zero cohomology map is therefore a bijection. By Lemma 1.1 it is the map on homotopy classes into the target complex. Thus $H(L) = \text{Hom}_K(L, I)$ descends to a contravariant functor on D . For $v : L \rightarrow I$ in D , send v to $H(v)(1_I)$. On homotopy classes this is plainly inverse to the canonical map in one direction. The other composite is a natural endomorphism of $\text{Hom}_D(-, I)$ which fixes 1_I . Naturality holds on ordinary arrows, on their inverted quasi-isomorphisms by inverting the naturality squares, and hence on every localized arrow. Applied to v , it forces that composite to fix v . The two maps are inverse. This will be the comparison mechanism for injective resolutions.

Injective modules, flasque sheaves and bounded-below derived functors

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A global lifting problem can be replaced by an exact complex of sheaves for which global lifting is easy. Injective modules make extensions of maps possible; flasque sheaves make extensions of sections possible. This lesson relates the two ideas and constructs the bounded-below resolutions used to define right derived functors.

The prerequisite is [Sheaves of modules on a ringed space](#), especially Theorem 2.1, Theorem 3.1 and Lemmas 5.1–5.2, together with [Complexes, cones and localization](#), Lemmas 1.1–1.2, 2.4, Theorems 3.1, 4.2, 5.1 and Proposition 5.2. We work with left modules over an arbitrary sheaf $\mathcal{O}$ of associative unital rings. No separation condition is imposed on the space. Commutativity is unnecessary throughout this lesson.

The construction sources are the Stacks project authors' *Injectives, Derived Categories and Cohomology of Sheaves*, in the [AI Integrated Stacks Project edition](#). The flasque extension argument corresponds to Tag 09SY. Spaltenstein's *Resolutions of unbounded complexes* provides historical background. Source attribution and the licence for adapted passages appear in the [course notice](#).

1. Extending maps and extending sections

An object I of an abelian category is **injective** if each map from a subobject $M \subset N$ into I extends to N . Equivalently, the contravariant functor $\text{Hom}(-, I)$ is exact: surjectivity at $\text{Hom}(M, I)$ is precisely the extension condition, while exactness at the earlier two terms follows from kernels and cokernels. A sheaf F is **flasque**, or **flabby**, if every restriction $F(V) \rightarrow F(U)$, for opens $U \subset V$, is onto. It suffices that every $F(X) \rightarrow F(U)$ be onto, since this map factors through $F(V)$.

Lemma 1.1 (injectives are flasque and restrict to injectives). An injective module sheaf is flasque. Its restriction to any open subset is injective in the module category of that open subset.

Proof. For opens $U \subset V$, Lemma 5.1 of the preceding lesson constructs a monomorphism $j_{U!}\mathcal{O}_U \rightarrow j_{V!}\mathcal{O}_V$. Under the isomorphisms $\text{Hom}(j_{W!}\mathcal{O}_W, I) = I(W)$, precomposition is restriction $I(V) \rightarrow I(U)$. Injectivity therefore makes this restriction onto. For the second claim, extension by zero is exact and left adjoint to restriction. Consequently

$$\text{Hom}_{\mathcal{O}_U}(-, I|_U) \cong \text{Hom}_{\mathcal{O}}(j_{U!}(-), I)$$

is a composite of exact functors. This is the required injectivity. $\square$

Lemma 1.2 (sections lift across a flasque kernel). Suppose

$$0 \rightarrow F \rightarrow G \rightarrow H \rightarrow 0$$

is an exact sequence of module sheaves and F is flasque. For every open U , the sequence $0 \rightarrow F(U) \rightarrow G(U) \rightarrow H(U) \rightarrow 0$ is exact. If G is also flasque, then H is flasque.

Proof. Kernels are computed on sections, so only surjectivity on sections requires proof. Fix $s \in H(U)$. Consider pairs (V, t) where $V \subset U$ is open and $t \in G(V)$ maps to $s|_V$. Order the pairs by extension: $(V, t) \leq (V', t')$ if $V \subset V'$ and $t'|_V = t$. This is a set, because opens and sections are sets. It is nonempty, containing the zero section on the empty open. A chain of pairs has an upper bound. Take the union of its opens and glue its compatible sections there. Any two chain members are comparable, which gives compatibility on their intersection. The sheaf condition gives the glued section and its uniqueness. Zorn's lemma gives a maximal pair (V, t) .

Choose $x \in U$. The epimorphism of sheaves $G \rightarrow H$ is onto on stalks. Lift the germ of s at x , represent the lift on an open neighbourhood, and shrink until its image equals s there. We obtain an open $W \subset U$ containing x and a section $t' \in G(W)$ mapping to $s|_W$. On $W \cap V$, the difference $t' - t$ has zero image in H . Since F is the kernel sheaf, it is the image of a unique section $r' \in F(W \cap V)$. Flasqueness extends r' to $r \in F(W)$. Replace t' by $t' - r$, where r is mapped into $G(W)$. Its image in H is unchanged, and it now agrees with t on $W \cap V$. Glue the two sections to a lift of s on $W \cup V$. Maximality forces $W \cup V = V$; thus $x \in V$. Since x was arbitrary, $V = U$. This proves surjectivity.

If G is flasque, take opens $U \subset V$ and $s \in H(U)$. Lift it to $G(U)$ by the first part. Extend that lift to $G(V)$, then map it to $H(V)$. The result restricts to s . Thus H is flasque. The proof uses neither commutativity of $\mathcal{O}$ nor separation of X . $\square$

The first assertion needs only the kernel to be flasque. This is useful even when the middle sheaf is not flasque. In particular, an exact sequence beginning with a flasque sheaf has an exact section sequence on every open; an arbitrary exact sheaf sequence need not, as the exponential example of the preceding lesson shows.

Proposition 1.3 (flasque resolutions compute zero higher cohomology of a flasque sheaf). Suppose module sheaves have enough injectives. Resolve a flasque sheaf F by injectives,

$$0 \longrightarrow F \longrightarrow I^0 \longrightarrow I^1 \longrightarrow \dots$$

For every open U , the augmented complex of sections is exact. For every continuous map with a compatible coefficient-ring morphism $f : X \rightarrow Y$, the augmented complex $0 \rightarrow f_*F \rightarrow f_*I^0 \rightarrow f_*I^1 \rightarrow \dots$ is exact. Consequently F is acyclic for $\Gamma(U, -)$ and f_* , once these right derived functors are constructed below.

Proof. Set $F^0 = F$ and $F^{n+1} = \text{coker}(F^n \rightarrow I^n)$. The resolution gives short exact sequences $0 \rightarrow F^n \rightarrow I^n \rightarrow F^{n+1} \rightarrow 0$. By Lemma 1.1, I^n is flasque. Starting from F^0 , Lemma 1.2 inductively makes each F^{n+1} flasque and makes each of these short exact sequences exact on U . The image of $I^{n-1}(U) \rightarrow I^n(U)$ is therefore $F^n(U)$, which is also the kernel of the next map. This proves exactness of the section complex.

For an open $V \subset Y$, apply the same argument to $U = f^{-1}V$. These section modules are exactly $(f_*I^n)(V)$. Thus the augmented pushforward complex is exact even on every open's sections, and in particular is exact as a complex of sheaves. Applying the later definition $R^q\Phi(F) = H^q(\Phi(I^\bullet))$ gives zero for $q > 0$ for either functor Φ . There is no need for a description of higher direct images to prove this consequence, so it does not depend on a later local-description theorem. $\square$

The existence assumption in Proposition 1.3 will be discharged by the explicit functorial embedding construction in the next section. Its statement separates an elementary lifting argument from the choice of a resolution. It is not an appeal to an unproved acyclicity theorem.

2. Functorial injective embeddings

We now discharge the existence assumption of Proposition 1.3. First we prove an extension test for modules over a ring. Then we construct a functorial embedding using character modules. Choosing injective hulls independently would not supply this functoriality.

Lemma 2.1 (Baer's criterion). A left module E over a ring A is injective if every A -linear map from a left ideal of A to E extends to A . **Proof.** Injectivity immediately gives the stated extension property. Conversely, suppose that property holds. Let $N \subset M$ be left A -modules and $f : N \rightarrow E$ linear. Order its extensions (N', f') , with $N \subset N' \subset M$, by restriction. A chain has an upper bound: take the union of its submodules and the compatible union of its maps. Zorn's lemma gives a maximal extension (N', f') . If $x \in M \setminus N'$, put $J = \{a \in A : ax \in N'\}$. This is a left ideal, and $a \mapsto f'(ax)$ is a linear map $J \rightarrow E$. By assumption it extends to $b : A \rightarrow E$. Define

$$f''(n + ax) = f'(n) + b(a) \quad (n \in N', a \in A).$$

If $n + ax = n' + a'x$, then $(a - a')x = n' - n$, so $a - a' \in J$ and $b(a - a') = f'(n' - n)$. Thus the formula is well defined. It is linear and extends f' to the strictly larger submodule $N' + Ax$, contradicting maximality.

Hence $N' = M$, and every map from a submodule extends. This is injectivity. Only the usual axiom of choice, in its Zorn form, is used; no existence theorem for injectives is assumed. $\square$

Lemma 2.2 (functorial injective embeddings of modules). Let A be a ring. Put $E_A = \text{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z})$, a left A -module by $(a\varphi)(b) = \varphi(ba)$. For a left A -module M put

$$I_A(M) = \prod_{\varphi \in \text{Hom}_A(M, E_A)} E_A, \quad \iota_M(m) = (\varphi(m))_{\varphi}. \quad (2.1)$$

Then $I_A(M)$ is injective and ι_M is injective. For $u : M \rightarrow N$, the formula $I_A(u)((y_{\varphi})_{\varphi}) = (y_{\chi \circ u})_{\chi}$, with χ running over $\text{Hom}_A(N, E_A)$, makes I_A a functor and ι a natural transformation.

Proof. E_A is a left module: $((aa')\varphi)(b) = \varphi(baa') = (a(a'\varphi))(b)$. The maps

$$\Phi(\varphi) = (m \mapsto \varphi(m)(1)), \quad \Psi(\psi) = (m \mapsto (b \mapsto \psi(bm)))$$

are inverse bijections between $\text{Hom}_A(M, E_A)$ and $\text{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})$, natural in M . Here $\Psi(\psi)$ is A -linear, since $\Psi(\psi)(am)(b) = \psi(bam) = (a\Psi(\psi)(m))(b)$. The group $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$ -module. To see this, use Baer's criterion (Lemma 2.1): the ideals of $\mathbb{Z}$ are the $n\mathbb{Z}$. For $n = 0$ there is nothing to extend. For $n \geq 1$, a map $n\mathbb{Z} \rightarrow \mathbb{Q}/\mathbb{Z}$ sending n to q extends to $\mathbb{Z}$ by sending 1 to any q' with $nq' = q$, which exists since $\mathbb{Q}/\mathbb{Z}$ is divisible. Exact sequences of A -modules are exact as groups. So $\text{Hom}_A(-, E_A) \cong \text{Hom}_{\mathbb{Z}}(-, \mathbb{Q}/\mathbb{Z})$ is exact, and E_A is injective. A product of injectives is injective, because $\text{Hom}(-, \prod E) = \prod \text{Hom}(-, E)$ and products of exact sequences of groups are exact: injectivity and kernels are coordinatewise, and a preimage of each coordinate of a target element can be chosen independently. This last assertion uses choice for a set-indexed family, and concerns groups, not exactness of products of sheaves. So $I_A(M)$ is injective. If $m \neq 0$, define ψ on the subgroup $\mathbb{Z}m$ by $\psi(m) = 1/n + \mathbb{Z}$ if m has finite order $n \geq 2$, and $\psi(m) = 1/2 + \mathbb{Z}$ if m has infinite order. Extend ψ to M , as $\mathbb{Q}/\mathbb{Z}$ is injective. Then $\varphi = \Psi(\psi)$ has $\varphi(m)(1) = \psi(m) \neq 0$, so $\iota_M(m) \neq 0$. Finally $I_A(u)(\iota_M(m)) = (\chi(u(m)))_{\chi} = \iota_N(u(m))$, and the displayed formula gives $I_A(vu) = I_A(v)I_A(u)$ and $I_A(\text{id}) = \text{id}$. $\square$

Theorem 2.3 (functorial injective embeddings). Let $\mathcal{O}$ be any sheaf of rings on X . Put

$$E(F) = \prod_{x \in X} x_*(I_{\mathcal{O}_x}(F_x)),$$

where $I_{\mathcal{O}_x}$ is the functor (2.1) for the ring $\mathcal{O}_x$, and let $\iota_F : F \rightarrow E(F)$ have x -component the morphism that corresponds under (5.2) of the preceding lesson to $\iota_{F_x} : F_x \rightarrow I_{\mathcal{O}_x}(F_x)$. Then $E(F)$ is injective, ι_F is a monomorphism, and E is a functor with ι natural. So $\text{Mod}(\mathcal{O})$ has enough injectives, with functorial injective embeddings.

Source: the Stacks project, [Tag 01DI in its AI Integrated Stacks Project edition](#), gives the commutative-ring construction. The full character-module proof here also checks the associative unital-ring generality; proof contributions by GPT-6 Astra at Ultra are edited by GPT-6.1 Sol at Ultra.

Proof. Products exist and are computed on sections ([Theorem 3.1 of the preceding lesson\(1\)](#)), and a product of injectives is injective, as in the proof of Lemma 2.2. Each factor is injective: for an injective stalk module J , the adjunction of Lemma 5.2 of the preceding lesson identifies $\text{Hom}(-, x_*J)$ with Hom over the stalk ring from the exact stalk functor into J . This is an exact contravariant functor. A product of injective sheaves is injective because its Hom functor is the product of their exact group-valued Hom functors; coordinatewise preimages prove surjectivity. Thus no assertion that products of sheaves are exact is needed. Explicitly, ι_F sends $s \in F(V)$ to $(\iota_{F_x}(s_x))_{x \in V}$. If this is zero, each $s_x = 0$, as ι_{F_x} is injective, so $s = 0$. Kernels are computed on sections ([Theorem 2.1 of the preceding lesson\(1\)](#)), so ι_F is a monomorphism. For $u : F \rightarrow G$ put $E(u) = \prod_x x_*(I_{\mathcal{O}_x}(u_x))$. Naturality of ι holds factor by factor, by Lemma 2.2. $\square$

The construction gives an injective embedding, not an injective hull. For example, over the field with two elements, the character module of the ring has two elements. For the one-dimensional module, there are two linear maps into that character module, including the zero map. The indicated product is two-dimensional, and evaluation embeds the original one-dimensional module into it. A different one-dimensional subspace

has zero intersection with the embedded image. Thus the embedding is not essential, while a hull would require essentiality. The functorial construction is useful precisely without a claim of minimality.

The embedding functor need not be additive in the morphism variable. Its product coordinates are indexed by all maps into a fixed injective cogenerator, and a module map changes coordinates by precomposition. Identities, composition and naturality of the embedding hold by the explicit formula. The later passage to derived functors uses comparison maps unique up to chain homotopy, not additivity of this embedding functor.

3. Complexes that admit comparison maps

A complex I is **K-injective** if every chain map $A \rightarrow I$ from an acyclic complex is null-homotopic. Equivalently, $\text{Hom}^\bullet(A, I)$ is acyclic for every acyclic A : by Lemma 1.1 of the common reading, its degree- r cohomology is $\text{Hom}_K(A, I[r])$, and shifting the acyclic source gives the equivalence. This definition concerns a complex, not just its individual terms.

Lemma 3.1 (closure properties). Shifts, cones of maps between K-injectives, homotopy-equivalent complexes and degreewise products of K-injective complexes are K-injective, whenever the indicated products exist.

Proof. Shifting the source in (1.2) of the common reading proves the shift assertion. For a map $I \rightarrow J$, the covariant Hom-cone identity of its Lemma 2.4 identifies $\text{Hom}^\bullet(A, C(I \rightarrow J))$ with the cone of a map between two acyclic group complexes. It is therefore acyclic. A homotopy equivalence induces inverse maps up to homotopy after applying Hom, so it preserves the condition. Finally,

$$\text{Hom}^\bullet\left(A, \prod_{\lambda} I_{\lambda}\right) \cong \prod_{\lambda} \text{Hom}^\bullet(A, I_{\lambda}).$$

Products of exact sequences of groups are exact by coordinatewise lifting and choice, as proved in Lemma 2.2. The product on the right is consequently acyclic. This uses exactness of products of groups, and does not presume that products of module sheaves are exact. $\square$

Lemma 3.2 (bounded-below complexes of injectives are K-injective). Let I be a complex of injectives with $I^n = 0$ for $n < a$, and let A be acyclic. Then every chain map $f : A \rightarrow I$ is null-homotopic.

Proof. We build maps $s^n : A^n \rightarrow I^{n-1}$ with

$$f^n = d s^n + s^{n+1} d \quad \text{for all } n. \quad (3.1)$$

Put $s^n = 0$ for $n \leq a$. Then (3.1) holds for $n < a$, where both sides are 0. Suppose s^k is built for $k \leq n$ and (3.1) holds for all indices below n . Put $g = f^n - d s^n : A^n \rightarrow I^n$. Using (3.1) at $n-1$, and $d^2 = 0$,

$$g d = f^n d - d s^n d = d f^{n-1} - d(f^{n-1} - d s^{n-1}) = 0.$$

So g vanishes on the image of $d : A^{n-1} \rightarrow A^n$. As A is acyclic, that image is the kernel of $d : A^n \rightarrow A^{n+1}$. Thus g factors uniquely through the epimorphism from A^n onto $\text{im } d$, giving a map from that subobject of A^{n+1} to I^n . Since I^n is injective, the factored map extends to $s^{n+1} : A^{n+1} \rightarrow I^n$. Then $s^{n+1} d = g$, which is (3.1) at n . $\square$

The induction needs a lower bound. An acyclic K-injective complex is contractible, since its own identity is one of the maps to which the defining vanishing condition applies. A termwise injective complex without a lower bound need not have this property; the singular-ring example below gives a counterexample.

Theorem 3.3 (maps into a K-injective complex). For a K-injective I , a quasi-isomorphism $s : M \rightarrow N$ induces a bijection

$$s^* : \text{Hom}_{K(\mathcal{A})}(N, I) \longrightarrow \text{Hom}_{K(\mathcal{A})}(M, I).$$

Consequently the natural map from either homotopy-category Hom group to the corresponding derived-category Hom group with target I is bijective.

Proof. The complex $\text{Hom}^\bullet(A, I)$ is acyclic when A is acyclic: its degree- r cohomology is $\text{Hom}_{K(\mathcal{A})}(A, I[r])$, which vanishes because a shift of A is again acyclic. Put $U = \text{Hom}^\bullet(N, I)$, $V = \text{Hom}^\bullet(M, I)$, and $u = s^* : U \rightarrow V$. In degree r , write a map out of $\text{Cone}(s)$ as $(a, b) \in U^r \oplus V^{r-1}$. Its differential is

$$(a, b) \mapsto (d_U a, d_V b + (-1)^{r+1} u(a)).$$

Thus $(a, b) \mapsto ((-1)^{r-1} b, a)$ is a chain isomorphism onto $\text{Cone}(u)[-1]$: its differential is $(v, a) \mapsto (-d_V v - u(a), d_U a)$. This explicitly identifies the Hom complex out of $\text{Cone}(s)$ with the shifted cone of

$$\text{Hom}^\bullet(N, I) \longrightarrow \text{Hom}^\bullet(M, I).$$

The cone of s is acyclic. Thus this map of Hom complexes is a quasi-isomorphism; its degree-zero cohomology gives the displayed bijection.

Here is a direct use of the localization universal property that avoids assuming a choice of roofs. The contravariant functor $H(L) = \text{Hom}_K(L, I)$ sends every quasi-isomorphism to an isomorphism by the first paragraph, so it factors through $D(\mathcal{A})^{\text{op}}$. For $v : L \rightarrow I$ in D , apply this factored functor to v and to the element 1_I ; this gives an element of $\text{Hom}_K(L, I)$. The resulting map is inverse to the natural map into $\text{Hom}_D(L, I)$: one composite fixes every chain-homotopy class, and the other is a natural endomorphism of the representable functor $\text{Hom}_D(-, I)$ fixing 1_I . Naturality for inverted quasi-isomorphisms follows by inverting their naturality squares. Every derived morphism is generated by ordinary morphisms and these inverses, so the second composite fixes all morphisms too. $\square$

Corollary 3.4 (the full characterization). For a complex I , the following are equivalent: it is K-injective; $\text{Hom}_K(A, I) = 0$ for every acyclic A ; every quasi-isomorphism into the source induces a bijection on homotopy classes of maps to I ; the natural map $\text{Hom}_K(L, I) \rightarrow \text{Hom}_D(L, I)$ is bijective for every L .

Proof. The first two conditions are the definition. Theorem 3.3 proves the third and fourth from them. Conversely, the third condition applied to $0 \rightarrow A$ makes $\text{Hom}_K(A, I) = 0$. The fourth does too, since any acyclic A becomes a zero object in D by the quasi-isomorphism $0 \rightarrow A$. Thus all conditions are equivalent. A quasi-isomorphism between K-injectives is a homotopy equivalence: the third condition supplies an inverse on one side, and its injectivity forces the other composite to be the identity. $\square$

4. Right derived functors on bounded-below complexes

For an additive functor Φ , applying it degreewise gives a functor on homotopy categories, but it need not preserve quasi-isomorphisms. We seek $R\Phi : D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$ with a comparison on $K^+(\mathcal{A})$. Its **right-derived universal property** is this: every natural comparison $\Phi \rightarrow T \circ Q$, with T a functor on $D^+(\mathcal{A})$, factors uniquely as $\Phi \rightarrow R\Phi \circ Q \rightarrow T \circ Q$. This makes $R\Phi$ the initial comparison out of Φ .

Theorem 4.1. Let $\mathcal{A}$ and $\mathcal{B}$ be abelian, and assume that every object of $\mathcal{A}$ embeds in an injective object. Let $\Phi : \mathcal{A} \rightarrow \mathcal{B}$ be additive.

1. Every bounded-below complex K has a qis $K \rightarrow I$ into a bounded-below complex of injectives. If $K^n = 0$ for $n < a$, it can be chosen injective in each degree, with $I^n = 0$ for $n < a$.
2. Such an I is K-injective (Lemma 3.2). Hence $\text{Hom}_{K(\mathcal{A})}(L, I) = \text{Hom}_{D(\mathcal{A})}(L, I)$ for every L . A morphism into I extends along a qis, uniquely up to homotopy.
3. The rule $R\Phi(K) = \Phi(I)$ gives a triangulated functor $D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$, and this functor is the right derived functor $R\Phi$. If Φ is left exact, $R^0\Phi = \Phi$ and $(R^i\Phi)_i$ is a universal δ -functor.
4. (Leray) If Φ is left exact and J is a bounded-below complex with $R^i\Phi(J^n) = 0$ for all $i > 0$ and all n , then $\Phi(J) \rightarrow R\Phi(J)$ is an isomorphism. Left exactness cannot be dropped. For a general additive Φ , the terms must also satisfy $\Phi(J^n) \cong R^0\Phi(J^n)$, which holds by (3) when Φ is left exact. For $\Phi = - \otimes_{\mathbb{Z}} \mathbb{Z}/2$ on $\text{Mod}(\mathbb{Z})$ and $J = \mathbb{Z}$ in degree 0, the injective resolution $\mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z}$ gives $R\Phi(\mathbb{Z}) = 0$, because $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are divisible, while $\Phi(\mathbb{Z}) = \mathbb{Z}/2$.
5. For $\mathcal{A} = \text{Mod}(\mathcal{O})$, with $\mathcal{O}$ any sheaf of rings, (1)–(4) hold by Theorem 2.3. This gives Rf_* , $R\Gamma(U, -)$, and $R\text{Hom}_{\mathcal{O}}(G, -)$ on D^+ . The support functor is treated in *Sections with support and the localization triangle*.

We have proved the comparison assertion in Section 3. We now construct resolutions and establish the derived-functor properties. The localization, triangulated structure and cohomology exact sequences used in these proofs are Theorems 3.1, 4.2, 5.1 and Lemma 1.2 of *Complexes, cones and localization*. The full proofs of [injective resolutions](#), [K-injective comparison](#) and [acyclic terms](#) in AI Integrated Stacks Project are proof sources. The explicit construction and expanded checks are by GPT-6 Astra at Ultra, checked and edited by GPT-6.1 Sol at Ultra, under GFDL-1.2-or-later.

Constructing the resolution one degree at a time

Proof of Theorem 4.1(1). The elementary operation is to replace one term of a complex by an injective object. Suppose C is a complex and choose a monomorphism $u : C^n \rightarrow E$, with E injective. Form the pushout

$$P = (E \oplus C^{n+1}) / \text{im}(u, -d_C^n).$$

Replace C^n by E and C^{n+1} by P . The incoming differential is ud_C^{n-1} , the new differential $E \rightarrow P$ sends e to $[e, 0]$, and the outgoing differential sends $[e, c]$ to $d_C^{n+1}c$. These maps square to zero: $[u(c), 0] = [0, d_C^n c]$, and $d_C^2 = 0$. This gives a complex C' and a chain map $C \rightarrow C'$.

The map $C^{n+1} \rightarrow P$ is a monomorphism. Indeed, the pullback of the monomorphism $(u, -d_C^n) : C^n \rightarrow E \oplus C^{n+1}$ along the second-summand inclusion is zero, because its projection to E has kernel $\ker u = 0$. Thus that summand has zero intersection with the subobject being quotiented. This is an argument by kernels and pullbacks in an arbitrary abelian category. The map of complexes is termwise monic. Quotienting in degree $n + 1$ also by the image of C^{n+1} leaves $E/u(C^n)$. The induced differential from degree n is the identity of that quotient. Consequently the quotient complex is

$$E/u(C^n) \xrightarrow{1} E/u(C^n)$$

in degrees $n, n + 1$, and zero in other degrees. It is contractible, so the cohomology exact sequence proves that $C \rightarrow C'$ is a quasi-isomorphism.

Start at $n = a$, then perform the operation at $a + 1, a + 2, \dots$. Once degree n has been replaced, later operations leave that object unchanged; once degree $n + 1$ has been replaced, the differential out of degree n is unchanged as well. Define I using these eventual objects and differentials. This uses no infinite colimit in $\mathcal{A}$: each degree is defined at a finite stage. All its terms are injective, it is zero below a , and the map $K \rightarrow I$ is termwise monic. For fixed r , the three terms and two differentials computing H^r have stabilized after finitely many steps. Every step was a quasi-isomorphism, so $H^r(K) \rightarrow H^r(I)$ is an isomorphism. This proves (1). $\square$

Comparison, triangulation and universality

Proof of Theorem 4.1(2)–(3). Choose a resolution $q_K : K \rightarrow I_K$. Use part (1). A chain map $f : K \rightarrow L$ has a unique homotopy-class extension $\bar{f} : I_K \rightarrow I_L$ satisfying $\bar{f}q_K = q_L f$ in $K(\mathcal{A})$, by Theorem 3.3. Uniqueness proves compatibility with composition and identities. If f is a quasi-isomorphism, $\bar{f}$ is one too. A quasi-isomorphism between K-injectives is a homotopy equivalence: apply Theorem 3.3 first to obtain a left inverse and again to show the other composite is the identity.

An additive Φ preserves chain homotopies, shifts and mapping cones, since these involve only finite direct sums, differentials and sums of maps. Therefore $\Phi(\bar{f})$ is invertible in $K(\mathcal{B})$ whenever $\bar{f}$ is a homotopy equivalence. The assignment consequently factors through the localization of $K^+(\mathcal{A})$. This localization identifies with the full $D^+(\mathcal{A})$: Proposition 5.2 of the common reading gives its bounded-below representatives, while Theorem 3.3 applied both in K^+ and in K computes all maps between such representatives by the same maps into their bounded-below injective resolutions. Thus inclusion is fully faithful as well as essentially surjective. All images lie in $D^+(\mathcal{B})$. Different choices of I_K give a unique natural comparison isomorphism: extend the identity between resolutions and apply the preceding argument. These comparisons satisfy the cocycle identity by uniqueness.

The functor is additive because the unique extension of $f + g$ is the sum of the unique extensions of f and g . It is triangulated as follows. A triangle in D^+ is an image of a cone triangle up to isomorphism, by Theorem

5.1 of the common reading. Replace its first two objects by bounded-below injective resolutions. Theorem 3.3 extends its first map, and Lemma 2.1 of the common reading compares its cone with the cone of the extended map. The comparison is a quasi-isomorphism by the cohomology exact sequences. Shifts and cones of K-injectives are K-injective by Lemma 3.1, and those resolutions' cones remain bounded below with injective terms, since finite sums of injectives are injective. Thus the triangle can be computed by this triangle of resolutions. Applying Φ preserves its cone and shift, proving the claim.

We verify the stated universal property. Applying Φ to q_K gives a natural comparison $\eta_K : \Phi(K) \rightarrow R\Phi(K)$ in $D(\mathcal{B})$. If $T : D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$ is another functor with a natural map $\theta_K : \Phi(K) \rightarrow T(K)$ on $K^+(\mathcal{A})$, the only possible induced map $R\Phi(K) \rightarrow T(K)$ is

$$\Phi(I_K) \xrightarrow{\theta_{I_K}} T(I_K) \xrightarrow{T(q_K)^{-1}} T(K).$$

It is natural by the comparison equation for $\bar{f}$, it composes with η to θ , and it is forced by that equation at q_K . To see uniqueness explicitly, at a K-injective resolution we may use the identity as its resolution, so the comparison there is the identity. Naturality at q_K then forces the displayed formula. The result is independent of that choice by the unique resolution comparison. This is the right-derived-functor property.

Define $R^i\Phi(M) = H^i(R\Phi(M[0]))$. If Φ is left exact, the initial exact sequence $0 \rightarrow M \rightarrow I^0 \rightarrow I^1$ gives $R^0\Phi(M) = \Phi(M)$. The resolution has no negative terms, so $R^i\Phi = 0$ for $i < 0$. Short exact sequences give distinguished triangles by Proposition 5.2 of the common reading. Their images under $R\Phi$ give natural long exact cohomology sequences. Thus the nonnegative functors $R^i\Phi$, with these connecting maps, form a cohomological δ -functor. Universality means that a natural transformation in degree zero into any other such δ -functor extends uniquely to every degree, commuting with all connecting maps.

Here are the details of universality of that δ -functor. Each monomorphism $M \rightarrow E$ into an injective kills $R^i\Phi(M)$ after mapping to $R^i\Phi(E)$ for $i > 0$, because $E[0]$ is already a resolution. Let T^i be any other cohomological δ -functor and choose a natural map $R^0\Phi \rightarrow T^0$. In $0 \rightarrow M \rightarrow E \rightarrow C \rightarrow 0$, exactness says that

$$R^0\Phi(C) \rightarrow R^1\Phi(M) \text{ is onto,} \quad R^{i-1}\Phi(C) \xrightarrow{\sim} R^i\Phi(M) \quad (i > 1).$$

For $i = 1$, compose the chosen degree-zero map at C with the boundary into $T^1(M)$. This vanishes on the image of $R^0\Phi(E)$, by naturality and exactness of T , so it descends uniquely. Inductively, compose the degree- $(i-1)$ map at C with the boundary into $T^i(M)$ and the inverse of the displayed isomorphism. Given a map $M \rightarrow M'$, extend its composite into an injective containing M' across $M \rightarrow E$; this gives a morphism of the two short exact sequences. Induction and naturality of boundary maps prove naturality of the constructed degree- i map. Two embeddings of the same M are compared with their diagonal embedding into the finite sum of the two injectives. The same argument proves independence of the embedding. Compatibility with arbitrary connecting maps follows by embedding the middle term of a short exact sequence into an injective and using the induced diagram of cokernels; the defining construction is precisely the connecting-map square in that diagram. Finally the displayed surjection and isomorphisms force every degree of any extension. Thus the extension exists uniquely as a morphism of δ -functors. This proves universality. $\square$

Why acyclic terms suffice in the bounded-below case

In the universality argument above, compatibility with an arbitrary boundary can be checked without any further resolution theorem. For $0 \rightarrow M \rightarrow N \rightarrow P \rightarrow 0$, embed N in an injective E and put $C = E/M$. The identity on M , the embedding $N \rightarrow E$, and the induced map $P \rightarrow C$ give a morphism to $0 \rightarrow M \rightarrow E \rightarrow C \rightarrow 0$. Naturality of each δ -functor gives

$$\delta_{M,N,P}^i = \delta_{M,E,C}^i \circ R^i\Phi(P \rightarrow C),$$

and the same formula for T . The constructed maps commute with the second boundary by their definition and with $P \rightarrow C$ by their established naturality. They therefore commute with the first boundary as well.

Proof of Theorem 4.1(4)–(5). Call an object A right Φ -acyclic when the comparison $\Phi(A)[0] \rightarrow R\Phi(A[0])$ is an isomorphism. For left exact Φ , part (3) shows that this is equivalent to $R^i\Phi(A) = 0$ for $i > 0$. For a

general additive functor the degree-zero comparison must additionally be an isomorphism, as the statement requires.

First let J have only finitely many nonzero terms, all right Φ -acyclic. Remove its bottom term using the degreewise split sequence

$$0 \longrightarrow \sigma_{\geq a+1} J \longrightarrow J \longrightarrow J^a[-a] \longrightarrow 0,$$

where σ denotes brutal truncation, with the indicated boundary differential set to zero. Applying Φ still gives a degreewise split exact sequence. The associated cone triangles and the natural comparison with $R\Phi$ show, by induction on the number of terms and the cohomology exact sequences, that $\Phi(J) \rightarrow R\Phi(J)$ is an isomorphism.

For a bounded-below J , fix an integer r and use

$$0 \longrightarrow \sigma_{\geq r+2} J \longrightarrow J \longrightarrow \sigma_{\leq r+1} J \longrightarrow 0.$$

The last complex is bounded. The first, and also its image under Φ , vanish in degrees below $r + 2$. By (1) it has an injective resolution vanishing below $r + 2$, so its derived image also has no cohomology below $r + 2$. The two long exact sequences therefore identify degree- r cohomology of both middle terms with that of the bounded last terms. The already proved bounded case gives the desired isomorphism in degree r . This holds for every r , proving (4). Part (5) now follows by applying the construction with the injective embeddings of Theorem 2.3. No countability, commutativity, spectral sequence, or exactness of infinite products of sheaves was used. $\square$

5. Examples: ordinary, flasque and unbounded resolutions

On a one-point space with coefficient ring R , sheaves are simply R -modules. Theorem 2.3 reduces to the character-module product of Lemma 2.2. It is a functorial injective embedding. It is not generally a hull, as the example over the field with two elements in Section 2 demonstrates. For $R = \mathbb{Z}$, an ordinary injective resolution of $\mathbb{Z}$ is

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0.$$

Both nonzero resolution terms are divisible, and Baer's criterion makes them injective. Thus, for $\Phi = \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n, -)$, the degree-zero term after applying Φ is zero and the degree-one term is the subgroup of $\mathbb{Q}/\mathbb{Z}$ killed by n . It is cyclic of order n , by the map $1 \mapsto 1/n + \mathbb{Z}$. Hence $R^1\Phi(\mathbb{Z}) = \mathbb{Z}/n$ and all other nonnegative derived values vanish. The right derived functor detects the extension information absent from the original Hom group.

For a different additive functor, $\Phi = - \otimes_{\mathbb{Z}} \mathbb{Z}/2$, the same resolution has zero image: a divisible group has zero quotient by twice itself. Thus its right derived value at $\mathbb{Z}[0]$ is zero, although $\Phi(\mathbb{Z}) = \mathbb{Z}/2$. This is precisely why higher vanishing alone does not define right acyclicity for a general additive functor. Later, left resolutions will give the derived tensor functor appropriate to tensor's right exactness.

The constant sheaf $\underline{\mathbb{Z}}$ on $[0, 1]$ is not flasque. On the disconnected open $(0, 1/3) \cup (2/3, 1)$, the section with value zero on the first component and one on the second cannot extend to a locally constant function on the connected interval. There is nevertheless a direct flasque resolution. Put

$$C^0(F) = \prod_{x \in X} x_* F_x.$$

Its sections on U are $\prod_{x \in U} F_x$. Restriction forgets coordinates, so it is onto: fill missing coordinates with zero. These sections form a sheaf because compatible families glue coordinate by coordinate. The germ map $F \rightarrow C^0(F)$ is monic by the stalk criterion. Let F^1 be its cokernel, form $C^0(F^1)$, and continue with successive cokernels. The resulting augmented complex

$$0 \rightarrow F \rightarrow C^0(F) \rightarrow C^0(F^1) \rightarrow C^0(F^2) \rightarrow \dots$$

is exact by its construction, and all its resolution terms are flasque. It is the Godement flasque resolution. Proposition 1.3 and Theorem 4.1(4) show that its section or pushforward complex computes the corresponding derived functor. Its terms need not be injective; acyclicity suffices because the resolution is bounded below.

An unbounded K-injective example goes in the other direction. On a point, put

$$I = \prod_{m \geq 0} (\mathbb{Q}/\mathbb{Z})[m].$$

Each factor is bounded below and injective, hence K-injective by Lemma 3.2. Lemma 3.1 makes the product K-injective. In degree $n \leq 0$, exactly the factor with $m = -n$ is nonzero, so $I^n = \mathbb{Q}/\mathbb{Z}$; in positive degrees it is zero. Its differentials are zero. Thus it is not bounded below. In fact this product represents the product in the derived category too: Theorem 3.3 and the Hom-product identity of Lemma 3.1 give

$$\mathrm{Hom}_D(K, I) \cong \prod_{m \geq 0} \mathrm{Hom}_D(K, (\mathbb{Q}/\mathbb{Z})[m])$$

for every K .

Termwise injectivity alone does not give such a conclusion. Let $R = k[\epsilon]/(\epsilon^2)$ for a field k . Its only proper nonzero ideal is (ϵ) : an element with nonzero constant term is a unit, while a nonzero scalar multiple of ϵ generates that ideal. An R -linear map $(\epsilon) \rightarrow R$ sends ϵ to $c\epsilon$, because its image is killed by ϵ . Multiplication by c extends it to R . Baer's criterion therefore proves that R is injective over itself. The bi-infinite complex

$$\dots \xrightarrow{\epsilon} R \xrightarrow{\epsilon} R \xrightarrow{\epsilon} R \xrightarrow{\epsilon} \dots$$

is termwise injective and acyclic, since $\ker(\epsilon) = \epsilon R = \mathrm{im}(\epsilon)$. A contraction would require $1 = \epsilon s^n + s^{n+1}\epsilon$ in every degree. Each R -endomorphism is multiplication by an element of R , so the right side has image in ϵR and cannot be the identity. Its identity is a non-null-homotopic map from an acyclic complex into itself. It is not K-injective. The failure comes from the missing starting degree in Lemma 3.2's induction.

6. Exercises with complete solutions

Exercise 1 (easy). At an arbitrary point x , prove directly that a skyscraper x_*J of an injective $\mathcal{O}_x$ -module is injective. Do not assume that x is a closed point.

Solution. For a monomorphism $F \rightarrow G$, exactness of stalks gives $F_x \rightarrow G_x$ monic. A map $F \rightarrow x_*J$ corresponds, under the stalk-skyscraper adjunction in Lemma 5.2 of the preceding lesson, to $F_x \rightarrow J$. Extend it to $G_x \rightarrow J$ by injectivity and transport back through that adjunction. This extends the original sheaf map. No assertion about the closure of x occurs in the proof; the sheaf's values and the adjunction already apply to every point.

Exercise 2 (medium). Give a second proof that injective module sheaves are flasque, using their embedding in Theorem 2.3 and a splitting, instead of extension by zero.

Solution. An injective I embeds into $E(I)$. Extend 1_I across this monomorphism to a retraction $r : E(I) \rightarrow I$. Each $E(I)(U)$ is a product of stalk injectives indexed by $x \in U$; its restrictions are onto by filling missing coordinates with zero. For $U \subset V$, include a section $s \in I(U)$ in $E(I)(U)$, extend it to $E(I)(V)$, and apply $r(V)$. Naturality of the retraction makes its restriction equal s . This proves flasqueness.

Exercise 3 (medium). Use dimension shifting to prove that a flasque F has $H^q(U, F) = 0$ for $q > 0$, where $H^q(U, F) = R^q\Gamma(U, F)$.

Solution. Embed F into an injective I and put $C = I/F$. By Lemmas 1.1 and 1.2, C is flasque and $I(U) \rightarrow C(U)$ is onto. Since I is already its own injective resolution, all its positive derived section functors vanish. The long exact sequence therefore makes $H^1(U, F)$ the cokernel of $I(U) \rightarrow C(U)$, hence zero, and gives $H^q(U, F) \cong H^{q-1}(U, C)$ for $q > 1$. Induct on q , applying the same first-step proof to each flasque quotient. This proves vanishing on every open, without an assumption of compactness or separation.

Exercise 4 (hard). Prove all implications in Corollary 3.4, including injectivity and surjectivity of the map to derived Hom, using the roofs of the common reading.

Solution. Assume every map from an acyclic complex to I is null-homotopic. Shifts have the same property, so both outer groups in the contravariant cone exact sequence for a quasi-isomorphism $s : M \rightarrow N$ vanish. Thus $s^* : \operatorname{Hom}_K(N, I) \rightarrow \operatorname{Hom}_K(M, I)$ is bijective. A derived map $L \rightarrow I$ has a roof $L \xleftarrow{s} M \xrightarrow{f} I$. The unique homotopy class $g : L \rightarrow I$ with $gs = f$ maps to this roof, proving surjectivity. If two homotopy classes have the same derived image, the equality criterion in Theorem 4.2 of the common reading equalizes them after some quasi-isomorphism into L . Injectivity of its induced s^* equalizes the two classes before localization, proving injectivity. Conversely, bijectivity to derived Hom implies vanishing from acyclic sources because those sources are zero in D . The source-comparison condition implies the same vanishing by applying it to $0 \rightarrow A$ for acyclic A . Vanishing is exactly K-injectivity, completing the equivalences.

Exercise 5 (hard). Suppose Φ is left exact, J is bounded below, and every term J^n is Φ -acyclic. Explain precisely why one cannot prove the comparison in degree r by truncating only at degree r . Give the correct truncation and prove the comparison.

Solution. A quotient brutal truncation ending at degree r deletes d^r , which is needed to determine $\ker d^r$ and hence H^r . Moreover the removed tail starts at degree $r + 1$, whose derived cohomology in degree $r + 1$ can occur as the next term of the long exact sequence. Instead use the split sequence with tail $\sigma_{\geq r+2}J$ and quotient $\sigma_{\leq r+1}J$. Both the tail's ordinary image and its derived image have zero cohomology in degrees $r, r+1$, because the bounded-below resolution preserves its lower term bound. The long exact sequences identify the degree- r middle cohomologies with those of the bounded quotient. Induction on the quotient's finitely many terms, using the split bottom-term sequence, proves its comparison. Thus the degree- r comparison for J is an isomorphism. The two-degree buffer is the reason this argument works without a spectral sequence.

The next lesson proves the existence of resolutions for all complexes in a Grothendieck abelian category. The unbounded existence proof supplies a replacement for the lower-bound induction; the comparison theorem already established here continues to apply unchanged.

K-injective resolutions in Grothendieck abelian categories

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A bounded-below resolution has a first degree. For an unbounded complex, starting farther and farther to the left does not by itself produce a compatible resolution. The solution developed here is to enforce the desired extension and homotopy properties on a set of small tests, then take a sufficiently long increasing union. The size estimates ensure that every test into the union appeared at an earlier stage where the construction dealt with it.

Read [Sheaves of modules on a ringed space](#), Theorems 2.1 and 3.1 and Lemmas 5.1–5.2; [Injective modules and bounded-below derived functors](#), Section 3 and Theorem 4.1; and [Complexes, cones and localization](#), Theorems 3.1, 4.2 and 5.1. Those readings supply abelian categories, exact filtered colimits of module sheaves, Hom complexes, cones, localization, and the comparison theorem for K-injectives. We use sets and choice in a fixed universe. A larger universe temporarily accommodates localization; this lesson proves that its Hom sets lie in the original universe.

The construction follows the Stacks project authors' *Injectives*, sections on Grothendieck categories and K-injective complexes, and *Derived Categories*, in the [AI Integrated Stacks Project edition](#). The chapter credits

Christian Serpé and the relation to Nicolas Spaltenstein's *Resolutions of unbounded complexes*. The size bounds and transfinite construction are developed below before their derived-functor applications. Source attribution and the licence for adapted passages appear in the [course notice](#).

1. The Grothendieck hypotheses

An abelian category has **AB3** if it has all small coproducts, **AB4** if those coproducts are exact, and **AB5** if it has coproducts and exact filtered colimits. The starred conditions refer respectively to products, exact products, and exact cofiltered limits. A **generator** U means that $\text{Hom}(U, -)$ is faithful: distinct morphisms become distinct after precomposition with some map from U . A locally small abelian category satisfying AB5 and having a generator is a **Grothendieck abelian category**. “Locally small” means that each Hom collection is a set in the chosen universe.

AB3 gives every small colimit. For a diagram (A_i) , take the cokernel of the map from the sum over its arrows to the sum over its objects, whose component for $i \rightarrow j$ is the inclusion at i minus the diagram map followed by the inclusion at j . Maps out of this cokernel are exactly compatible families of maps out of the diagram, which proves its universal property. AB5 implies AB4: a coproduct of short exact sequences is the filtered colimit of the exact finite subsums. Exact filtered colimits also commute with cohomology, since they preserve the kernels and cokernels defining it. These facts will be used at successor and limit stages.

In an abelian category with coproducts the generator condition has two useful equivalent forms. Every proper subobject $N \subsetneq M$ is escaped by some map $U \rightarrow M$; and every object is a quotient of a coproduct of copies of U . Faithfulness implies the first form by applying it to the nonzero quotient map $M \rightarrow M/N$. That form implies the second: sum all maps from U into M ; their image cannot be proper. The second implies faithfulness: if a morphism is nonzero, an epimorphism from a coproduct of copies of U cannot kill it on every summand. We use all three forms without assuming that U is projective.

Proposition 1.1 (a generator). Let $\mathcal{O}$ be any sheaf of rings on X , and $G = \bigoplus_U j_{U!}\mathcal{O}_U$, the sum over all open sets U . Then:

- (a) for every subsheaf $N \subsetneq M$ there is a morphism $G \rightarrow M$ that does not factor through N ;
- (b) $\text{Hom}_{\mathcal{O}}(G, -)$ is faithful: a nonzero morphism $M \rightarrow M'$ stays nonzero after composing with some $G \rightarrow M$;
- (c) every M receives an epimorphism from a direct sum of copies of G .

Together with Theorem 3.1 of the first lesson, which gives all colimits and exact filtered colimits, this makes $\text{Mod}(\mathcal{O})$ a Grothendieck abelian category as defined above.

Reference: [AI Integrated Stacks Project, unbounded complexes](#) states this generator for commutative $\mathcal{O}$.

Proof. A section $s \in M(U)$ gives $e_s : j_{U!}\mathcal{O}_U \rightarrow M$ with $e_s(1) = s$ (Lemma 5.1 of the first lesson). Composing with the projection $G \rightarrow j_{U!}\mathcal{O}_U$ gives $\tilde{e}_s : G \rightarrow M$. (a) A subobject of M in $\text{Mod}(\mathcal{O})$ is a subsheaf, since kernels are computed on sections. If $N \neq M$, there are U and $s \in M(U) \setminus N(U)$. If $\tilde{e}_s$ factored through N , then $s = e_s(1)$ would lie in $N(U)$. (b) If $\varphi : M \rightarrow M'$ is nonzero, then $\varphi(s) \neq 0$ for some section $s \in M(U)$. So $\varphi \circ \tilde{e}_s \neq 0$: on the summand $j_{U!}\mathcal{O}_U$ it sends 1 to $\varphi(s)$. (c) The sum of all $\tilde{e}_s$, over all pairs (U, s) , is a morphism $\bigoplus_{(U,s)} G \rightarrow M$ whose image contains every section. So it is onto on stalks, hence an epimorphism. $\square$

The three forms of generator are equivalent by the argument preceding the proposition.

2. Sizes and functorial injective embeddings

In the bounded-below case there was a first degree from which to construct a homotopy. An unbounded complex has no such starting point. Instead we will arrange that every map from an acyclic complex is null-homotopic. There are too many acyclic complexes to attach them all at once. The first part of the proof reduces the tests to a set of small complexes; the second kills those tests and makes the final terms injective.

For the construction let $\mathcal{A}$ be any locally small Grothendieck abelian category: it has small coproducts, exact filtered colimits, and a generator U . Here “generator” means that $\text{Hom}(U, -)$ detects distinct morphisms.

Proposition 1.1 and Theorem 3.1 of the first lesson give these properties for our module sheaves. Sets, well-orderings and the axiom of choice are understood in the fixed universe. The argument uses elementary abelian-category kernels, images, quotients and pushouts, but not the construction of the derived category.

We use two consequences of the filtered-colimit axiom. Coproducts are exact: a coproduct is the filtered colimit of its finite subsums, and finite direct sums are exact. Also filtered colimits commute with cohomology of complexes: exactness preserves kernels and cokernels in the formula for cohomology. Thus a filtered union of acyclic complexes is acyclic, and a filtered composite of monic quasi-isomorphisms is a monic quasi-isomorphism.

Size bounds and maps into increasing unions

Lemma 2.1 (a set of subobjects and bounded representatives). For every object B , its subobjects form a set. Write $s(B)$ for the cardinality of this set. A subobject or quotient of B has size at most $s(B)$. For each infinite cardinal μ , there is a set of representatives of the isomorphism classes of objects with $s(B) \leq \mu$.

Proof. Associate to a subobject $V \subset B$ the subset of $\text{Hom}(U, B)$ consisting of maps that factor through V . This subset determines V : the images of all maps $U \rightarrow V$ fill V . Indeed, if their sum had a proper image $W \subsetneq V$, the nonzero quotient map $V \rightarrow V/W$ would, by the generator property, have nonzero composite with some map $U \rightarrow V$, a contradiction. Hence subobjects inject into a power set of a Hom set.

Subobjects of a subobject remain subobjects of B . Subobjects of a quotient correspond, by inverse image, to subobjects of B containing its kernel. This proves the size inequalities.

For each proper subobject $V \subsetneq B$, choose a map $a_V : U \rightarrow B$ not factoring through V . Such a map exists by applying the generator property to $B \rightarrow B/V$. The sum of these maps is onto: a proper image would be one of the subobjects the chosen maps were required to escape. There are at most $s(B)$ summands. Therefore, if $s(B) \leq \mu$, B is a quotient of $U^{(\mu)}$, after adding zero summands if necessary. Its isomorphism class is represented by the quotient by one of the set of subobjects of $U^{(\mu)}$. This proves the last assertion. $\square$

Lemma 2.2 (smallness for monic systems). Suppose that $(B_\beta)_{\beta < \lambda}$ is an increasing ordinal-indexed system of subobjects with union B , and $\text{cf}(\lambda) > s(A)$. Every map $A \rightarrow B$ factors through some B_β .

Proof. The inverse images $A_\beta = A \times_B B_\beta$ form an increasing family of subobjects of A , with union A . To see the last equality without using elements, take filtered colimits of the kernel diagrams for

$$A \oplus B_\beta \longrightarrow B, \quad (a, b) \longmapsto f(a) - b.$$

Exactness identifies the resulting kernel with $A \times_B B = A$. At most $s(A)$ distinct subobjects occur among the A_β . Choose an index for each distinct one. The cofinality assumption bounds all those indices by a single $\gamma < \lambda$, so A_γ contains every A_β , and is A . The pullback property gives the required factorization. $\square$

Recall that the cofinality of a limit ordinal is the least cardinality of an unbounded subset of it. We will always choose a limit ordinal with cofinality larger than a specified infinite cardinal μ ; one may take the initial ordinal of μ^+ . Indeed, a family of at most μ ordinals below μ^+ has union of cardinality at most μ , hence is bounded below μ^+ . This also explains why countably many factorization stages below such an ordinal have a common upper bound.

Enough injectives, with a functorial choice

Lemma 2.3 (the generator extension test). An object E is injective if and only if every map $V \rightarrow E$, where $V \subset U$, extends to U .

Proof. Necessity is the definition of injectivity. Conversely, take a monomorphism $A \subset B$ and a map $A \rightarrow E$. Order its extensions to subobjects of B by restriction. This is a set by Lemma 2.1 and local smallness. The union of a chain, with the compatible map to E , is an upper bound, so Zorn's lemma gives a maximal extension $a : A' \rightarrow E$.

If $A' \neq B$, the nonzero quotient $B \rightarrow B/A'$ has nonzero composite with some $\psi : U \rightarrow B$. Put $V = \psi^{-1}(A')$. The map $a\psi : V \rightarrow E$ extends to $b : U \rightarrow E$ by assumption. It vanishes on $\ker \psi$, since its restriction there

is $a\psi = 0$, so it descends to $\text{im } \psi$. This descended map and a agree on $A' \cap \text{im } \psi$. They therefore give a map $A' + \text{im } \psi \rightarrow E$, strictly extending a , a contradiction. Thus $A' = B$. $\square$

Theorem 2.4 (functorial injective embeddings). Every object M has a natural monomorphism into an injective object $E(M)$.

Proof. For an object M , form the pushout

$$\begin{array}{ccc} \bigoplus_{(V,a)} V & \longrightarrow & M \\ \downarrow & & \downarrow \\ \bigoplus_{(V,a)} U & \longrightarrow & \mathcal{E}(M), \end{array} \quad (V, a) : V \subset U, \quad a : V \rightarrow M. \quad (2.1)$$

This is a set-indexed coproduct by Lemma 2.1. The left arrow is monic because coproducts are exact; its pushout $M \rightarrow \mathcal{E}(M)$ is monic. Every selected map $V \rightarrow M$ extends to $U \rightarrow \mathcal{E}(M)$. A map $M \rightarrow M'$ takes the summand labelled (V, a) to the summand labelled by its composite, so $\mathcal{E}$ and the embedding are functorial.

Iterate $\mathcal{E}$, taking unions at limit stages. Choose a fixed λ_0 with $\text{cf}(\lambda_0) > s(U)$, and call the union $E(M)$. Given $V \subset U$ and a map $V \rightarrow E(M)$, Lemma 2.2 makes it factor through a stage before λ_0 , since $s(V) \leq s(U)$. At the next stage (2.1) extends it to U . Lemma 2.3 now proves that $E(M)$ is injective. We have constructed functorial monomorphisms $M \rightarrow E(M)$. The functor E need not be additive; no subsequent argument assumes that it is. $\square$

3. Testing on a set of acyclic complexes

Lemma 3.1 (bounded small lifts). There is an infinite cardinal κ , depending only on $\mathcal{A}$ and U , such that every nonzero acyclic complex contains a nonzero bounded-above acyclic subcomplex all of whose terms have size at most κ .

Proof. For an infinite cardinal μ , set

$$c(\mu) = \max\{\mu, \aleph_0, s(U^{(\mu)})\}.$$

This function is nondecreasing: an inclusion of index sets gives a split monomorphism between the corresponding coproducts.

First consider an epimorphism $p : B \rightarrow C$, with $s(C) \leq \mu$. For each proper subobject $C' \subset C$, the map $B \rightarrow C/C'$ is nonzero. Choose $b_{C'} : U \rightarrow B$ whose composite is nonzero. The image B' of their sum maps onto C , because its image escapes every proper C' . It is a quotient of at most μ copies of U , and hence $s(B') \leq c(\mu)$. Notice that this argument does not assert that maps from U lift through epimorphisms: U was not assumed projective.

Put $\kappa_0 = \max\{s(U), \aleph_0\}$, $\kappa_{r+1} = c(\kappa_r)$, and $\kappa = \sup_{r < \omega} \kappa_r$. Let $A^\bullet$ be acyclic and let $\psi : U \rightarrow A^n$. Set

$$N^n = \text{im } \psi, \quad N^{n+1} = d(N^n), \quad N^j = 0 \quad (j > n+1).$$

The resulting complex is already exact at and above degree $n+1$. Both nonzero terms have size at most κ_0 . If N^j has been constructed for $j \geq k$, use the epimorphism

$$(d^{k-1})^{-1}(\ker(N^k \rightarrow N^{k+1})) \rightarrow \ker(N^k \rightarrow N^{k+1}).$$

It is onto because $A^\bullet$ is acyclic. The small-lift construction gives a subobject N^{k-1} mapping onto that kernel. If $s(N^k) \leq \kappa_r$, it can be chosen with $s(N^{k-1}) \leq \kappa_{r+1}$. Descending through all integers constructs a bounded-above acyclic subcomplex containing $\text{im } \psi$, with every term of size at most κ . Only finite iterates of c are used at each degree; we do not assume $c(\kappa) = \kappa$.

If $A^\bullet \neq 0$, choose a degree with $A^n \neq 0$ and a nonzero $\psi : U \rightarrow A^n$. The constructed subcomplex is nonzero. $\square$

Choose a set $\mathcal{S}$ of representatives of all bounded-above acyclic complexes with term sizes at most κ . Such a set exists: Lemma 2.1 supplies a set of choices for each term, and local smallness supplies a set of differentials between each pair of choices; there are only countably many degrees. Restrict that set of complexes to the ones satisfying $d^2 = 0$, acyclicity and boundedness above.

Lemma 3.2 (testing K-injectivity on a set). Suppose that every term of $I^\bullet$ is injective and every chain map $S^\bullet \rightarrow I^\bullet$, for $S^\bullet \in \mathcal{S}$, is null-homotopic. Then $I^\bullet$ is K-injective.

Proof. Take an acyclic $A^\bullet$. Starting with $F_0 = 0$, build an increasing continuous filtration by acyclic subcomplexes. If $F_\beta \neq A$, the quotient A/F_β is nonzero and acyclic. Choose in it a nonzero small acyclic subcomplex from Lemma 3.1, and let $F_{\beta+1}$ be its inverse image. At a limit stage take the union. Successor exactness follows from the short exact sequence of complexes; limit exactness follows from exact filtered colimits. The construction exhausts A : the subcomplexes of A form a set, since they are among the sequences of subobjects of its terms. A strictly increasing chain longer than that set is impossible. Every nonzero successor quotient $F_{\beta+1}/F_\beta$ is isomorphic to an element of $\mathcal{S}$.

Let $f : A \rightarrow I$ be a chain map. We construct null-homotopies h_β of $f|_{F_\beta}$ that agree on earlier stages. At a successor, extend each component $h_\beta^n : F_\beta^n \rightarrow I^{n-1}$ to a map $\tilde{h}^n : F_{\beta+1}^n \rightarrow I^{n-1}$, using injectivity. The map

$$g = f|_{F_{\beta+1}} - d_I \tilde{h} - \tilde{h} d_F$$

is a chain map, as $d_I^2 = d_F^2 = 0$. It vanishes on F_β , and so descends to a chain map $\bar{g} : F_{\beta+1}/F_\beta \rightarrow I$. By the test assumption $\bar{g} = d_I k + k d$ for a degree-minus-one map k . With $\pi : F_{\beta+1} \rightarrow F_{\beta+1}/F_\beta$, put $h_{\beta+1} = \tilde{h} + k\pi$. This extends h_β and satisfies $f = d_I h_{\beta+1} + h_{\beta+1} d_F$.

At a limit stage the compatible components define maps out of the union, and the same identity holds because it holds on every earlier stage. At exhaustion they give a null-homotopy of f . Thus every map from every acyclic complex to I is null-homotopic, exactly the definition of K-injectivity. This proof does not pass cohomology through an inverse limit. $\square$

4. Constructing the unbounded resolution

Theorem 4.1. In a locally small Grothendieck abelian category, every complex M has a functorial, termwise monic quasi-isomorphism $M \rightarrow I(M)$, where each term of $I(M)$ is injective and $I(M)$ is K-injective.

Proof.

There are two operations, with different purposes.

For $S \in \mathcal{S}$, let $C(S) = \text{Cone}(1_S)$, with

$$C(S)^n = S^n \oplus S^{n+1}, \quad d(a, b) = (da + b, -db).$$

The map $S \rightarrow C(S)$, $a \mapsto (a, 0)$, is monic and null-homotopic: $a \mapsto (0, a)$ is a homotopy for it. The same formula contracts $C(S)$. Its quotient by S is $S[1]$, which is acyclic. Define $P(M)$ by pushing out all these inclusions along all chain maps $S \rightarrow M$:

$$\begin{array}{ccc} \bigoplus_{(S,w)} S & \xrightarrow{\Sigma w} & M \\ \downarrow & & \downarrow \\ \bigoplus_{(S,w)} C(S) & \longrightarrow & P(M). \end{array} \quad (4.1)$$

Then $M \rightarrow P(M)$ is termwise monic. Its quotient is the coproduct of the acyclic complexes $S[1]$, hence is acyclic. It is therefore a quasi-isomorphism. Every chosen w becomes null-homotopic in $P(M)$, using the corresponding cone. Indexing by all maps makes P and its embedding functorial.

The second operation uses the embeddings $i_A : A \rightarrow E(A)$ constructed above. Set

$$J(M)^n = E(M^n) \oplus E(M^{n+1}), \quad d_J(a, b) = (b, 0), \quad u^n = (i_{M^n}, i_{M^{n+1}} d_M). \quad (4.2)$$

The equation $d_J u = u d_M$ follows from $d_M^2 = 0$, and u is monic because its first component is monic. Let $Q = J(M)/u(M)$, with quotient map $v : J(M) \rightarrow Q$, and define

$$\begin{aligned} N(M) &= \text{Cone}(v)[-1], & N(M)^n &= Q^{n-1} \oplus J(M)^n, \\ d_N(q, j) &= (-d_Q q - v(j), d_J j), & M &\longrightarrow N(M), \quad m \longmapsto (0, u(m)). \end{aligned} \quad (4.3)$$

These formulas are identities of component morphisms in the abelian category; they do not require objects to have underlying sets of elements. They show directly that the last map is a monic chain map. Its degree- n map factors through the injective subobject $0 \oplus J(M)^n$ of $N(M)^n$. A finite direct sum of injectives is injective, since a map into it extends component by component.

The quotient $N(M)/M$ has terms $Q^{n-1} \oplus Q^n$, differential

$$D(q, a) = (-dq - a, da),$$

and contraction $h(q, a) = (0, -q)$: substitution gives $Dh + hD = 1$. Thus $M \rightarrow N(M)$ is a quasi-isomorphism. Naturality of i_A gives functoriality of every map in (4.2)–(4.3), even though E need not be additive.

The roles of the two operations are worth separating:

$$\begin{aligned} T_\beta &\xrightarrow{\text{kill test maps}} P(T_\beta), \\ P(T_\beta) &\xrightarrow{\text{insert injectives}} T_{\beta+1}. \end{aligned} \quad (4.4)$$

Both arrows are monic quasi-isomorphisms. Neither intermediate complex is asserted to be K-injective.

The final union and its two checks

Choose once and for all a limit ordinal λ of cofinality greater than κ . Define

$$T_0(M) = M, \quad T_{\beta+1}(M) = N(P(T_\beta(M))), \quad T_\gamma(M) = \varinjlim_{\beta < \gamma} T_\beta(M)$$

at limit stages, and put $I(M) = T_\lambda(M)$. All transitions are monic quasi-isomorphisms, including at limits by exactness of filtered colimits. Hence $M \rightarrow I(M)$ is monic and a quasi-isomorphism. The choices of $U, \mathcal{S}, \lambda_0, \lambda$ are fixed independently of M , so the construction is functorial in M .

First, every term $I(M)^n$ is injective. Given $V \subset U$ and a map $V \rightarrow I(M)^n$, smallness factors it through $T_\beta(M)^n$ for some $\beta < \lambda$. Its composite into $T_{\beta+1}(M)^n$ factors through the injective object $J(P(T_\beta(M)))^n$ by (4.3). Extend $V \rightarrow J(P(T_\beta(M)))^n$ to U , and compose into $I(M)^n$. Lemma 2.3 proves injectivity. This is a proof about the final terms, not an assertion that filtered colimits of injectives are automatically injective.

Second, let $S \in \mathcal{S}$ and $w : S \rightarrow I(M)$ be a chain map. Every component factors through a stage by Lemma 2.2, since $s(S^n) \leq \kappa$. There are countably many components. As $\text{cf}(\lambda) > \kappa \geq \aleph_0$, all components factor through a single $T_\beta(M)$. They already commute with the differentials there: their commutators vanish after the monomorphism $T_\beta(M)^{n+1} \rightarrow I(M)^{n+1}$, and hence vanish before it. Thus w factors as a chain map. Operation P makes that map null-homotopic at the next stage, so it is null-homotopic in $I(M)$. Lemma 3.2 proves K-injectivity.

Apply this construction to $\mathcal{A} = \text{Mod}(\mathcal{O})$. It proves the resolution assertion of Theorem 4.1, including termwise injectivity, naturality and noncommutative coefficients. The unbounded derived-functor consequences are proved in Section 5 below. The derived category was constructed in the common reading; the existence proof here only uses complexes and homotopies. $\square$

5. Derived functors, adjoints and products

Let $Q : K(\mathcal{A}) \rightarrow D(\mathcal{A})$ denote localization. By Theorem 3.3 of the preceding lesson, maps into a K-injective complex satisfy

$$\mathrm{Hom}_K(K, I) = \mathrm{Hom}_D(QK, QI). \quad (5.1)$$

That theorem also proves that a quasi-isomorphism between K-injectives is a homotopy equivalence. We now use it without boundedness restrictions. Although the construction of $I(M)$ is functorial on chain maps, its functor need not be additive. The additive functor on the derived category is obtained from (5.1), rather than by claiming additivity of the transfinite construction.

Theorem 5.1 (right derived functors on all complexes). Suppose every complex of an abelian category $\mathcal{A}$ has a quasi-isomorphism $q_K : K \rightarrow I_K$ into a K-injective complex. For every exact functor of triangulated categories $F : K(\mathcal{A}) \rightarrow \mathcal{T}$, there is an exact right derived functor

$$RF : D(\mathcal{A}) \longrightarrow \mathcal{T}, \quad RF(K) = F(I_K).$$

Its comparison $\eta : F \rightarrow RF \circ Q$ is initial among natural comparisons from F to a functor on $D(\mathcal{A})$. In particular, for any additive $\Phi : \mathcal{A} \rightarrow \mathcal{B}$, with $\mathcal{B}$ abelian, $R\Phi : D(\mathcal{A}) \rightarrow D(\mathcal{B})$ is computed by applying Φ degreewise to a K-injective resolution.

Proof. The full subcategory $K_{\mathrm{inj}} \subset K(\mathcal{A})$ of K-injectives is closed under shifts and cones by Lemma 3.1 of the preceding lesson. Equation (5.1) shows that Q restricted to this subcategory is fully faithful, and the resolution hypothesis makes it essentially surjective. Define an inverse R explicitly: send K to I_K , and a derived morphism $v : QK \rightarrow QL$ to the unique homotopy class $Rv : I_K \rightarrow I_L$ satisfying

$$Q(Rv) = Q(q_L) v Q(q_K)^{-1}. \quad (5.2)$$

Existence and uniqueness follow from (5.1). The same formula proves preservation of identities, composition and addition. For an ordinary homotopy class $f : K \rightarrow L$, injectivity of the map in (5.1) gives $(Rf)q_K = q_L f$ already in $K(\mathcal{A})$.

The inverse R is exact. Its shift comparison is the unique isomorphism $I_{K[1]} \cong I_K[1]$ inducing the corresponding isomorphism in D . A morphism between K-injectives has a K-injective cone. For any distinguished triangle in D , compare its first arrow with the unique arrow between the chosen K-injectives. The cone triangle of that arrow maps under Q to a distinguished triangle with the same first arrow. The triangle morphism axiom and the long exact representable sequences, proved in the common reading, supply an isomorphism on the third objects compatible with all arrows. Full faithfulness lifts this isomorphism and its equations to K_{inj} . Thus R transports distinguished triangles to distinguished triangles, with the stated shift comparison. Set $RF = F \circ R$ and $\eta_K = F(q_K)$; the equality just proved gives naturality.

To verify universality, take any functor $T : D(\mathcal{A}) \rightarrow \mathcal{T}$ and natural $\theta : F \rightarrow TQ$. Define

$$\alpha_K = T(Qq_K)^{-1} \theta_{I_K} : F(I_K) \longrightarrow T(QK). \quad (5.3)$$

Naturality of θ for the representative Rv , together with (5.2), proves naturality of α for every derived morphism v . Naturality for q_K gives $\alpha_K \eta_K = \theta_K$. This factorization is unique: at a K-injective object I , its resolution map q_I is a homotopy equivalence, so $F(q_I)$ is invertible and forces α_I . Naturality for the isomorphism Qq_K then forces α_K . Consequently the displayed construction has exactly the right-derived universal property. It also proves independence of the chosen resolutions and comparisons.

Finally, an additive Φ preserves chain homotopies, shifts, and finite direct sums, and carries the defining matrix of a cone to that of the cone of the image map. Composing its functor on homotopy categories with $K(\mathcal{B}) \rightarrow D(\mathcal{B})$ is therefore exact. Apply the result to this F . On bounded-below cohomology it agrees with

the preceding lesson: its bounded-below injective resolution is K-injective, and (5.1) uniquely compares it with any unbounded resolution. $\square$

For a Grothendieck category, (5.1) also settles the size of derived morphisms. $\text{Hom}_K(K, I_L)$ is the quotient of the set of chain maps by the set-valued homotopy relation; chain maps and homotopies are subsets of countable products of original-universe Hom sets. Thus $D(\mathcal{A})$ is locally small in the original universe, including when $\mathcal{A}$ is a large category of sheaves.

Proposition 5.2 (exact left adjoints). Let $u : \mathcal{A} \rightarrow \mathcal{B}$ be exact and left adjoint to an additive $v : \mathcal{B} \rightarrow \mathcal{A}$. Then v preserves K-injectives.

Proof. The adjunction on objects identifies chain maps and their homotopies componentwise, hence

$$\text{Hom}_{K(\mathcal{A})}(A, vI) = \text{Hom}_{K(\mathcal{B})}(uA, I).$$

If A is acyclic, exactness of u makes uA acyclic and K-injectivity makes the right side zero. This is the definition of K-injectivity of vI . $\square$

In particular, restriction j^{-1} to an open subset preserves K-injectives because its left adjoint $j_!$ is exact. For a continuous $f : X \rightarrow Y$ with $\mathcal{O}_X = f^{-1}\mathcal{O}_Y$, the exact left adjoint f^{-1} shows that f_* preserves K-injectives. For an arbitrary morphism of ringed spaces, Rf_* still exists by Theorems 4.1 and 5.1 applied to module sheaves; preservation by the underived f_* requires an exact left adjoint, for example flat module pullback. Section 6 gives a failure when pullback is not exact.

Proposition 5.3 (existence of products in $\mathcal{A}$). A locally small abelian category with a generator and small coproducts has all small products. In particular a Grothendieck category satisfies AB3*.

Proof. Fix a family $(A_i)_{i \in J}$. For any cone $(f_i : T \rightarrow A_i)$, let $N \subset T$ be the sum of all subobjects on which every f_i vanishes. This is a set of subobjects by Lemma 2.1, so its sum exists as the image of their coproduct. Each f_i vanishes on the sum. The induced family on $B = T/N$ is **jointly monic**, meaning that a map killed by all its components is zero. Indeed, if a subobject of B is killed by every component, its inverse image in T would enlarge N ; hence it is zero. Apply this to the image of a map killed by the family. This constructs a jointly monic quotient of every cone, without using products or intersections to define it.

For a jointly monic cone on B , the map of sets

$$\text{Hom}(U, B) \longrightarrow \prod_{i \in J} \text{Hom}(U, A_i)$$

is injective: the difference of two maps with equal images is killed by every component. Let ν be the cardinality of the product on the right and set $\mu = \max(2^\nu, \aleph_0)$. The subobject-to-subset argument in Lemma 2.1 gives $s(B) \leq \mu$. That lemma supplies a set of representatives of all possible B , and local smallness supplies a set of all their cones to (A_i) . Take the coproduct C of all these jointly monic cones, with the induced maps $C \rightarrow A_i$, and divide by their common vanishing subobject as in the first paragraph. Call the result P .

Every cone on T factors through its jointly monic quotient B , through a representative included as a summand of C , and then through P . Thus a map $T \rightarrow P$ inducing the given cone exists. It is unique because the cone on P is jointly monic. This is the universal property of $\prod_i A_i$. The empty product is zero and the construction still works: its only jointly monic cone has zero source. $\square$

The proposition proves existence, not exactness, of products. Products of sheaves already existed in the first lesson, but their epimorphisms can fail to be onto on stalks.

Proposition 5.4 (products and coproducts in the derived category). A Grothendieck category has all small derived products and coproducts. A coproduct is represented by the termwise sum of any representatives. A product is represented by the termwise product of K-injective resolutions.

Proof. Put $P = \prod_i I_i$ for K-injectives I_i . Componentwise projections identify the complex of abelian groups $\text{Hom}^\bullet(A, P)$ with $\prod_i \text{Hom}^\bullet(A, I_i)$. Products of abelian groups are exact: kernels are coordinatewise and

coordinatewise lifts prove surjectivity using choice. Therefore if A is acyclic, the product Hom complex is acyclic. Thus P is K-injective. For any K , exactness of those group products and (5.1) give

$$\mathrm{Hom}_D(K, P) = \prod_i \mathrm{Hom}_D(K, I_i),$$

which proves the product property. The proof assumes exact products in abelian groups, not in $\mathcal{A}$.

For the sum $S = \bigoplus_i K_i$, and any K-injective I , the Hom complex identifies with $\prod_i \mathrm{Hom}^\bullet(K_i, I)$. The same exactness gives $\mathrm{Hom}_K(S, I) = \prod_i \mathrm{Hom}_K(K_i, I)$. Equation (5.1) and a K-injective resolution of every target prove the coproduct property in D . AB4 ensures that changing representatives by quasi-isomorphisms changes their sum by a quasi-isomorphism. $\square$

6. Examples that isolate the hypotheses

Unbounded termwise injectivity. Over $R = k[\epsilon]/(\epsilon^2)$, the complex with R in every integer degree and differential multiplication by ϵ is acyclic, because $\ker(\epsilon) = \epsilon R = \mathrm{im}(\epsilon)$. Its terms are injective: the only ideals are $0, \epsilon R, R$, and a map $\epsilon R \rightarrow R$ sends ϵ to an element ϵa , so it extends by sending 1 to a ; the Baer criterion was proved in the preceding lesson. Its identity is not null-homotopic, because each expression $\epsilon h^n + h^{n+1} \epsilon$ has image in ϵR and cannot equal 1_R . Taking the acyclic complex itself as a source proves that it is not K-injective. Theorem 4.1 must change this complex, although its terms are already injective.

Pushforward can lose K-injectivity. Take one-point ringed spaces with ring map $\mathbb{Z} \rightarrow \mathbb{F}_2$. The module pushforward is restriction of scalars. The complex $\mathbb{F}_2[0]$ is K-injective over $\mathbb{F}_2$: a linear map from a subspace extends after choosing a basis, so every vector space is injective, and the bounded-below lemma applies. Consider the acyclic complex of abelian groups

$$A^\bullet = (\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{\mathrm{mod} 2} \mathbb{F}_2), \quad \text{in degrees } 0, 1, 2.$$

The map $A \rightarrow \mathbb{F}_2[0]$ that is reduction modulo two in degree zero and zero elsewhere is a chain map. A null-homotopy would imply $\mathrm{mod} 2 = h^1 \circ 2$, impossible for a homomorphism $h^1 : \mathbb{Z} \rightarrow \mathbb{F}_2$. Thus the pushed-forward complex is not K-injective. Its left adjoint $\mathbb{F}_2 \otimes_{\mathbb{Z}} -$ is not exact, since it sends the injection $2 : \mathbb{Z} \rightarrow \mathbb{Z}$ to zero. This matches the hypothesis of Proposition 5.2 exactly.

An unbounded derived product. The injective group $\mathbb{Q}/\mathbb{Z}$ placed in degree $-m$ is $(\mathbb{Q}/\mathbb{Z})[m]$. The product of these complexes for $m \geq 0$ has one copy of $\mathbb{Q}/\mathbb{Z}$ in every nonpositive degree, zero elsewhere, and zero differential. Proposition 5.4 makes it K-injective and the product in $D(\mathbb{Z})$. It has nonzero cohomology in arbitrarily negative degrees, so it is not bounded below. This also shows why resolving only bounded-below targets would miss objects of the unbounded category.

7. Exercises with checked solutions

Exercise 1 (easy: a concrete generator). For a discrete finite space X with constant ring R , identify the generator G of Proposition 1.1 and prove directly that it detects morphisms. Does the proposition require R to be commutative?

Solution. A sheaf of modules is a tuple $(M_x)_{x \in X}$, and $j_{U!} \mathcal{O}_U$ has stalk R at points of U and zero elsewhere. Thus $G_x = \bigoplus_{U \ni x} R$. If a tuple of maps has a nonzero component at x , choose an element there with nonzero image. The summand belonging to the singleton $\{x\}$ gives a map into the source tuple detecting that element. For a left module, $R \rightarrow M_x, a \mapsto am$, is left linear for every associative unital R . No commutativity is used. The empty-space module category has only its zero object, and its zero generator is faithful there.

Exercise 2 (medium: AB3* without AB4*). Give the size bound that makes the product construction in Proposition 5.3 a set-sized construction. Explain why it proves no exactness statement about products of sheaves.

Solution. With $H_i = \text{Hom}(U, A_i)$, every jointly monic cone on B embeds $\text{Hom}(U, B)$ into the set $\prod_i H_i$. A subobject of B is determined by the subset of maps from U that factor through it. Hence $s(B) \leq 2^{|\prod_i H_i|}$, and Lemma 2.1 provides a set of representatives. For each representative, the cones form a subset of $\prod_i \text{Hom}(B, A_i)$, also a set. Their union is a set, so their coproduct is permitted by AB3. This argument concerns a universal property. It supplies no common open neighbourhood on which infinitely many stalkwise lifts exist. The product-of-epimorphisms counterexample in Section 6 of the first lesson therefore remains compatible with AB3*.

Exercise 3 (medium: the unbounded direct image). For an arbitrary morphism of ringed spaces, construct Rf_* , explain its agreement with bounded-below Rf_* , and say when f_*I can itself be used as a K-injective complex on the target.

Solution. The module categories are Grothendieck by Proposition 1.1 and the first lesson. Choose a K-injective resolution $K \rightarrow I_K$ by Theorem 4.1. Define Rf_*K as f_*I_K in the target derived category, with morphisms obtained by applying f_* to the unique comparison homotopy classes. Additivity of f_* makes it preserve homotopies and cones, so Theorem 5.1 proves existence, exactness and uniqueness. A bounded-below injective resolution is K-injective and is uniquely homotopy equivalent to any chosen K-injective resolution of the same object, giving agreement. If the left adjoint f^* is exact, Proposition 5.2 proves that f_*I_K is K-injective on the target. The one-point example in Section 6 proves that this final claim cannot be made for arbitrary ring maps.

Exercise 4 (hard: what a transfinite step does for groups). Work in $\text{Mod}(\mathbb{Z})$, with generator $U = \mathbb{Z}$. For $M = \mathbb{Z}[0]$, describe an explicit nonzero small acyclic complex that occurs among the tests, and write the contraction that kills any map from it in $P(M)$. Then check the contraction of $N(P(M))/P(M)$. Explain why a countable sequence of steps has not been shown sufficient.

Solution. The complex $S = (\mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}/2)$ in degrees 0, 1, 2 is bounded above and acyclic, with terms having at most countably many subobjects. After replacing it by its representative it belongs to $\mathcal{S}$. Every selected chain map $w : S \rightarrow M$ has an attached copy of $C(S)$. If $e : C(S) \rightarrow P(M)$ is its pushout map, then the degree-minus-one map $a \mapsto e(0, a)$ satisfies $dh + hd = jw$, because $d(0, a) = (a, -da)$. For this particular M , $w^0 : \mathbb{Z} \rightarrow \mathbb{Z}$ can be multiplication by any integer and the other components vanish; the same formula kills all these maps.

Put $L = P(M)$, $Q = J(L)/L$. In $N(L)/L$, write a term as $(q, a) \in Q^{n-1} \oplus Q^n$. Its differential is $D(q, a) = (-dq - a, da)$ and $h(q, a) = (0, -q)$. Direct substitution gives $Dh(q, a) = (q, -dq)$ and $hD(q, a) = (0, dq + a)$, whose sum is (q, a) . Thus this quotient is contractible. Finally, a test has countably many components. Their first factorization stages could be unbounded in ω ; no single finite stage would then contain the chain map. The chosen cofinality greater than $\kappa \geq \aleph_0$ ensures a common stage and makes the next-step killing argument valid.

Exercise 5 (hard: compatible homotopies at a limit). In Lemma 3.2, explain why choosing an unrelated null-homotopy at each filtration stage is insufficient. Derive the successor correction that makes the chosen homotopies agree.

Solution. A map out of a union is induced by compatible maps on its subobjects. Unrelated null-homotopies need not restrict to one another, so they define no homotopy on the union. Suppose h_β has been fixed. Termwise injectivity extends it to an arbitrary graded map $\tilde{h}$ on $F_{\beta+1}$. The error $g = f - d\tilde{h} - \tilde{h}d$ is a chain map and restricts to zero on F_β . It factors through $\pi : F_{\beta+1} \rightarrow F_{\beta+1}/F_\beta$. This quotient is a small acyclic test, so its error map is $dk + kd$. Define $h_{\beta+1} = \tilde{h} + k\pi$. Then $dh_{\beta+1} + h_{\beta+1}d = f$, and its restriction is the prescribed h_β , because π vanishes there. At a limit, the universal property of the colimit assembles these compatible components and their equations. This uses exact filtered colimits to keep the filtration acyclic, but no exact inverse-limit assertion.

The next lesson constructs K-flat resolutions by a different method. This theorem is not needed for their existence; the two resolution tools address different variables of derived operations.

Sources and licensing: the Stacks project authors, *Injectives*, [the complete pinned chapter](#), especially the proofs

corresponding to Tags 0E8N and 079C–079P; *Derived Categories*, [the pinned K-injective section](#), for comparison, derived functors, products and exact adjoints. The elementary large-cofinality argument in Section 2 is also the proof of [Stacks, Tag 05N3, in the pinned AI Integrated edition](#). These adapted passages retain GFDL-1.2-or-later. N. Spaltenstein, *Resolutions of unbounded complexes*, Compositio Mathematica 65 (1988), 121–154, is [available from Numdam](#). See the [combined licence notice](#) and [GNU FDL text](#).

Flat modules and K-flat resolutions

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Tensoring a resolution should preserve the cohomology information it was meant to carry. Flat modules solve this problem for a single tensor factor in one degree. Unbounded complexes require a stronger condition, K-flatness. The two conditions are independent for unbounded complexes: a contractible complex can have nonflat terms, and an acyclic complex of free terms can fail to be K-flat.

The prerequisites are [Sheaves of modules on a ringed space](#), Theorems 2.1, 3.1 and 4.4 and Lemmas 5.1–5.2, and [Complexes, cones and localization](#), Lemmas 1.1–1.2, 2.4 and Proposition 5.2. We construct K-flat resolutions independently of the K-injective existence theorem. This avoids a dependency cycle when derived tensor products and Hom are developed.

Throughout this lesson $\mathcal{O}$ is a sheaf of **commutative** unital rings on an arbitrary topological space X . Tensor totalizations use direct sums, including for unbounded complexes. We impose no separation, compactness, Noetherian or finite homological-dimension assumption.

The construction sources are the Stacks project authors’ *Modules on Ringed Spaces*, “Flat modules”, and *Cohomology of Sheaves*, “Flat resolutions”, in the [AI Integrated Stacks Project edition](#). We develop flat-module calculations, cycle attachments and compatible bounded resolutions. Spaltenstein’s *Resolutions of unbounded complexes* provides historical background. Source attribution and the licence for adapted passages appear in the [course notice](#).

1. Flat sheaves and exact tensor sequences

A sheaf M is **flat** if $-\otimes_{\mathcal{O}} M$ is exact. The tensor sheaf is the sheafification of the presheaf $U \mapsto F(U) \otimes_{\mathcal{O}(U)} M(U)$. Its balanced universal property follows by gluing the presheaf balanced maps, using structured sheafification from the first lesson. Tensor is right exact: a balanced map on a quotient is precisely a balanced map on the original module vanishing on the relations defining that quotient. Thus flatness amounts to preservation of injections.

Flat building blocks

Lemma 1.1 (local covers). Every module sheaf is a quotient of a direct sum of sheaves $B_U = j_{U!}\mathcal{O}_U$.

Proof. For an open $U \subset X$, Lemma 5.1 of the first lesson gives

$$\mathrm{Hom}_{\mathcal{O}}(B_U, M) = M(U). \quad (1.1)$$

For every module sheaf M , form the set-indexed sum

$$L(M) = \bigoplus_{U \subset X \text{ open}} \bigoplus_{s \in M(U)} B_U. \quad (1.2)$$

The map $\lambda_M : L(M) \rightarrow M$ sends the generator indexed by s to s , using (1.1). Every germ is represented by such a local section, so λ_M is an epimorphism. $\square$

This uses local sections on all opens. A sheaf need not be generated by its global sections.

Lemma 1.2 (flatness and stalks). A module sheaf M is flat if and only if every M_x is flat over $\mathcal{O}_x$. In particular B_U , $L(M)$, and direct sums of flat module sheaves are flat.

Proof. The tensor sheaf is the sheafification of the sectionwise tensor presheaf. Its stalk is

$$(F \otimes_{\mathcal{O}} M)_x = F_x \otimes_{\mathcal{O}_x} M_x. \quad (1.3)$$

Here is the relevant finite-relation argument. A tensor is a finite sum of elementary tensors. Represent all its factors on one neighbourhood of x . An equality between tensors is generated by finitely many additive and balancing relations. Represent their scalar and module entries on a common smaller neighbourhood; each equality of germs in this finite certificate holds after a further shrinking. This proves both surjectivity and injectivity of (1.3), even though the section rings vary with the neighbourhood.

If every M_x is flat, (1.3) and Theorem 2.1 of the first lesson show that tensoring with M preserves exact sequences. Conversely, an injection $A \rightarrow B$ of $\mathcal{O}_x$ -modules gives an injection $x_*A \rightarrow x_*B$ of module sheaves: these maps are injective on sections. If M is flat, tensoring preserves this injection, and its stalk at x is $A \otimes M_x \rightarrow B \otimes M_x$. Tensor products are right exact, by their presentation with generators and relations, so this proves that M_x is flat.

The stalks of B_U are $\mathcal{O}_x$ or zero (Lemma 5.1 of the first lesson); both are flat. Stalks commute with direct sums. A direct sum of flat modules is flat because its tensor functor is the direct sum of their exact tensor functors, and direct sums of exact module sequences remain exact coordinatewise. This proves the final assertions. $\square$

Lemma 1.3 (flat quotients and flat exact sequences). If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact and M'' is flat, then tensoring with any sheaf N preserves that exact sequence. Consequently flat sheaves are closed under extensions and kernels of epimorphisms between flat sheaves. Filtered colimits of flat sheaves are flat, and a direct summand of a flat sheaf is flat.

Proof. We first prove the flat-quotient assertion over a ring R . Resolve a test module N by a complex $F^\bullet$ of free modules in degrees at most zero. Choose a free module on the elements of N , map it onto N , choose a free module onto that kernel, and repeat. This construction gives $H^0(F) = N$ and zero negative cohomology. Flatness of M'' makes $F \otimes_R M''$ exact in negative degrees, by tensoring the short exact sequences of cycles. Free terms make

$$0 \rightarrow F \otimes M' \rightarrow F \otimes M \rightarrow F \otimes M'' \rightarrow 0$$

termwise exact, because a free tensor is a direct sum of the original exact sequence. The long exact sequence of cohomology gives an injection $H^0(F \otimes M') \rightarrow H^0(F \otimes M)$, since the preceding $H^{-1}(F \otimes M'')$ is zero. Right exactness identifies these degree-zero groups with $N \otimes M'$ and $N \otimes M$. This proves the assertion without presupposing a Tor theorem. Apply it at each stalk, using Lemma 1.2, to obtain the sheaf assertion.

For an injection $N \rightarrow N_1$, compare the two exact tensor rows supplied by this assertion. If the outside sheaves M', M'' are flat, their vertical maps are monic and so is the middle one: an element in its kernel maps to zero in the right column and then is forced to be zero by the left column. This is the same kernel argument on every stalk. If M, M'' are flat, the injection $N \otimes M' \rightarrow N \otimes M$, followed by the monomorphism into $N_1 \otimes M$, shows that the vertical map for M' is monic as well.

Tensor commutes with colimits by its balanced universal property. Exactness of filtered colimits, proved in the first lesson, therefore proves flatness of a filtered colimit of flats. Finally $N \otimes (M_1 \oplus M_2) = (N \otimes M_1) \oplus (N \otimes M_2)$; if this map preserves each injection, so does each summand. $\square$

The flat-quotient assertion is often phrased as vanishing of first Tor against a flat module. The proof here establishes the exact sequence needed before Tor is defined in the next lesson.

2. K-flat complexes and their closure properties

For cochain complexes define

$$(A \otimes P)^n = \bigoplus_{p+q=n} A^p \otimes P^q, \quad d(a \otimes p) = d_A a \otimes p + (-1)^{|a|} a \otimes d_P p. \quad (2.1)$$

Two applications of d have opposite mixed terms, while the remaining terms contain d_A^2 and d_P^2 ; hence $d^2 = 0$. A complex P is **K-flat** if $A \otimes P$ is acyclic for every acyclic A . The graded flip $a \otimes p \mapsto (-1)^{|a||p|} p \otimes a$ is an isomorphism of complexes: on the two differential terms the sign exponents differ by exactly the degree increased in that term. It interchanges the tensor variables.

Lemma 2.1 (tensor respects homotopies and triangles). For any fixed complex A , tensoring with A is an exact functor on the homotopy category. A K-flat P therefore carries quasi-isomorphisms in the other variable to quasi-isomorphisms.

Proof. If $f - g = dh + hd$, then $H(a \otimes p) = (-1)^{|a|} a \otimes h(p)$ gives $1_A \otimes (f - g) = dH + Hd$; expanding the mixed terms cancels them. The shift comparison $A \otimes P[1] \rightarrow (A \otimes P)[1]$ multiplies a component in A^p by $(-1)^p$. For the cone convention of the common reading, the same factor on the shifted second summand gives

$$A \otimes \text{Cone}(f) \cong \text{Cone}(1_A \otimes f). \quad (2.2)$$

Indeed, the off-diagonal term on the left is $(-1)^p f$; the factor $(-1)^p$ makes it the off-diagonal term on the right. On the second diagonal block, the resulting differential is the negative of the total differential on $A \otimes P$. The comparison respects the cone inclusion and the negative projection used in the distinguished triangle. Thus shifts and cone triangles are preserved. The graded flip gives the assertion in the other variable. A quasi-isomorphism has acyclic cone. If P is K-flat, its tensor with that cone is acyclic, and (2.2) proves the last assertion. $\square$

Lemma 2.2 (stalks and pullbacks). A complex P of module sheaves is K-flat if and only if every stalk P_x is K-flat over $\mathcal{O}_x$. Pullback along any morphism of commutative ringed spaces preserves flat sheaves and K-flat complexes.

Proof. Stalks commute with the tensor sheaf by Lemma 1.2 and with direct sums by the first lesson. Hence $(A \otimes P)_x = A_x \otimes P_x$. If the stalks of P are K-flat, an acyclic A has acyclic tensor on all stalks, and the tensor sheaf complex is acyclic. Conversely, for an acyclic complex C over $\mathcal{O}_x$, the complex of skyscrapers $x_* C$ is acyclic: the skyscraper functor is exact on sections, as established in Lemma 5.2 of the first lesson. Its stalk at x is C . Applying K-flatness of P to this complex and taking that stalk proves K-flatness of P_x .

For a ring map $R \rightarrow S$ and K-flat P over R , any acyclic complex C of S -modules remains acyclic after restriction of scalars, and

$$C \otimes_S (S \otimes_R P) = C \otimes_R P$$

is acyclic. The isomorphism sends $c \otimes (s \otimes p)$ to $sc \otimes p$; its inverse sends $c \otimes p$ to $c \otimes (1 \otimes p)$, and both commute with the total differentials. This proves base change of K-flatness. The identical one-degree argument proves base change of flat modules. The stalk formula for pullback from the first lesson, with $R = \mathcal{O}_{Y, f(x)}$ and $S = \mathcal{O}_{X, x}$, proves the sheaf statements. No flatness of S over R is needed. $\square$

Lemma 2.3 (operations preserving K-flatness). A flat sheaf in a single degree is K-flat. Direct sums and filtered colimits of K-flat complexes are K-flat. If

$$0 \longrightarrow P \longrightarrow Q \longrightarrow R \longrightarrow 0 \quad (2.3)$$

is a degreewise split exact sequence of complexes and P, R are K-flat, then Q is K-flat. Every bounded-above complex of flat sheaves is K-flat.

Proof. If A is acyclic and M is flat, tensor the exact sequences $0 \rightarrow Z^n A \rightarrow A^n \rightarrow Z^{n+1} A \rightarrow 0$ with M . They identify the kernels and images in $A \otimes M$, proving its acyclicity. Shifting a complex does not change this conclusion.

Tensor totalization commutes with direct sums and filtered colimits. These operations are exact for module sheaves by Theorem 3.1 of the first lesson (an arbitrary sum is the filtered colimit of its finite partial sums). They therefore commute with cohomology: kernels and images in an exact filtered colimit are the colimits of the corresponding kernels and images. Apply this to $A \otimes P_i$ to obtain the assertions about sums and colimits.

After tensoring (2.3) with any complex A , the sequence remains degreewise split exact, including after totalization. Its long exact cohomology sequence proves the extension assertion. The splitting is only a splitting of the underlying graded sheaves; it need not respect the differentials.

A bounded complex of flat sheaves is K-flat by induction on its length, using its bottom-degree term as the quotient and the remaining tail as the subcomplex in (2.3). If P is merely bounded above, its brutal lower truncations $P^{\geq m}$ are bounded complexes and are subcomplexes of P . As m decreases their union is P . The filtered-colimit assertion proves the last statement. $\square$

Lemma 2.4 (tensor products and two out of three). The tensor product of two K-flat complexes is K-flat. Shifts, direct summands, and homotopy equivalent complexes preserve K-flatness. Two out of three objects in a distinguished triangle of the homotopy category being K-flat forces the third to be K-flat. The same two-out-of-three statement holds in a short exact sequence of complexes whose quotient has flat terms.

Proof. Direct-sum reindexing identifies $A \otimes (P \otimes Q)$ with $(A \otimes P) \otimes Q$. On a term of degrees a, p, q , both differentials have signs $1, (-1)^a, (-1)^{a+p}$. If A is acyclic, successively applying K-flatness of P and Q gives acyclicity. Shifts and homotopy equivalences follow from Lemma 2.1. A direct summand of an acyclic tensor complex is acyclic, giving the summand assertion. Tensoring a distinguished triangle gives a distinguished triangle by that lemma, and its long exact cohomology sequence gives two out of three. For a short exact sequence with flat quotient terms, Lemma 1.3 makes its tensor with each term of any complex exact. Direct sums preserve exactness, so totalization gives a short exact sequence of tensor complexes. Its long exact sequence supplies the last claim. $\square$

Only **filtered** colimits occur in Lemma 2.3. An arbitrary cokernel of K-flat complexes need not be K-flat; Section 5 supplies the smallest counterexample.

3. A surjective resolution by attaching cycles

Theorem 3.1 (K-flat resolutions). Every complex G of $\mathcal{O}$ -modules has a quasi-isomorphism $P \rightarrow G$, surjective in each degree, where P is K-flat and every P^n is flat.

Proof. For a sheaf M , let $D_n(M)$ have M in degrees $n, n+1$, identity differential between them, and zero elsewhere. A map $M \rightarrow G^n$ extends uniquely to a chain map $D_n(M) \rightarrow G$, with upper component its composite with d_G . Start with

$$P_0 = \bigoplus_{n \in \mathbb{Z}} D_n(L(G^n)), \quad (3.1)$$

$$\epsilon_0 : P_0 \rightarrow G.$$

Use λ_{G^n} on the lower term of the n -th summand. The map is onto in every degree. Each disk is contractible. Tensoring a contraction h on the second factor with a complex A gives the contraction $a \otimes p \mapsto (-1)^{|a|} a \otimes h(p)$. Thus P_0 is K-flat; its terms are flat by Lemma 1.2.

Suppose $P_s \rightarrow G$ has been built, and put $K_s = \ker(\epsilon_s)$, a subcomplex. For every integer q , open U , and section

$$z \in Z^q(K_s)(U),$$

add a copy of B_U in degree $q-1$. Write its generator locally as e_z , and prescribe

$$de_z = z, \quad \epsilon_{s+1}(e_z) = 0. \quad (3.2)$$

These are morphisms of sheaves by (1.1). The two conditions needed for them to define a complex over G hold: $dz = 0$ and $\epsilon_s(z) = 0$. Explicitly, if V_s^n is the sum of the newly added B_U 's in degree n , the underlying graded sheaf is $P_s \oplus V_s$, with differential

$$d(p, v) = (d_{P_s} p + \alpha_s(v), 0),$$

where $\alpha_s : V_s^n \rightarrow P_s^{n+1}$ sends each generator to its chosen cycle. Since $d_{P_s} \alpha_s = 0$, its square is zero.

The inclusion $P_s \rightarrow P_{s+1}$ is degreewise split, and the quotient is V_s with zero differential. It is a direct sum of shifts of the flat sheaves B_U , hence K-flat. Lemma 2.3 proves by induction that every P_s is K-flat. Its terms remain direct sums of the flat building blocks.

Set $P = \varinjlim_{s \geq 0} P_s$. The maps to G agree, and the map $\epsilon : P \rightarrow G$ is termwise onto already from P_0 . Its terms are flat, and P is K-flat by Lemma 2.3. Exactness of filtered colimits gives

$$\ker(\epsilon) = \varinjlim_s K_s, \quad H^q(\ker \epsilon) = \varinjlim_s H^q(K_s). \quad (3.3)$$

Each transition $H^q(K_s) \rightarrow H^q(K_{s+1})$ is zero. Indeed, a stalk class is locally represented by a cycle section z ; the next stage supplies e_z in the kernel with $de_z = z$. Thus every term in the colimit on the right of (3.3) maps to zero at the next stage, and the colimit vanishes. The kernel of ϵ is acyclic. The long exact sequence for $0 \rightarrow \ker \epsilon \rightarrow P \rightarrow G \rightarrow 0$ proves that ϵ is a quasi-isomorphism. $\square$

The stages can create new cycles; the next stage kills them. A germ involves a finite stage, so a countable sequence of these stages suffices. Nothing in the construction asks sections to lift through an arbitrary sheaf epimorphism: generators are added for sections that actually exist on each open.

Lemma 3.2 (making another resolution termwise surjective). If $Q \rightarrow G$ is a quasi-isomorphism with Q K-flat, then $Q \oplus P_0 \rightarrow G$, the sum of that map and (3.1), is a termwise-surjective K-flat resolution. If Q also has flat terms, so does this enlargement.

Proof. The new map is onto because ϵ_0 is. The complex P_0 is contractible, and the inclusion $Q \rightarrow Q \oplus P_0$ is a quasi-isomorphism commuting with the maps to G . The new map is therefore a quasi-isomorphism. The direct sum is K-flat by Lemma 2.3; this uses no termwise-flatness assumption on Q . If its terms are flat, Lemma 1.2 gives flatness of the enlarged terms as well. $\square$

Remark 3.3 (what the construction does not prove). Extensions by zero of structure sheaves need not be projective objects in the category of module sheaves. The construction gives a resolution suitable for **tensor products**, not a K-projective resolution for computing arbitrary Hom functors. Neither an inverse limit of injective truncations nor exactness of products of sheaves was used.

4. Compatible bounded resolutions and factorization

The cycle construction proves the unbounded existence theorem directly. We also prove the stronger compatible bounded-above form used in the Stacks treatment, with every term a direct sum of the flat building blocks B_U .

Lemma 4.1 (bounded-above resolutions). If C vanishes in degrees greater than b , there is a termwise-surjective quasi-isomorphism $E \rightarrow C$, with E vanishing above b and its terms direct sums of B_U .

Proof. At degree n , replace the current term C^n by its surjective cover $u : L(C^n) \rightarrow C^n$. Replace degree $n-1$ by the pullback $C^{n-1} \times_{C^n} L(C^n)$, with differential into degree n its second projection. The next differential is $d_C u$; the preceding differential sends c to $(d_C c, 0)$. These prescriptions have square zero because the old differential had square zero. They give a termwise epimorphic chain map to the old complex; the pullback projection is epic because it is a pullback of u . Its kernel is $\ker u$ in degrees $n-1, n$, with identity differential, so the replacement is a quasi-isomorphism.

Perform these replacements for $n = b, b-1, b-2, \dots$. A fixed term changes at most twice and is finally a chosen cover $L(-)$. Define E , its differential and its map to the original complex by these stabilized components. The finite compositions into any fixed original term are epic. For cohomology in degree r , the three terms and two differentials in degrees $r-1, r, r+1$ stabilize after finitely many replacements. At that point the finite composition is a quasi-isomorphism, so the stabilized map is an isomorphism on H^r . This proves the result without assuming exact inverse limits. $\square$

Lemma 4.2 (an explicit extension over a cone). Given $a : P \rightarrow H$ and a termwise-surjective quasi-isomorphism $u : Q \rightarrow C(a)$, write $u(q) = (u_H q, u_P q)$, where $u_P : Q^n \rightarrow P^{n+1}$. Define

$$\begin{aligned}
N^n &= P^n \oplus Q^n, \\
d_N(p, q) &= (d_P p - u_P q, d_Q q), \\
f(p, q) &= a(p) + u_H q.
\end{aligned} \tag{4.1}$$

Then $P \rightarrow N$ is a degreewise split inclusion, its quotient is Q , $f : N \rightarrow H$ is a termwise-surjective quasi-isomorphism, and $f|_P = a$.

Proof. The chain-map equation for u gives $d_P u_P + u_P d_Q = 0$ and $d_H u_H + a u_P = u_H d_Q$. These identities give $d_N^2 = 0$ and $d_H f = f d_N$ by substitution. Surjectivity of u followed by the graded projection $C(a)^n \rightarrow H^n$ proves surjectivity of u_H , and hence of f .

Compare the triangle from $0 \rightarrow P \rightarrow N \rightarrow Q \rightarrow 0$ with the cone triangle of a . The maps on P, N, Q are respectively $1, f, u$. The first square commutes strictly. In the second square, $u\pi - if$ is the homotopy boundary of $h(p, q) = (0, -p)$ into $C(a)^{n-1}$, because

$$d_C h + h d_N = (-a(p), u_P q) = u\pi - if.$$

The boundary of the split extension is $-u_P : Q \rightarrow P[1]$, since that is its off-diagonal differential, and it equals the negative cone projection composed with u . Thus all squares commute on cohomology. The outside vertical maps are an identity and a quasi-isomorphism; the long exact cohomology sequences and the Five Lemma prove that f is a quasi-isomorphism. $\square$

Proposition 4.3 (compatible bounded-above system). For any complex G , there are termwise-surjective quasi-isomorphisms $E_n \rightarrow \tau_{\leq n} G$, for $n = 1, 2, \dots$, with E_n bounded above and all its terms sums of B_U . They form a commutative direct system whose transitions are termwise split injections with quotients termwise sums of B_U . The induced map $\lim_{\rightarrow} E_n \rightarrow G$ is a termwise-surjective quasi-isomorphism from a K-flat complex of flat terms.

Proof. Recall that $\tau_{\leq n} G$ agrees with G below n , has $\ker d_G^n$ at n , and vanishes above n . Its inclusions into the next truncation are chain maps, and their union is G . Lemma 4.1 supplies E_1 . Suppose $E_n \rightarrow \tau_{\leq n} G$ has been constructed. Compose it into $H = \tau_{\leq n+1} G$, and take the cone $C(a)$. This cone is bounded above. Resolve it by a termwise-surjective $Q \rightarrow C(a)$ as in Lemma 4.1 and apply Lemma 4.2. Set $E_{n+1} = N$ and its map to H equal to f . Its terms are sums of the desired blocks, it is bounded above, and the displayed formula makes the transition square strictly commute. Its graded quotient by E_n is Q , with the required terms.

Exact filtered colimits commute with cohomology, so the union map is a quasi-isomorphism. It is onto in each degree because the truncations exhaust G . Each E_n is K-flat by Lemma 2.3, so their filtered colimit is K-flat. Its terms are flat by Lemma 1.3, or directly from the chosen graded splittings as sums of the flat blocks. $\square$

Corollary 4.4 (factoring a map from a K-flat complex). If $a : P \rightarrow H$ and P is K-flat, then a factors through a K-flat $N \rightarrow H$ that is a termwise-surjective quasi-isomorphism. If P has flat terms, N can have flat terms too.

Proof. Resolve $C(a)$ by Theorem 3.1 and apply Lemma 4.2. The split exact sequence $0 \rightarrow P \rightarrow N \rightarrow Q \rightarrow 0$ makes N K-flat by Lemma 2.3. Its terms $P^n \oplus Q^n$ are flat when the terms of P are. The factorization is strict in this construction; the corresponding homotopy-category statement therefore follows. $\square$

Lemma 4.5 (comparison after tensoring). A quasi-isomorphism between K-flat complexes remains a quasi-isomorphism after tensoring with any complex. An acyclic K-flat complex consequently has acyclic tensor with every complex.

Proof. For a complex A , choose a K-flat resolution $F \rightarrow A$ by Theorem 3.1. If $P \rightarrow Q$ is a quasi-isomorphism between K-flats, in the tensor square with rows $F \otimes P \rightarrow F \otimes Q$ and $A \otimes P \rightarrow A \otimes Q$, the vertical maps are quasi-isomorphisms by K-flatness of P, Q , and the top map is one by K-flatness of F . Two out of three forces the bottom map to be one. Apply this with $P \rightarrow 0$ for the last assertion. $\square$

5. Examples and failed shortcuts

K-flat with nonflat terms. The disk $D_0(\mathbb{Z}/2)$ is contractible. The contraction used for the disks in Theorem 3.1 makes its tensor with every complex contractible, so it is K-flat. Its nonzero terms are not flat over $\mathbb{Z}$: tensoring the injection $2 : \mathbb{Z} \rightarrow \mathbb{Z}$ gives zero on the nonzero group $\mathbb{Z}/2$.

Free terms without K-flatness. Over $R = k[\epsilon]/(\epsilon^2)$, let P have R in every integer degree and differential ϵ . It is acyclic since both kernel and image are ϵR , and its terms are free. Let F be its left half in degrees at most zero, with augmentation $F \rightarrow k = R/(\epsilon)$. Then F resolves $k[0]$: the negative degrees are exact and the degree-zero cokernel is k . It is K-flat by the bounded-above lemma. Its tensor with the acyclic P is therefore acyclic. But $k \otimes P$ has one copy of k in every degree and zero differential. If P were K-flat, it would carry $F \rightarrow k[0]$ to a quasi-isomorphism, which is impossible. More explicitly, the acyclic test $C(F \rightarrow k[0])$ has tensor with P of cohomology k in every degree, by the cone exact sequence. This displays the test that fails.

Why a general colimit is different. The cokernel of $2 : \mathbb{Z}[0] \rightarrow \mathbb{Z}[0]$ is $(\mathbb{Z}/2)[0]$. Both source and target are K-flat, but the cokernel is not: tensoring the acyclic complex $\mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}/2$ in degrees 0, 1, 2 gives a complex with zero first differential and nonzero degree-zero cohomology. Thus the word “filtered” in the colimit theorem cannot be dropped.

Local generators need not be projective. On $[0, 1]$ with the constant ring sheaf $\mathbb{Z}$, let $j : (0, 1) \hookrightarrow [0, 1]$. The sequence $0 \rightarrow j_! \mathbb{Z} \rightarrow \mathbb{Z}_{[0,1]} \rightarrow i_* \mathbb{Z}_{\{0,1\}} \rightarrow 0$ is exact on stalks. Its global-section map in the last two terms is the diagonal $\mathbb{Z} \rightarrow \mathbb{Z}^2$, so it is not onto. Therefore $\text{Hom}(B_X, -) = \Gamma(X, -)$ is not exact and the flat building block $B_X = \mathcal{O}$ is not projective. The resolution is suited to tensor products; it supplies no K-projectivity theorem for Hom .

6. Exercises with checked solutions

Exercise 1 (easy: summands). Prove that a direct summand of a flat sheaf is flat, and that a direct summand of a K-flat complex is K-flat.

Solution. Tensor distributes over the finite direct sum defining the splitting. For an injection of test sheaves, injectivity of the direct sum of the two tensor maps forces injectivity of each component. Right exactness gives flatness of each sheaf summand. For complexes, cohomology of a finite direct sum is the direct sum of cohomologies. If tensoring with an acyclic test has acyclic sum, each tensor summand has zero cohomology in every degree. That is K-flatness.

Exercise 2 (medium: the converse stalk test). Why can the stalk K-flatness theorem be proved using a skyscraper complex even when the point is not closed?

Solution. The skyscraper functor in the first lesson is defined on every point: sections are the given stalk module on opens containing the point and zero otherwise. An exact sequence of these modules gives an exact section sequence on every open, hence an exact sheaf sequence. Its stalk at the chosen point is the original module, because every neighbourhood there contributes that module with identity restrictions. No assertion about stalks at other points is needed. For an acyclic stalk complex C , use $x_* C$ as the sheaf test, then take its tensor stalk $C \otimes P_x$. Thus the proof applies without a closed-point assumption.

Exercise 3 (medium: the bound direction). Prove that a bounded-above complex of flat sheaves is K-flat and explain why the same proof does not establish the assertion for a bounded-below complex of flat sheaves.

Solution. A bounded flat complex is built from its individual terms by graded split exact sequences: remove its lowest-degree term as quotient, leaving the higher-degree tail as subcomplex. Lemma 2.3 gives induction on length. If P is bounded above, the brutal lower truncations $P^{\geq m}$ are bounded subcomplexes, because the differential raises degree. Their inclusions as m decreases form a filtered system with union P , so filtered closure proves the assertion. Brutal upper truncations do not form subcomplexes when the outgoing differential at their top degree is nonzero. Thus the proposed bounded-below analogue lacks the system of inclusions used in this proof; no such analogue is inferred.

Exercise 4 (hard: a two-term Sierpiński example). Let $X = \{\eta, s\}$ have opens $\emptyset, \{\eta\}, X$ and constant

structure ring $\mathbb{Z}$. Set $B = B_X$, $E = B_{\{\eta\}}$, and let $i : E \rightarrow B$ be the canonical inclusion. Let M have section diagram $M(X) = \mathbb{Z}/2 \rightarrow M(\{\eta\}) = 0$, and $T = j_{\eta!}(\mathbb{Z}/2)$. Construct a bounded-above resolution by sums of B, E of the two-term complex $G = (M \xrightarrow{0} T)$ in degrees 0, 1, and the compatible system of Proposition 4.3 for this G .

Solution. On this space a module sheaf is exactly its restriction map from X to $\{\eta\}$; the only other sheaf condition is on the empty set. Here B has diagram $\mathbb{Z} \xrightarrow{1} \mathbb{Z}$, and E has $0 \rightarrow \mathbb{Z}$. Define

$$P^{-2} = E, \quad P^{-1} = E \oplus B, \quad P^0 = B \oplus E, \quad P^1 = E,$$

with zero other terms and differentials

$$t \mapsto (-2t, i(t)), \quad (a, b) \mapsto (i(a) + 2b, 0), \quad (b, c) \mapsto 2c.$$

Both adjacent composites vanish. The augmentation reduces the B coordinate in degree zero onto M and reduces the degree-one E onto T ; it is epic on every stalk. At s , the E stalks vanish and the complex is $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ in degrees $-1, 0$, resolving M_s . At η , the first three components contributed by M are $\mathbb{Z} \rightarrow \mathbb{Z}^2 \rightarrow \mathbb{Z}$, with maps $(-2, 1)$ and $(1, 2)$. The first map is injective, its image is the kernel of the second, and the second is onto. The remaining E coordinates give $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ in degrees 0, 1, resolving T_η in degree one. Thus the augmentation is a quasi-isomorphism. Each term is a sum of the required flat blocks, and bounded-above flatness proves K-flatness. Since $\tau_{\leq n} G = G$ for all $n \geq 1$, take every $E_n = P$ and every transition the identity. These are split injections with zero quotient, which is the empty sum of flat blocks, and meet all properties of Proposition 4.3.

Exercise 5 (hard: the extension signs). In Lemma 4.2, verify both the chain-map equation for f and the homotopy in the second triangle square. What breaks if only the sign of u_P in d_N is reversed?

Solution. The first component of the equation $d_C u = u d_Q$ is $d_H u_H + a u_P = u_H d_Q$. Therefore

$$d_H f(p, q) = a d_P p + u_H d_Q q - a u_P q = f d_N(p, q).$$

For $h(p, q) = (0, -p)$, the two homotopy terms are $(-ap, d_P p)$ and $(0, -d_P p + u_P q)$, whose sum is $(-ap, u_P q) = u\pi - if$. Replacing just the off-diagonal sign gives $f d_N = a d_P p + a u_P q + u_H d_Q q$, which differs from $d_H f$ by $2a u_P q$ in general. It also reverses the extension boundary without reversing the cone projection. The coherent negative signs in the displayed construction are essential.

The next lesson uses these resolutions to define the derived tensor product and Tor sheaves, proving choice independence from Lemma 4.5 and the localization universal property.

Sources and licensing: Stacks project authors, [Modules on Ringed Spaces, flat modules](#); [Cohomology of Sheaves, flat resolutions](#); and [Derived Categories, compatible direct systems](#). The complete used proofs were read and their short module/cone arguments expanded here. The cycle construction is an expository alternative to the source's compatible bounded resolutions, not a new existence theorem. N. Spaltenstein, *Resolutions of unbounded complexes*, Compositio Mathematica 65 (1988), 121–154, is [available from Numdam](#). See the [combined licence notice](#) and [GNU FDL text](#).

The derived tensor product and Tor sheaves

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Ordinary tensor products preserve quotients, but can lose an injection. For example, tensoring $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ with $\mathbb{Z}/2$ turns an injection into the zero map. A resolution keeps the lost information in negative cohomological

degrees. The derived tensor product packages this information without depending on a chosen resolution, and Tor sheaves name its cohomology for two module sheaves.

The prerequisite is [Flat modules and K-flat resolutions](#), Lemmas 1.1–1.3 and 2.1–2.4, Theorem 3.1, Lemmas 3.2 and 4.1, and Lemma 4.5. We also use its earlier prerequisite, [Complexes, cones and localization](#), Theorems 4.2 and 5.1 and Proposition 5.2. In particular, roofs and their common refinements have already been proved, rather than assumed.

Throughout, X is an arbitrary topological space and $\mathcal{O}$ is a sheaf of **commutative unital rings**. Complexes are cohomologically indexed and can be unbounded in both directions. Their tensor product means the direct-sum totalization

$$\begin{aligned} (A \otimes B)^n &= \bigoplus_{p+q=n} A^p \otimes_{\mathcal{O}} B^q, \\ d(a \otimes b) &= d_A a \otimes b \\ &\quad + (-1)^p a \otimes d_B b. \end{aligned} \tag{0.1}$$

We write $K(\mathcal{O})$ for the homotopy category, $D(\mathcal{O})$ for its localization at quasi-isomorphisms, and Q for localization. Cone triangles have the convention $(f, i, -p)$ of the common reading. No compactness, separation, finite dimension or Noetherian assumption enters the construction.

The construction follows the Stacks project authors’ *Cohomology of Sheaves*, “Flat resolutions”, especially Tags 06YG, 06YH, 08BP and 08BQ, in the [AI Integrated Stacks Project edition](#). We prove the comparison and coherence maps before defining Tor sheaves. Source attribution and the licence for adapted passages appear in the [course notice](#).

1. The two tensor comparisons

A K-flat complex P has $A \otimes P$ acyclic whenever A is acyclic. The two variables in a tensor comparison play different roles.

Lemma 1.1 (tensor invariance). If P is K-flat and $u : A \rightarrow B$ is a quasi-isomorphism, then $u \otimes 1_P$ is a quasi-isomorphism. If $v : P \rightarrow P'$ is a quasi-isomorphism **between K-flat complexes**, then $1_G \otimes v$ is a quasi-isomorphism for every complex G .

Proof. Lemma 2.1 of the K-flat lesson identifies the tensor of a cone with the cone of the tensored map. When the cone is in the second variable, the shifted summand acquires the factor $(-1)^{|a|}$; the graded flip gives the other variable. The cone of u is acyclic, so its tensor with P is acyclic. This proves the first assertion.

For the second, choose a K-flat resolution $R \rightarrow G$, using Theorem 3.1 of that lesson. Consider the commuting square

$$\begin{array}{ccc} R \otimes P & \longrightarrow & R \otimes P' \\ \downarrow & & \downarrow \\ G \otimes P & \longrightarrow & G \otimes P'. \end{array} \tag{1.1}$$

Both vertical maps are quasi-isomorphisms by the first assertion, using respectively P and P' . The top map is one because R is K-flat, after interchanging the tensor factors. On every cohomology sheaf, three arrows are isomorphisms and the square commutes; the fourth is therefore an isomorphism. This proves the second assertion for arbitrary, possibly unbounded and nonflat, G . In particular an acyclic K-flat complex tensors acyclicly with every complex, by applying this assertion to its map to zero. $\square$

The second assertion would be false for an arbitrary quasi-isomorphism in place of v . For example $(\mathbb{Z} \xrightarrow{2} \mathbb{Z}) \rightarrow \mathbb{Z}/2[0]$, in degrees $-1, 0$, becomes a map from a zero-differential two-term complex to a one-term complex after tensoring with $\mathbb{Z}/2$. It loses an isomorphism in degree -1 . Its target is not K-flat.

Theorem 1.2 (one-variable definition and independence). For a K-flat resolution $\epsilon : P \rightarrow F$, the rule $G \mapsto G \otimes P$ defines an exact functor

$$- \otimes_{\mathcal{O}}^L F : D(\mathcal{O}) \rightarrow D(\mathcal{O}). \quad (1.2)$$

The functors from different resolutions have canonical natural comparison isomorphisms satisfying the identity and cocycle laws. If $R \rightarrow G$ is K-flat, both arrows in

$$G \otimes P \longleftarrow R \otimes P \longrightarrow R \otimes F \quad (1.3)$$

are quasi-isomorphisms. Thus either variable, or both variables, may be resolved.

Proof. Tensor respects homotopies, shifts and cone triangles by Lemma 2.1 of the K-flat lesson. Lemma 1.1 shows that $Q(- \otimes P)$ inverts quasi-isomorphisms, so the localization universal property, Theorem 4.2 of the common reading, gives (1.2). Theorem 5.1 there identifies distinguished triangles with images of cone triangles; their tensor images are distinguished, so the functor is exact.

For the comparison, first let $\epsilon_i : P_i \rightarrow F$, $i = 1, 2$, be termwise-surjective K-flat resolutions. Form the ordinary fibre product of complexes

$$W = P_1 \times_F P_2. \quad (1.4)$$

Its projection onto P_1 is termwise surjective with kernel $\ker \epsilon_2$; similarly for the other projection. Each kernel is acyclic, by the short exact cohomology sequence. Hence both projections are quasi-isomorphisms. Choose a termwise-surjective K-flat resolution $T \rightarrow W$. Its maps $r_i : T \rightarrow P_i$ are quasi-isomorphisms between K-flats. Lemma 1.1 gives a natural comparison $c_{12}(G)$ from $Q(G \otimes P_1)$ to $Q(G \otimes P_2)$, given by

$$c_{12}(G) = Q(1 \otimes r_2)Q(1 \otimes r_1)^{-1}. \quad (1.5)$$

Two choices $T, T' \rightarrow W$ have a common refinement: resolve $T \times_W T'$ by a termwise-surjective K-flat complex. The projections are again surjective quasi-isomorphisms. All maps to the P_i commute strictly, so after tensoring and inverting the refinement arrows the two formulas (1.5) agree.

For three resolutions, resolve $P_1 \times_F P_2 \times_F P_3$. Its projections to the pairwise products are surjective quasi-isomorphisms, with acyclic kernels. The previous refinement argument allows every pairwise comparison to be computed through this one model. Cancelling the middle invertible arrow proves the cocycle law.

For the identity law put $W = P \times_F P$. Its diagonal $\delta : P \rightarrow W$ is a quasi-isomorphism, because either projection is one and its composite with δ is the identity. Lemma 3.2 of the K-flat lesson enlarges δ to a termwise-surjective K-flat resolution $P \oplus D \rightarrow W$, with D contractible. The two maps from this resolution to P agree after precomposition with $P \rightarrow P \oplus D$. Tensoring that inclusion with G gives an invertible derived map by Lemma 1.1. The two tensor maps therefore agree in the derived category, proving the identity law. The maps are natural in G , first for chain maps and homotopies and then for roofs by localization.

If a chosen resolution is not termwise surjective, Lemma 3.2 of the K-flat lesson enlarges it by a contractible disk complex. Its inclusion into the enlargement is a quasi-isomorphism between K-flats over F . Lemma 1.1 supplies the comparison to the enlargement. Here is compatibility for two enlargements E_1, E_2 of P . The inclusions give $b : P \rightarrow W = E_1 \times_F E_2$; it is a quasi-isomorphism because $W \rightarrow E_1$ and $P \rightarrow E_1$ are. Resolve W by a surjective $T \rightarrow W$. The projection $T \times_W P \rightarrow P$ is a surjective quasi-isomorphism, with kernel $\ker(T \rightarrow W)$. Its other projection to T is a quasi-isomorphism by two out of three after composing into W . Resolve this fibre product by a K-flat complex. Its commuting maps to P, T, E_1, E_2 , all quasi-isomorphisms between K-flats, show after tensoring that the comparison of the enlargements intertwines their inclusions from P . This proves independence of the enlargement.

Finally the left arrow of (1.3) is a quasi-isomorphism because P is K-flat; the right one is a quasi-isomorphism because R is K-flat. This proves balancing. $\square$

2. Morphisms, associativity and symmetry

Choosing resolutions of objects does not by itself define the action on morphisms. We now justify that action and its compatibility with every comparison above.

Lemma 2.1 (localizing the K-flat models). Let $K_{\text{fl}}(\mathcal{O})$ be the full subcategory of $K(\mathcal{O})$ on K-flat complexes, and let S_{fl} consist of its quasi-isomorphisms. The inclusion induces an equivalence

$$K_{\text{fl}}(\mathcal{O})[S_{\text{fl}}^{-1}] \simeq D(\mathcal{O}). \quad (2.1)$$

Proof. Lemma 2.4 of the K-flat lesson shows that this full subcategory contains zero and is closed under shifts, finite sums and cones. All fraction squares and cancellation constructions in Lemma 4.1 of the common reading use these operations. When their starting objects are K-flat, their new cone or shifted cone is therefore K-flat. Consequently the roof construction and equivalence by common refinements in Theorem 4.2 there apply within K_{fl} .

Every object of $D(\mathcal{O})$ is isomorphic to a K-flat resolution, proving essential surjectivity. A roof $P \leftarrow Z \rightarrow R$ in the ambient homotopy category, with P, R K-flat, can be refined by a K-flat resolution $T \rightarrow Z$. The composite $T \rightarrow P$ is a quasi-isomorphism. Thus it has a roof entirely in K_{fl} , proving fullness. If two such roofs are equal in $D(\mathcal{O})$, Theorem 4.2 supplies a common refinement, possibly with a nonflat middle complex Z . Resolve that Z by a K-flat T . The composites are a common refinement within K_{fl} , proving faithfulness. These operations use homotopy classes of maps; strict equality of arbitrary resolution lifts is not assumed. $\square$

Theorem 2.2 (the tensor bifunctor). There is a bifunctor

$$\otimes_{\mathcal{O}}^{\text{L}} : D(\mathcal{O}) \times D(\mathcal{O}) \longrightarrow D(\mathcal{O}) \quad (2.2)$$

exact in each variable, whose value is represented by $P \otimes R$ for K-flat models of its two arguments. It has canonical associativity, symmetry and unit isomorphisms, and commutes with restriction to any open subset.

Proof. On $K_{\text{fl}} \times K_{\text{fl}}$ use ordinary total tensor. It respects homotopies and sends a quasi-isomorphism in either variable to one by Lemma 1.1. Localize successively in each variable. To check that the two resulting morphism actions commute, first note that $f \otimes 1$ and $1 \otimes g$ commute for chain maps of degree zero. They therefore commute with inverse denominators too: invert either side of the commuting square. A general roof is a numerator followed by an inverse denominator. This proves the commuting rule for all pairs of roofs and defines a bifunctor on the two localizations. Lemma 2.1 transfers it to $D(\mathcal{O}) \times D(\mathcal{O})$.

Cones of maps between K-flats are K-flat. Cone squares and tensor shift comparisons thus prove exactness on their localization. More explicitly, any roof between K-flats can be written $Qf(Qs)^{-1}$ with the middle object K-flat; its cone triangle is isomorphic to the image of the cone triangle of f . Tensor preserves the latter, so it preserves the former. Theorem 1.2 identifies this bifunctor with the one-variable construction, including its resolution comparisons.

For three K-flat models P, R, T , the ordinary association map is

$$\begin{aligned} (P \otimes R) \otimes T &\longrightarrow P \otimes (R \otimes T), \\ (p \otimes r) \otimes t &\longmapsto p \otimes (r \otimes t). \end{aligned} \quad (2.3)$$

Its inverse removes the parentheses. Both totalizations have the same direct sum indexed by $i + j + k = n$. Their differentials on an elementary tensor have the three terms with signs $1, (-1)^i, (-1)^{i+j}$, so (2.3) is a chain isomorphism. The tensor product of two K-flats is K-flat by Lemma 2.4 of the previous lesson. Thus the left and right sides compute the two iterated derived products. Naturality for chain maps extends to roofs and inverse denominators as above, proving a natural derived associativity isomorphism.

For $|p| = i$ and $|r| = j$, symmetry is represented by

$$c_{P,R}(p \otimes r) = (-1)^{ij} r \otimes p. \quad (2.4)$$

For the $d_P p$ term the two routes have sign exponents $(i+1)j$ and $ij+j$; for the $d_R r$ term they have exponents $i+i(j+1)$ and ij , equal modulo two. This checks the chain-map equation. Its square has sign $(-1)^{2ij} = 1$, giving its inverse. The unit is $\mathcal{O}[0]$, which is K-flat; the usual multiplication $P \otimes \mathcal{O} \rightarrow P$ is a chain isomorphism.

These identifications are coherent. For four factors every path around the associativity pentagon sends a tensor to the same tensor with the final parentheses. The unit triangle does the same after multiplying by the degree-zero scalar. In the symmetry hexagon, moving a degree- i tensor past degrees j, k gives either $i(j+k)$ or $ij+ik$; the signs agree. These are equalities of chain maps, hence of derived maps. Naturality and (2.1) imply that replacing any model preserves these equalities.

Finally, restriction j^* to an open U is exact and commutes with tensor and direct sums, by the stalk and sheafification formulas of the first lesson. It preserves K-flatness by Lemma 2.2 of the K-flat lesson. Restricting resolutions $P \rightarrow F, R \rightarrow G$ therefore gives resolutions on U , and the ordinary tensor identification gives

$$(F \otimes_{\mathcal{O}}^{\mathbf{L}} G)|_U \cong F|_U \otimes_{\mathcal{O}_U}^{\mathbf{L}} G|_U. \quad (2.5)$$

On local elementary tensors this is the identity, so it commutes with the association, symmetry and unit maps and with all roof comparisons. $\square$

The tensor sign in (2.4) concerns the degrees of the tensors. A morphism in the homotopy or derived category has degree zero, so no additional sign is inserted when composing the two morphism actions. Shifts carry the signed identifications already checked in Lemma 2.1 of the K-flat lesson.

3. Tor sheaves and the flatness criterion

For module sheaves F, G , viewed in degree zero, and integers $p \geq 0$, define

$$\mathrm{Tor}_p^{\mathcal{O}}(F, G) = H^{-p}(F \otimes_{\mathcal{O}}^{\mathbf{L}} G). \quad (3.1)$$

They are sheaves of $\mathcal{O}$ -modules and are functorial in both arguments. The cohomological minus sign is essential: ordinary projective or flat resolutions have their resolving terms in nonpositive degrees.

Proposition 3.1 (degree zero and stalks). There is a natural identification $\mathrm{Tor}_0(F, G) = F \otimes G$. The derived tensor of two sheaves has no positive cohomology. Moreover

$$\mathrm{Tor}_p^{\mathcal{O}}(F, G)_x \cong \mathrm{Tor}_p^{\mathcal{O}_x}(F_x, G_x). \quad (3.2)$$

Proof. Lemma 4.1 of the K-flat lesson supplies a resolution $P \rightarrow F[0]$ by flat terms, with $P^n = 0$ for $n > 0$. It is K-flat by Lemma 2.3 there. Its degree-zero cohomology is F , so $\mathrm{coker}(d_P^{-1}) = F$. Tensoring this presentation with G is right exact and gives

$$H^0(P \otimes G) = \mathrm{coker}(d_P^{-1} \otimes 1_G) = F \otimes G.$$

The tensor has no positive terms, hence no positive cohomology. The augmentation induces the displayed degree-zero identification; naturality follows either by the roof construction or by the uniqueness of the degree-zero quotient map.

Stalks are exact and commute with tensor and direct sums, by Theorems 2.1 and 3.1 of the first lesson and Lemma 1.2 of the K-flat lesson. Hence $(P \otimes G)_x = P_x \otimes_{\mathcal{O}_x} G_x$ and taking stalks commutes with cohomology.

Lemma 2.2 of that lesson says P_x is K-flat, and its map to F_x is a quasi-isomorphism. It therefore computes the tensor over the stalk ring, proving (3.2). $\square$

Proposition 3.2 (long exact Tor sequence). A short exact sequence $0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0$ gives a natural long exact sequence

$$\begin{aligned} & \cdots \longrightarrow \mathrm{Tor}_p(F_1, G) \\ & \longrightarrow \mathrm{Tor}_p(F_2, G) \longrightarrow \mathrm{Tor}_p(F_3, G) \\ & \longrightarrow \mathrm{Tor}_{p-1}(F_1, G) \longrightarrow \cdots \\ & \longrightarrow \mathrm{Tor}_1(F_3, G) \longrightarrow F_1 \otimes G \\ & \longrightarrow F_2 \otimes G \longrightarrow F_3 \otimes G \longrightarrow 0. \end{aligned} \tag{3.3}$$

There is the analogous sequence in the second variable.

Proof. Proposition 5.2 of the common reading turns the short exact sequence into a distinguished triangle. The exact tensor functor of Theorem 2.2 gives a distinguished triangle after tensoring with G . Its long exact cohomology sequence exists by Theorem 5.1 of that reading. Replace H^{-p} by (3.1) and H^0 by Proposition 3.1; the subsequent positive cohomology is zero. The connecting map is the usual connecting cohomology map with the cone sign convention fixed above. The construction is natural for maps of short exact sequences. Symmetry gives the second-variable assertion. $\square$

Theorem 3.3 (Tor detects flatness). For a module sheaf F , the following are equivalent:

1. F is flat.
2. $\mathrm{Tor}_1(F, G) = 0$ for every module sheaf G .
3. $\mathrm{Tor}_p(F, G) = 0$ for every $p > 0$ and every module sheaf G .

Proof. If F is flat, $F[0]$ is K-flat by Lemma 2.3 of the preceding lesson. The one-variable construction therefore computes $F \otimes^L G$ as the ordinary sheaf $F \otimes G$ in degree zero. This proves $1 \Rightarrow 3 \Rightarrow 2$.

For $2 \Rightarrow 1$, take any injection $G \rightarrow H$, with cokernel C . The second-variable version of (3.3) contains

$$\mathrm{Tor}_1(F, C) \longrightarrow F \otimes G \longrightarrow F \otimes H.$$

The left sheaf is zero, so the right arrow is injective. Tensor is already right exact, so it is exact and F is flat. This proves all implications without a finiteness assumption. $\square$

The quantifier “every G ” cannot be replaced by the single test $G = F$. The continuous-function example below makes this failure explicit.

Quotients by ideals

Lemma 3.4. For ideals I, J in a commutative ring R ,

$$\begin{aligned} \mathrm{Tor}_0^R(R/I, R/J) &= R/(I + J), \\ \mathrm{Tor}_1^R(R/I, R/J) &= (I \cap J)/(IJ). \end{aligned} \tag{3.4}$$

For $p \geq 2$, $\mathrm{Tor}_p^R(R/I, R/J) \cong \mathrm{Tor}_{p-1}^R(I, R/J)$. The same assertions hold for ideal sheaves, with intersections, sums and products taken as sheaf ideals.

Proof. Apply (3.3) to $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$, with second factor R/J . Flatness of R annihilates all its positive Tor. The resulting first segment is

$$\begin{aligned}
0 \longrightarrow \mathrm{Tor}_1(R/I, R/J) &\longrightarrow I \otimes_R R/J \\
&\longrightarrow R/J \longrightarrow (R/I) \otimes_R R/J \longrightarrow 0.
\end{aligned}$$

The presentation of tensor by balancing relations identifies $I \otimes_R R/J$ with I/IJ : the map sends $i \otimes (r + J)$ to $ri + IJ$, and its inverse sends $i + IJ$ to $i \otimes 1$. The kernel of $I/IJ \rightarrow R/J$ is $(I \cap J)/IJ$, and the cokernel is $R/(I + J)$. The remaining exact segments give the dimension-shifting assertion. For sheaf ideals perform these calculations on every stalk; stalks preserve the relevant kernels and images, including ideal products, so they give the stated sheaf identities. $\square$

Formula (3.4) computes only the first two groups without extra information on I . The higher groups require a resolution or the displayed dimension shift. Exercise 1 supplies a complete all-degree computation for two integer ideals.

4. Two geometric calculations

Skyscrapers on a smooth one-dimensional local model

Take $X = \mathbb{R}$ with $\mathcal{O}$ the sheaf of real analytic functions, and let $k_a = a_* \mathbb{R}$, with the action given by evaluation at a . Multiplication by $t - a$ gives an exact sequence

$$0 \longrightarrow \mathcal{O} \xrightarrow{t-a} \mathcal{O} \longrightarrow k_a \longrightarrow 0. \quad (4.1)$$

Here is a stalk proof. At $b \neq a$, the germ $t - a$ is invertible. At a , a germ of an analytic function has a convergent power series in $t - a$. Vanishing at a means its constant coefficient is zero, so dividing by $t - a$ shifts that series and gives another convergent series on a smaller interval. Conversely every such multiple vanishes at a . Multiplication is injective because a continuous function with $(t - a)f(t) = 0$ vanishes away from a and hence also at a . Evaluation is surjective on germs, by constants. The first lesson's stalk criterion proves (4.1).

The two $\mathcal{O}$ terms give a bounded flat, hence K-flat, resolution in degrees $-1, 0$. For the same point a , tensoring it with k_a makes its differential zero. Therefore

$$\begin{aligned}
\mathrm{Tor}_0(k_a, k_a) &= k_a, \\
\mathrm{Tor}_1(k_a, k_a) &= k_a, \\
\mathrm{Tor}_p(k_a, k_a) &= 0 \quad (p \geq 2).
\end{aligned} \quad (4.2)$$

For $a \neq b$, tensoring (4.1) with k_b makes the differential multiplication by the nonzero real number $b - a$, an isomorphism. All Tor sheaves are then zero. The nonzero degree-one sheaf at a coincident point records the kernel lost by ordinary tensor; its location is visible directly from the stalk computation.

Continuous functions and two axes in the plane

Now let $X = \mathbb{R}^2$ and $\mathcal{O}$ be the sheaf of real-valued continuous functions. Set $Z = \{y = 0\}$, $Z' = \{x = 0\}$, and write $I_Z, I_{Z'}$ for the sheaf ideals of functions vanishing on these subsets. Define

$$\mathcal{O}_Z = \mathcal{O}/I_Z, \quad \mathcal{O}_{Z'} = \mathcal{O}/I_{Z'}.$$

These are the continuous-function sheaves along the axes, extended to X : a continuous function on an axis extends locally by projection onto that axis, proving surjectivity of the restriction map on germs. Its kernel is precisely the indicated vanishing ideal.

For these ideals

$$I_Z \cap I_{Z'} = I_Z I_{Z'}. \quad (4.3)$$

Indeed, for a real continuous function f vanishing on both axes, put $h = \sqrt{|f|}$ and define $g = f/\sqrt{|f|}$ where $f \neq 0$, with $g = 0$ on its zero set. Both functions are continuous: at the zero set their absolute values are $\sqrt{|f|}$, which tends to zero. Both vanish on each axis and $f = gh$. This proves inclusion into the product on every germ; the reverse inclusion follows from vanishing. Lemma 3.4 gives

$$\mathrm{Tor}_1(\mathcal{O}_Z, \mathcal{O}_{Z'}) = 0. \quad (4.4)$$

The ordinary tensor is the real skyscraper at the origin. Outside the origin at least one quotient has zero stalk. At the origin the sum of the two ideals is the maximal ideal of germs vanishing there. To prove this last assertion, decompose such a germ f as

$$f = \frac{x^2 f}{x^2 + y^2} + \frac{y^2 f}{x^2 + y^2}, \quad (4.5)$$

with both terms defined to be zero at the origin. Their absolute values are bounded by $|f|$, so they are continuous there. The first vanishes on Z' , and the second on Z . Thus $\mathcal{O}/(I_Z + I_{Z'}) = 0_* \mathbb{R}$, including its evaluation action. Equations (4.4) and (4.5) compute degree one and degree zero for the required closed-subset example.

This behaviour of first Tor does not imply flatness. Let $R = \mathcal{O}_0$, $\mathfrak{m}$ its evaluation kernel, and $k = R/\mathfrak{m} = \mathbb{R}$. The same square-root factorization gives $\mathfrak{m} = \mathfrak{m}^2$, so $\mathrm{Tor}_1^R(k, k) = 0$. But multiplication by the coordinate x is injective on R : a continuous germ annihilated by x vanishes on the dense complement of the vertical axis and therefore everywhere on a neighbourhood. The two-term free resolution of $R/(x)$ tensors with k to a zero-differential complex, giving

$$\mathrm{Tor}_1^R(k, R/(x)) = k \neq 0. \quad (4.6)$$

Thus k is nonflat by Theorem 3.3. Notice that $I_{Z',0} \neq (x)$: the continuous germ $\sqrt{|x|}$ vanishes on the vertical axis, but cannot be divided by x continuously at the origin. In the analytic curve calculation division by its parameter was possible. The ideals in these examples must therefore be calculated in their actual function rings.

Bounded inputs with an unbounded answer

On a one-point space put $R = k[\varepsilon]/(\varepsilon^2)$ and view $k = R/(\varepsilon)$ in degree zero. The resolution

$$\cdots \xrightarrow{\varepsilon} R \xrightarrow{\varepsilon} R \xrightarrow{\varepsilon} R \longrightarrow k \quad (4.7)$$

has its last R in degree zero. In each negative degree the kernel and image of multiplication by ε are the same ideal (ε) , and the degree-zero cokernel is k . It is a bounded-above free resolution, hence K-flat. Tensoring with k kills every differential, so $\mathrm{Tor}_p^R(k, k) = k$ for every $p \geq 0$. Even two sheaves in degree zero can therefore have derived tensor with infinitely many negative cohomology sheaves. The perfect-complex lesson later proves the finite bounds furnished by a finite locally free model.

5. Exercises with checked solutions

Exercise 1 (easy: ideals on a one-point space). Let R be a commutative ring and I, J ideals. Derive formulas for Tor in degrees zero and one. Then compute every $\mathrm{Tor}_p^{\mathbb{Z}}(\mathbb{Z}/m, \mathbb{Z}/n)$, for positive integers $m, n \geq 2$. Explain why the first formulas alone do not determine all higher groups for arbitrary ideals.

Solution. Tensor $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$ with R/J in the derived sense. The vanishing of positive Tor against R embeds first Tor as the kernel of $I/IJ \rightarrow R/J$. This kernel is $(I \cap J)/IJ$, and the cokernel is $R/(I + J)$. The next segments identify higher Tor with $\text{Tor}_{p-1}(I, R/J)$, which need not vanish.

For $\mathbb{Z}/m$, use the injective multiplication map in the free resolution $\mathbb{Z} \xrightarrow{m} \mathbb{Z}$, in degrees $-1, 0$. Tensor with $\mathbb{Z}/n$. Put $d = \gcd(m, n)$. Its cokernel is $\mathbb{Z}/d$, and its kernel consists of the multiples of n/d modulo n , also a cyclic group of order d . Thus

$$\text{Tor}_p^{\mathbb{Z}}(\mathbb{Z}/m, \mathbb{Z}/n) = \begin{cases} \mathbb{Z}/d, & p = 0, 1, \\ 0, & p \geq 2. \end{cases}$$

The first-degree ideal formula agrees: $I \cap J = \text{lcm}(m, n)\mathbb{Z}$ and $IJ = mn\mathbb{Z}$. In contrast, for the dual-number ring in (4.7), the ideals $I = J = (\varepsilon)$ have all positive Tor equal to k , by that explicit free resolution. Thus the higher-degree answer needs information about the actual ideal or resolution.

Exercise 2 (medium: associativity with changed models). Let P, R, T resolve F, G, H . Prove the derived associativity isomorphism, including independence when a different K-flat model $V \rightarrow P \otimes R$ is used for the intermediate object. Check the symmetry hexagon on tensors of degrees $1, 1, -1$.

Solution. The product $P \otimes R$ is K-flat, so $(P \otimes R) \otimes T$ computes the left-associated product without another resolution. The right-associated product is computed by $P \otimes (R \otimes T)$. Their summands are indexed by the same triples, and their differential terms have signs $1, (-1)^i, (-1)^{i+j}$. Reassociation is therefore a chain isomorphism with no extra sign.

The map $V \rightarrow P \otimes R$ is a quasi-isomorphism between K-flats. Lemma 1.1 makes $V \otimes T \rightarrow (P \otimes R) \otimes T$ a quasi-isomorphism. Its composite with reassociation gives the comparison from this alternative intermediate model. For two such choices, resolve their termwise-surjective enlargements' fibre product; the resulting comparisons agree by Theorem 1.2. Changing the individual P, R, T is handled by the same comparison and the naturality of the ordinary association map. Consequently the isomorphism belongs to the derived objects, rather than to the displayed choice alone.

For the stated degrees, moving the degree-one factor across the combined block has sign $(-1)^{1(1-1)} = 1$. Moving it successively past the two odd-degree factors gives $(-1)^{1 \cdot 1}(-1)^{1 \cdot (-1)} = (-1)(-1) = 1$. This checks that instance of the hexagon; the calculation $i(j+k) = ij + ik$ proves every instance.

Exercise 3 (medium: which Tor tests suffice?). Prove Theorem 3.3 from the derived construction. Show also that it is enough to test $\text{Tor}_1(F, x_*N) = 0$ for every point x and every $\mathcal{O}_x$ -module N . Explain the role of “every” using the continuous germ ring at the origin.

Solution. A flat F in degree zero is K-flat, so its ordinary tensor with $G[0]$ already computes the derived tensor and has no negative cohomology. Thus flatness implies all positive Tor vanish, which implies first Tor vanishes. Conversely, the exact sequence for an injection $G \rightarrow H$ has $\text{Tor}_1(F, \text{coker}(G \rightarrow H))$ immediately before $F \otimes G \rightarrow F \otimes H$. Vanishing of this first Tor makes that map injective. Right exactness then proves flatness.

For the point tests, take the stalk at x . Proposition 3.1 and $(x_*N)_x = N$, from Lemma 5.2 of the first lesson, give $\text{Tor}_1^{\mathcal{O}_x}(F_x, N) = 0$. For any injection of stalk-ring modules apply the same Tor exact sequence, proving that F_x is flat. The stalk criterion, Lemma 1.2 of the K-flat lesson, proves F flat.

In the continuous germ ring R , the residue k has $\text{Tor}_1(k, k) = \mathfrak{m}/\mathfrak{m}^2 = 0$, but $\text{Tor}_1(k, R/(x)) = k$. This particular additional test detects the failure. Testing against the residue itself therefore does not replace testing against all modules.

Exercise 4 (hard: only one resolution for an unbounded input). Over $\mathbb{Z}$, let $F = \mathbb{Z}/2[0]$, and let G have $\mathbb{Z}/2$ in every integer degree with zero differentials. Compute $G \otimes^{\mathbb{L}} F$ by resolving only F . Prove in general that the resulting one-resolution answer agrees with resolving both variables. Compare it with the ordinary tensor.

Solution. Use $P = (\mathbb{Z} \xrightarrow{2} \mathbb{Z})$ in degrees $-1, 0$, with its augmentation to F . This is a bounded free, hence K-flat, resolution. In total degree n , $G \otimes P$ has two summands, $G^n \otimes P^0$ and $G^{n+1} \otimes P^{-1}$. Both are $\mathbb{Z}/2$. The G differential is zero, and the contribution of 2 on P is zero after tensoring; the total differential is therefore zero. Hence

$$H^n(G \otimes^L F) = (\mathbb{Z}/2) \oplus (\mathbb{Z}/2),$$

for every $n \in \mathbb{Z}$.

No resolution of G was needed for this computation. To verify agreement in general, choose any K-flat $R \rightarrow G$. The map $R \otimes P \rightarrow G \otimes P$ is a quasi-isomorphism by K-flatness of P , and $R \otimes P \rightarrow R \otimes F$ is a quasi-isomorphism by K-flatness of R . This gives (1.3) with no boundedness assumption. If P is replaced by another K-flat resolution, the common-refinement comparison in Theorem 1.2 makes the resulting isomorphisms agree naturally.

The ordinary tensor $G \otimes F$ has only one $\mathbb{Z}/2$ in each degree. The augmentation map $G \otimes P \rightarrow G \otimes F$ is the projection to the P^0 summand, so it is not a quasi-isomorphism. The target F is not K-flat; it cannot serve as the target K-flat model in the second assertion of Lemma 1.1.

Exercise 5 (medium: where can tensor cohomology occur?). For complexes A, B , put $\Sigma(A) = \{x : H^n(A)_x \neq 0 \text{ for some } n\}$. Prove

$$\Sigma(A \otimes^L B) \subset \Sigma(A) \cap \Sigma(B).$$

Use this to compute the derived tensor of sheaves supported at two distinct points of a Hausdorff space.

Solution. Choose K-flat resolutions $P \rightarrow A, R \rightarrow B$. At a point outside $\Sigma(A)$, the stalk A_x is acyclic, so P_x is an acyclic K-flat complex. Lemma 1.1 says that its tensor with every complex, in particular R_x , is acyclic. Since stalks commute with total tensor and cohomology, all stalk cohomology of $P \otimes R$ vanishes there. The same argument applies outside $\Sigma(B)$, proving the inclusion.

For a sheaf supported at one point the only possibly nonzero cohomology stalk is at that point. If the two points are distinct the two sets are disjoint, so every cohomology stalk of the derived tensor is zero. Exactness detection by stalks makes the product zero in the derived category. On a Hausdorff space ordinary skyscrapers have precisely this stalk support: a different point has a neighbourhood avoiding the chosen point. This recovers the distinct-point part of the curve example without its parameter resolution.

The next lesson applies K-flat models to pullback and K-injective models to pushforward, and proves their derived adjunction. The tensor bifunctor and its comparisons established here will make the pullback's compatibility with tensor natural.

Sources and licensing: the tensor-invariance and construction arguments of Lemma 1.1 and Theorem 1.2 are taken from the Stacks project, tags 06YA, 06YG and 06YH, in its AI Integrated Stacks Project edition (GFDL), checked and edited by GPT-6.1 Sol (OpenAI), at Ultra, with the full fibre-product comparison included. The Tor definition and flatness proof are taken from the Stacks project, tags 08BP and 08BQ, in that edition (GFDL), checked and edited by the same writing AI; the explicit K-flat computation is given above. The pinned [complete source passage](#) and its preceding comparison lemmas were read. The localization, coherence, geometric calculations and exercises are expanded here with their actual conditions. See the [combined licence notice](#) and [GNU FDL text](#).

Derived pullback and pushforward

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Pullback changes both the space and the coefficient ring. Pushforward collects sections over inverse images of opens. Their ordinary adjunction survives passage to derived categories, but proving this requires two different resolutions. A K-flat model controls the tensor in pullback, and a K-injective model controls pushforward. The proof must also work when pushforward of that K-injective model is not itself K-injective.

The prerequisites are [The derived tensor product and Tor sheaves](#), Lemma 1.1, Theorem 1.2, Lemma 2.1 and Theorem 2.2, and [K-injective resolutions in Grothendieck categories](#), Theorems 4.1 and 5.1 and Proposition 5.2. Their earlier prerequisites supply the chain-level adjunction (first lesson, Theorem 4.4), the open-extension adjunction (Lemma 5.1 there), maps into K-injectives (bounded-derived lesson, Theorem 3.3), and roof cancellation (common reading, Theorem 4.2). For the spectral-sequence exercise we use the lower-bound preservation in Theorem 4.1 of the bounded-derived lesson.

Unless explicitly stated otherwise, all structure sheaves are commutative and unital, and all derived categories are unbounded. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. Write

$$f^*G = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}G. \quad (0.1)$$

We retain cohomological indexing, direct-sum tensor totalization and the common reading's negative cone-projection convention. These results require no separation, Noetherian, compactness or finite homological-dimension hypothesis.

The construction sources are the Stacks project authors' *Cohomology of Sheaves*, “Derived pullback”, “Cohomology of unbounded complexes” and “Some properties of K-injective complexes”, in the [AI Integrated Stacks Project edition](#). The proofs use resolutions, roofs and adjunction to establish the comparison maps with their actual coefficient hypotheses. Source attribution and the licence for adapted passages appear in the [course notice](#).

1. Derived pullback on K-flat models

Theorem 1.1. Choosing a K-flat resolution $P \rightarrow G$ and setting

$$Lf^*G = f^*P \quad (1.1)$$

defines a well-defined exact functor $Lf^* : D(\mathcal{O}_Y) \rightarrow D(\mathcal{O}_X)$. It carries K-flat models to K-flat models. For composable $f : X \rightarrow Y, g : Y \rightarrow Z$, there are canonical natural isomorphisms

$$Lf^*Lg^*G \cong L(gf)^*G. \quad (1.2)$$

Proof. Pullback preserves K-flatness by Lemma 2.2 of the K-flat lesson. It also preserves quasi-isomorphisms **between** K-flats. Indeed, at x their pullback map is

$$\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,f(x)}} P_{f(x)} \longrightarrow \mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,f(x)}} P'_{f(x)}.$$

The stalk models are K-flat; Lemma 1.1 of the tensor lesson applies to their quasi-isomorphism and the possibly nonflat module $\mathcal{O}_{X,x}$. Thus this map is a quasi-isomorphism on every stalk. Pullback is additive, so it preserves homotopies, shifts and cones. Lemma 2.1 of the tensor lesson identifies the localization of K-flat models with the full derived category. Localizing pullback on those models gives the claimed exact functor and canonical comparisons for all choices and roofs.

Choose P K-flat on Z . Then g^*P is K-flat on Y , so the iterated derived pullback is computed by f^*g^*P . Ordinary pullback composition identifies this with $(gf)^*P$. On stalks its scalar-extension map has the formula

$$\begin{aligned} s \otimes (t \otimes p) &\longmapsto sf^\sharp(t) \otimes p, \\ s \otimes p &\longmapsto s \otimes (1 \otimes p). \end{aligned}$$

Balancing proves that these are inverse maps; scalars have degree zero, so they commute with differentials. The same balanced sheaf maps, using inverse-image composition, define the identification before taking stalks. They are natural for chain maps, hence for inverse denominators and roofs. For three successive pullbacks every association multiplies the same successive scalar images; associativity proves coherence of (1.2). $\square$

Proposition 1.2 (tensor compatibility). There are canonical natural isomorphisms

$$\begin{aligned} & Lf^*(A \otimes_{\mathcal{O}_Y}^{\mathbf{L}} B) \\ & \cong Lf^*A \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*B, \\ & F \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*G \\ & \cong F \otimes_{f^{-1}\mathcal{O}_Y}^{\mathbf{L}} f^{-1}G. \end{aligned} \tag{1.3}$$

In the second line F is an $\mathcal{O}_X$ -complex; the right side retains its $\mathcal{O}_X$ -action from F . The isomorphisms respect the tensor associativity, symmetry, unit and pullback-composition comparisons.

Proof. For K-flat P, R on Y , their tensor is K-flat. Thus the first line is computed by the ordinary isomorphism $f^*(P \otimes R) \cong f^*P \otimes f^*R$. On a stalk, with coefficient map $R_0 \rightarrow S$, the inverse-direction map and its inverse are

$$\begin{aligned} (s \otimes p) \otimes (t \otimes r) &\longmapsto st \otimes (p \otimes r), \\ s \otimes (p \otimes r) &\longmapsto (s \otimes p) \otimes (1 \otimes r). \end{aligned}$$

Commutativity of the rings and balancing make the formulas well defined. Their differential signs are $1, (-1)^{|p|}$ on both sides. Inverse image and scalar extension commute with direct sums, so the same maps identify the totalizations. They commute with elementary reassociation, the Koszul flip and units; on four scalar factors every route is scalar multiplication in the same order. Their naturality extends to localized models.

For the second line choose only a K-flat $P \rightarrow G$. Then $f^{-1}P$ is K-flat over $f^{-1}\mathcal{O}_Y$, by the K-flat pullback criterion applied with unchanged coefficients, and f^*P is K-flat over $\mathcal{O}_X$. The one-resolution construction computes both sides by

$$F \otimes_{\mathcal{O}_X} (\mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}P) \cong F \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}P.$$

The map sends $a \otimes (s \otimes p)$ to $sa \otimes p$, with inverse $a \otimes p \mapsto a \otimes (1 \otimes p)$. This proves the asserted module action and naturality without assuming F flat. $\square$

There is also a natural comparison $Lf^*G \rightarrow f^*G$ for a specified complex G , induced by $f^*P \rightarrow f^*G$. It need not be an isomorphism. All comparisons to ordinary tensor commute with (1.3): on resolving tensors both routes send $(s \otimes p) \otimes (t \otimes r)$ to $st \otimes (\epsilon_P p \otimes \epsilon_R r)$. For the map $a : P \otimes R \rightarrow A \otimes B$, Corollary 4.4 of the K-flat lesson supplies a factorization $a = cb$ through a K-flat resolution $c : N \rightarrow A \otimes B$. Applying pullback gives $f^*c f^*b = f^*a$; the left side represents the route through $Lf^*(A \otimes B)$ and its ordinary comparison, and the right side is the same elementary tensor formula. This proves the complete comparison square, including the augmentation maps.

2. Derived pushforward and the adjunction

Choose $F \rightarrow I$ K-injective with injective terms, using Theorem 4.1 of the K-injective lesson. Theorem 5.1 there, applied to the additive functor f_* , gives an exact functor

$$\begin{aligned} Rf_* : D(\mathcal{O}_X) &\longrightarrow D(\mathcal{O}_Y), \\ Rf_*F &= f_*I. \end{aligned} \tag{2.1}$$

Two K-injective models are homotopy equivalent, and additive pushforward preserves these equivalences and their homotopies. Maps between the models, unique up to homotopy, define the functor on all derived morphisms. No claim about K-injectivity of f_*I is needed for this construction.

Lemma 2.1 (a relative Hom comparison). For P K-flat on Y and I K-injective on X , localization induces a bijection

$$\begin{aligned} & \text{Hom}_{K(\mathcal{O}_Y)}(P, f_*I) \\ & \cong \text{Hom}_{D(\mathcal{O}_Y)}(P, f_*I). \end{aligned} \tag{2.2}$$

Proof. Represent a derived map by a roof $P \leftarrow Z \rightarrow f_*I$. Resolve Z by a K-flat $T \rightarrow Z$. Its composite $s : T \rightarrow P$ is a quasi-isomorphism between K-flats. Theorem 1.1 makes f^*s a quasi-isomorphism. Maps into I consequently give a bijection

$$\text{Hom}_{K(\mathcal{O}_X)}(f^*P, I) \longrightarrow \text{Hom}_{K(\mathcal{O}_X)}(f^*T, I),$$

by Theorem 3.3 of the bounded-derived lesson. The chain-level adjunction identifies this with precomposition by s on maps into f_*I . Thus the roof's numerator comes from a unique homotopy class $P \rightarrow f_*I$, proving surjectivity of (2.2).

For injectivity, if such a homotopy class becomes zero after localization, roof cancellation (common reading, Theorem 4.2) gives a quasi-isomorphism $Z \rightarrow P$ making its composite zero in the homotopy category. Resolve Z by a K-flat T . Precomposition along the resulting $T \rightarrow P$ is the bijection just proved by the chain adjunction, so the original class was zero. Applying this to a difference also proves injectivity for equality of two classes. $\square$

Theorem 2.2 (unbounded adjunction). There is a bifunctorial bijection

$$\begin{aligned} & \text{Hom}_{D(\mathcal{O}_X)}(Lf^*G, F) \\ & \cong \text{Hom}_{D(\mathcal{O}_Y)}(G, Rf_*F). \end{aligned} \tag{2.3}$$

It gives $Lf^* \dashv Rf_*$, with natural unit $\eta : G \rightarrow Rf_*Lf^*G$ and counit $\epsilon : Lf^*Rf_*F \rightarrow F$.

Proof. Choose $P \rightarrow G$ K-flat and $F \rightarrow I$ K-injective. The left group is $\text{Hom}_K(f^*P, I)$, because the target is K-injective. The ordinary adjunction on complexes, including homotopies, identifies this with $\text{Hom}_K(P, f_*I)$. Lemma 2.1 identifies the latter with $\text{Hom}_D(P, f_*I)$; use $P \simeq G$ to obtain the right group.

For naturality in G , every derived map between its K-flat models is a roof with K-flat middle, by Lemma 2.1 of the tensor lesson. The chain adjunction and all Hom comparisons commute with its numerator and denominator, hence with the inverse denominator. For naturality in F , derived maps between K-injective models are homotopy classes of chain maps; the adjunction commutes with them. Replacing either model uses these same comparisons. Thus (2.3) is independent of choices and bifunctorial.

Define the unit as the transpose of 1_{Lf^*G} , and the counit as the inverse transpose of 1_{Rf_*F} . Write $L = Lf^*$, $R = Rf_*$, and α for the bijection. Naturality implies $\alpha(u) = R(u)\eta$ and $\alpha^{-1}(v) = \epsilon L(v)$. Applying these to $\eta = \alpha(1)$ and $\epsilon = \alpha^{-1}(1)$ gives

$$\begin{aligned} \epsilon_{LG}L(\eta_G) &= 1_{LG}, \\ R(\epsilon_F)\eta_{RF} &= 1_{RF}. \end{aligned} \tag{2.4}$$

These are the triangle identities, so the asserted maps give the adjunction. $\square$

Corollary 2.3 (unchanged coefficients, also noncommutative). If $\mathcal{O}_X = f^{-1}\mathcal{O}_Y$, for any unital sheaf of rings, $f^{-1} \dashv Rf_*$ on unbounded derived categories.

Proof. Inverse image is exact, and f_* is its right adjoint. Proposition 5.2 of the K-injective lesson makes f_*I K-injective. Therefore

$$\begin{aligned}\mathrm{Hom}_D(f^{-1}G, F) &= \mathrm{Hom}_K(f^{-1}G, I) \\ &= \mathrm{Hom}_K(G, f_*I) = \mathrm{Hom}_D(G, Rf_*F).\end{aligned}$$

Each equality is natural for maps and homotopies and extends to roofs. This proves the exact-coefficient adjunction without using commutative K-flat theory in the noncommutative case. $\square$

3. Composition, Leray and the local descriptions

Proposition 3.1. For composable morphisms $f : X \rightarrow Y$ and $g : Y \rightarrow Z$,

$$Rg_*Rf_*F \cong R(gf)_*F \quad (3.1)$$

canonically and naturally. In particular,

$$R\Gamma(Y, Rf_*F) \cong R\Gamma(X, F). \quad (3.2)$$

The second identification is in $D(\Gamma(Y, \mathcal{O}_Y))$, with the right side's scalars restricted along $\Gamma(Y, \mathcal{O}_Y) \rightarrow \Gamma(X, \mathcal{O}_X)$.

Proof. Successive adjunctions show that Rg_*Rf_* is right adjoint to Lf^*Lg^* . Theorem 1.1 identifies the latter with $L(gf)^*$, whose right adjoint is $R(gf)_*$.

For completeness, a right adjoint is unique in the required canonical sense. If $L \dashv R_1$ and $L \dashv R_2$, send the counit $LR_1B \rightarrow B$ through the second adjunction to a map $\beta_B : R_1B \rightarrow R_2B$. Reverse the roles to obtain γ_B . Naturality of the adjunctions and their triangle identities show that the transpose of $\gamma_B\beta_B$ under the first adjunction is its counit, so $\gamma_B\beta_B = 1$; likewise $\beta_B\gamma_B = 1$. Transposing maps after a morphism $B \rightarrow B'$ proves naturality. This is the unique isomorphism compatible with the given adjunctions, and its uniqueness also gives compatibility for three successive functors. It proves (3.1) without assuming f_*I is K-injective.

Put $A = \Gamma(Y, \mathcal{O}_Y)$. The canonical morphism $Y \rightarrow (\mathrm{pt}, A)$ has pushforward $\Gamma(Y, -)$. Its composite with f has pushforward $\Gamma(X, -)$ with the indicated A -action. Apply (3.1) to obtain (3.2). $\square$

Proposition 3.2 (opens and cohomology sheaves). Let $U \subset X$ and $V \subset Y$ be open, and for the restriction formula set $U = f^{-1}V$, $f_V : U \rightarrow V$. Then:

1. A K-injective I restricts to a K-injective $I|_U$, and $H^q(U, F) = H^q(U, F|_U)$.
2. The sheafification of $U \mapsto H^q(U, F)$ is $H^q(F)$.
3. $(Rf_*F)|_V \cong R(f_V)_*(F|_U)$.
4. $R\Gamma(V, Rf_*F) \cong R\Gamma(f^{-1}V, F)$, with scalar restriction understood.
5. $H^q(Rf_*F)$ is the sheaf associated to $V \mapsto H^q(f^{-1}V, F)$.

Proof. Extension by zero $j_!$ is exact and left adjoint to restriction, by Lemma 5.1 of the first lesson. Proposition 5.2 of the K-injective lesson makes its right adjoint, restriction, preserve K-injectives. Thus $I|_U$ resolves $F|_U$, and both definitions of the group in (1) use the same section complex $\Gamma(U, I)$.

For (2), this complex gives the presheaf

$$U \mapsto \frac{\ker(I^q(U) \rightarrow I^{q+1}(U))}{\mathrm{im}(I^{q-1}(U) \rightarrow I^q(U))}. \quad (3.3)$$

At a point, its filtered colimit is $H^q(I_x)$, because stalks and filtered colimits are exact. This is $H^q(F)_x$. The map sending a section cycle to its cohomology germ therefore induces an isomorphism after sheafification,

by the first lesson's stalk criterion. This argument works for any complex of sheaves; it does not require exactness of taking sections.

For (3), use the K-injective I on X and $I|_U$ on U . The ordinary sheaf identity $(f_*I)|_V = (f_V)_*(I|_U)$ follows by evaluating on opens of V , so it gives the derived identity. Apply (3.2) to f_V and use (1) and (3) to obtain (4). Finally Rf_*F is represented by f_*I ; apply the cohomology-presheaf argument to this complex. On an open V , its section cohomology is $H^q(\Gamma(f^{-1}V, I)) = H^q(f^{-1}V, F)$ by (1). Sheafification gives (5). $\square$

These assertions distinguish a cohomology sheaf from the cohomology of sections on a specified open. Sections of a quotient sheaf need not be the quotient of sections; sheafification in (2) and (5) is part of the statement.

4. Coefficient comparison and exact adjoints

Write F_{ab} for the underlying complex of abelian sheaves. Forgetting the module action is exact: kernels and images are the same underlying sheaf kernels and images. It therefore induces $D(\mathcal{O}_X) \rightarrow D(\mathbb{Z}_X)$, even when $\mathcal{O}_X$ has integer torsion.

Theorem 4.1 (modules and underlying abelian sheaves). The canonical comparison maps

$$\begin{aligned} R\Gamma(U, F) &\longrightarrow R\Gamma(U, F_{ab}), \\ (Rf_*F)_{ab} &\longrightarrow Rf_*^{ab}(F_{ab}) \end{aligned} \tag{4.1}$$

are isomorphisms, respectively in $D(\text{Ab})$ and $D(\mathbb{Z}_Y)$.

Proof. Choose $F \rightarrow I$ K-injective over $\mathcal{O}_X$, and a K-injective abelian-sheaf resolution $I_{ab} \rightarrow J$. The comparisons are represented by $\Gamma(U, I) \rightarrow \Gamma(U, J)$ and $f_*I_{ab} \rightarrow f_*J$. The choice of J changes them only by the canonical homotopy comparison between K-injective resolutions, so these are natural derived maps.

Consider the coefficient morphism $h : (X, \mathcal{O}_X) \rightarrow (X, \mathbb{Z}_X)$, the identity on spaces and the unit map on rings. Its ordinary pushforward is exactly forgetting the action, so its derived pushforward is also that exact functor. Also $Lh^*\mathbb{Z}_X = \mathcal{O}_X$, because $\mathbb{Z}_X[0]$ is K-flat over itself. The adjunction of Theorem 2.2 gives, for every integer n ,

$$\begin{aligned} \text{Hom}_{D(\mathcal{O}_X)}(\mathcal{O}_X, F[n]) \\ \cong \text{Hom}_{D(\mathbb{Z}_X)}(\mathbb{Z}_X, F_{ab}[n]). \end{aligned} \tag{4.2}$$

The left group is $H^n(\Gamma(X, I))$: maps from $\mathcal{O}_X$ into $I[n]$, modulo homotopy, are section cycles modulo section boundaries. The right group is $H^n(\Gamma(X, J))$. To check that (4.2) is the specified comparison, use Lemma 2.1 for h and the K-flat source $\mathbb{Z}_X$. Its intermediate chain adjunction sends a section of I to the same underlying section of I_{ab} ; localization followed by $I_{ab} \rightarrow J$ gives exactly the cohomology map of $\Gamma(X, I) \rightarrow \Gamma(X, J)$. Hence that map is an isomorphism in every degree. Repeat on U , using Proposition 3.2 to restrict the models, to prove the first line of (4.1).

For the second line, take the section cohomology of $f_*I_{ab} \rightarrow f_*J$ on every open V . It is the first comparison on $f^{-1}V$, so it is an isomorphism in every degree. Sheafifying these cohomology presheaves, as in Proposition 3.2, proves that the map is a quasi-isomorphism. No preservation of K-injectivity by forgetting coefficients was assumed. $\square$

Proposition 4.2 (open extension and flat pushforward). For an open inclusion $j : U \rightarrow X$, exact extension by zero is left adjoint to restriction on derived categories. For a flat morphism f , pushforward preserves K-injective complexes.

Proof. For $B \in D(\mathcal{O}_U)$ and $F \rightarrow I$ K-injective on X , restriction of I is K-injective. Exactness of $j_!$ and the chain adjunction give

$$\begin{aligned}
\mathrm{Hom}_{D(\mathcal{O}_X)}(j_! B, F) &= \mathrm{Hom}_{K(\mathcal{O}_X)}(j_! B, I) \\
&= \mathrm{Hom}_{K(\mathcal{O}_U)}(B, I|_U) \\
&= \mathrm{Hom}_{D(\mathcal{O}_U)}(B, F|_U).
\end{aligned}$$

Naturality is inherited from the chain adjunction and localization, so this is $j_! \dashv j^*$.

Flatness of f means that each $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$. The pullback stalk formula then makes f^* exact. If A is acyclic on Y , $f^* A$ is acyclic, and $\mathrm{Hom}_K(A, f_* I) = \mathrm{Hom}_K(f^* A, I) = 0$. This is the definition of K-injectivity of $f_* I$. It applies to every K-injective I , without a termwise-injectivity assumption. $\square$

5. Point maps and constant coefficients

For $f : X \rightarrow \mathrm{pt}$ with target ring A , pushforward is global sections with its A -action, so $Rf_* = R\Gamma(X, -)$ with that action. The pullback is

$$Lf^* G = \mathcal{O}_X \otimes_{A_X}^L G_X, \quad (5.1)$$

where A_X, G_X denote inverse-image constant sheaves. Resolve G by a K-flat complex over A ; its constant inverse image is K-flat, proving (5.1) by one-variable tensor. If $\mathcal{O}_X = A_X$ with the identity coefficient map, this is the exact constant sheaf functor, also for noncommutative A by Corollary 2.3. For the canonical map to $(\mathrm{pt}, \Gamma(X, \mathcal{O}_X))$, the structure sheaf need not be constant, and (5.1) gives the correct pullback.

For a point x , give its one-point source the ring $\mathcal{O}_{X,x}$. Pullback is then the exact stalk functor, since the coefficient map at that point is the identity. If the stalk ring is local and we instead give the point its residue field $\kappa(x)$, the ring map is the quotient, and

$$Li_x^* F = \kappa(x) \otimes_{\mathcal{O}_{X,x}}^L F_x. \quad (5.2)$$

Take stalks of a K-flat resolution to prove this formula. A residue field is part of this additional local-ring hypothesis, not data attached to every arbitrary ringed space.

On the analytic curve of the tensor lesson, pull back k_a to the residue point at a . Its two-term parameter resolution becomes $\mathbb{R} \xrightarrow{0} \mathbb{R}$ in degrees $-1, 0$. Thus the derived pullback has $\mathbb{R}$ in both degrees, whereas the ordinary pullback retains only degree zero. With the stalk-ring structure on the point, pullback has only $k_{a,a}$ in degree zero, because taking a stalk is exact. These are two different morphisms of ringed spaces on the same underlying point inclusion.

6. Exercises with checked solutions

Exercise 1 (easy: an open subspace). Prove $H^p(U, F) = H^p(U, F|_U)$, for an arbitrary open U and an unbounded complex. Explain why restriction of the resolution is admissible.

Solution. Resolve $F \rightarrow I$ by a K-injective. For an acyclic A on U , exactness of $j_!$ makes $j_! A$ acyclic, and $\mathrm{Hom}_K(A, I|_U) = \mathrm{Hom}_K(j_! A, I) = 0$. Thus $I|_U$ is K-injective. Both groups are the cohomology of the same complex of sections $\Gamma(U, I)$. No bound on I , nor on the cohomological dimension of U , was used.

Exercise 2 (medium: the adjunction without projective models). Let $P \rightarrow G$ be K-flat and $F \rightarrow I$ K-injective. Prove that a roof $P \leftarrow Z \rightarrow f_* I$ comes from a unique homotopy class $f^* P \rightarrow I$. Deduce the unbounded adjunction and its naturality, without assuming that P is K-projective or $f_* I$ K-injective.

Solution. Resolve Z by K-flat T . The denominator $s : T \rightarrow P$ is a quasi-isomorphism between K-flats, so $f^* s$ is a quasi-isomorphism. Transpose the numerator $a : T \rightarrow f_* I$ to $f^* T \rightarrow I$. Since I is K-injective, there is a unique homotopy class $b : f^* P \rightarrow I$ whose composite with $f^* s$ is that numerator. Transposing back makes the roof the localization of a map $P \rightarrow f_* I$.

If two maps $P \rightarrow f_* I$ give the same derived class, cancellation supplies a quasi-isomorphism into P annihilating their difference. Resolve its source by a K-flat; the same precomposition bijection forces the original difference to be zero in the homotopy category. This proves uniqueness. Maps $Lf^*G \rightarrow F$ are maps $f^*P \rightarrow I$ modulo homotopy, so we obtain (2.3). The constructions commute with chain maps. All source derived morphisms can be expressed as roofs between K-flat models; commuting with their arrows also gives commuting with their inverse denominators. Target morphisms are homotopy classes between K-injective models. This proves bifactoriality.

Exercise 3 (medium to hard: Leray and convergence). For a module sheaf M , prove

$$\begin{aligned} E_2^{p,q} &= H^p(Y, R^q f_* M) \\ &\implies H^{p+q}(X, M), \\ p, q &\geq 0, \end{aligned} \tag{6.1}$$

including the construction, differential bidegrees and convergence. This exercise concerns sheaves in degree zero; it imposes no finite dimension on X or Y .

Solution. Choose a bounded-below injective resolution of M , with zero terms below zero. It is K-injective by Lemma 3.2 of the bounded-derived lesson. Put $K = f_* I$, so K represents $Rf_* M$, has no negative terms and has $H^q(K) = R^q f_* M$.

Let $A_q = R\Gamma(Y, \tau_{\leq q} K)$, with $A_q = 0$ for $q < 0$. Good upper truncations give triangles

$$\begin{aligned} A_{q-1} &\longrightarrow A_q \\ &\longrightarrow R\Gamma(Y, H^q K)[-q] \\ &\longrightarrow A_{q-1}[1]. \end{aligned} \tag{6.2}$$

To check the cofiber before applying $R\Gamma$, the quotient of $\tau_{\leq q} K$ by $\tau_{\leq q-1} K$ has the injection $\text{im } d^{q-1} \rightarrow \ker d^q$ as its two possibly nonzero terms. Its quotient map to $H^q K[-q]$ is a quasi-isomorphism. The common reading's short-exact-sequence triangle therefore proves (6.2).

Write $D_q^n = H^n(A_q)$ and $E_q^n = H^{n-q}(Y, H^q K)$. The long exact sequences give maps

$$\begin{aligned} i : D_q^n &\longrightarrow D_{q+1}^n, \\ j : D_q^n &\longrightarrow E_q^n, \\ k : E_q^n &\longrightarrow D_{q-1}^{n+1}. \end{aligned} \tag{6.3}$$

They form an exact couple: $\ker j = \text{im } i$, $\ker k = \text{im } j$ and $\ker i = \text{im } k$, with the adjacent indices understood. This is simply exactness at the three positions of (6.2).

Here is an explicit iteration, including the proof that it gives successive pages. For $r \geq 2$, define

$$\begin{aligned} Z_r^{n,q} &= k^{-1}(\text{im}(i^{r-2})), \\ B_r^{n,q} &= j \ker(i^{r-2}). \end{aligned} \tag{6.4}$$

In the first line the indicated power maps D_{q-r+1}^{n+1} into D_{q-1}^{n+1} ; in the second it maps D_q^n into D_{q+r-2}^n . The image of j lies in every Z_r , since $kj = 0$, so $B_r \subset Z_r$. Set $\bar{E}_r^{p,q} = Z_r^{p+q,q} / B_r^{p+q,q}$. For $r = 2$ the power is the identity, so this is precisely $H^p(Y, H^q K)$.

Given $e \in Z_r^{n,q}$, choose $x \in D_{q-r+1}^{n+1}$ with $i^{r-2}x = ke$, and define $d_r[e] = [jx]$. Changing x by an element of the power's kernel changes jx by B_r . Changing e by an element of B_r leaves ke unchanged, so this is well defined. Its output has total degree $n + 1$ and second index $q - r + 1$; hence its bidegree is $(r, 1 - r)$. Since $kj = 0$, its square is zero.

To verify the next page, suppose $d_r[e] = 0$. Then $jx = jy$ for a y killed by i^{r-2} . Exactness gives $x - y = iz$, and therefore $ke = i^{r-1}z$. Thus $e \in Z_{r+1}$; the converse follows by taking $x = iz$. A differential image jx has $i^{r-1}x = ike = 0$, so it lies in B_{r+1} . Conversely, if $i^{r-1}x = 0$, then $i^{r-2}x \in \ker i = \operatorname{im} k$. Choose e with $ke = i^{r-2}x$; it lies in Z_r , and its differential is jx . Consequently cycles modulo differential images are Z_{r+1}/B_{r+1} , proving $E_{r+1} = H(E_r, d_r)$. This constructs the spectral sequence directly from the triangles.

It remains to identify its limit. The tail $\tau_{\geq q+1}K$ is represented in degrees at least $q+1$. The lower-bound preserving injective resolution of the bounded-derived lesson shows that its $R\Gamma$ also has no cohomology below $q+1$. The truncation triangle therefore gives $H^n(A_q) \cong H^n(R\Gamma(Y, K))$ whenever $q \geq n$. Put this stable group T^n , and filter it by

$$F_q T^n = \operatorname{im}(D_q^n \rightarrow T^n). \quad (6.5)$$

For $r > q+1$, the source D_{q-r+1}^{n+1} in (6.4) is zero, so $Z_r = \ker k = \operatorname{im} j$. For sufficiently large r , the target of the power in the second line is already the stable T^n . Hence

$$\begin{aligned} E_{\infty}^{n-q,q} &= \operatorname{im} j / j \ker(D_q^n \rightarrow T^n) \\ &\cong F_q T^n / F_{q-1} T^n. \end{aligned} \quad (6.6)$$

The last identification follows from $\ker j = \operatorname{im}(D_{q-1}^n \rightarrow D_q^n)$ and the quotient theorem for the map $D_q^n \rightarrow T^n$. The filtration starts with $F_{-1} = 0$ and ends with $F_n = T^n$. All $E_2^{p,q}$ with $p < 0$ or $q < 0$ vanish, because sheaf cohomology has no negative degrees and K has no negative cohomology. Thus each total degree has a finite filtration, and each position has finitely many possible incoming and outgoing differential degrees. This proves convergence. Finally Proposition 3.1 identifies $T^n = H^n(R\Gamma(Y, Rf_* M))$ with $H^n(X, M)$, proving (6.1) with the stated scalar action.

Exercise 4 (hard: the flatness hypothesis for K-injectives). Prove preservation of K-injectives by a flat pushforward using its defining test. Show that the statement can fail for the one-point map with coefficient map $\mathbb{Z} \rightarrow \mathbb{F}_2$.

Solution. An acyclic A pulls back to an acyclic f^*A when f is flat. For K-injective I , the chain adjunction gives $\operatorname{Hom}_K(A, f_* I) = \operatorname{Hom}_K(f^* A, I) = 0$. This proves the preservation assertion.

For the indicated nonflat map, $I = \mathbb{F}_2[0]$ is K-injective over $\mathbb{F}_2$: the field is injective by Baer's criterion, whose only ideal tests are zero and the field itself, and a bounded complex of injectives is K-injective. Its pushforward is the same complex as a $\mathbb{Z}$ -module. The complex $A = (\mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{F}_2)$ in degrees 0, 1, 2 is acyclic. It has a chain map to I , reduction modulo two in degree zero and zero elsewhere. A null homotopy in degree zero would express this nonzero reduction as $h^1 \circ 2$, but every homomorphism $\mathbb{Z} \rightarrow \mathbb{F}_2$ kills multiples of two. It cannot be null-homotopic, so the pushforward is not K-injective. Nevertheless the general derived adjunction holds by Theorem 2.2.

Exercise 5 (hard: torsion coefficients and comparison). Let $\mathcal{O}_X$ have arbitrary integer torsion. Prove that $\Gamma(U, I) \rightarrow \Gamma(U, J)$ is a quasi-isomorphism when I is K-injective over $\mathcal{O}_X$ and $I_{ab} \rightarrow J$ is K-injective over $\mathbb{Z}_X$. Explain why exactness of forgetting alone does not prove that I_{ab} is K-injective.

Solution. For the coefficient morphism $h : (U, \mathcal{O}_U) \rightarrow (U, \mathbb{Z}_U)$, the right derived pushforward is exact forgetting, while $Lh^* \mathbb{Z}_U = \mathcal{O}_U$. Theorem 2.2 identifies maps from $\mathcal{O}_U$ into $I[n]_U$ with derived maps from $\mathbb{Z}_U$ into $I_{ab}[n]_U$. Lemma 2.1 identifies these with chain maps into the same underlying I_{ab} ; passing to $J|_U$ computes the latter derived group by a K-injective. On section cycles the comparison sends a cycle to its image in J , and on boundaries it sends boundaries to boundaries. Thus the induced map is precisely $H^n \Gamma(U, I) \rightarrow H^n \Gamma(U, J)$, and is an isomorphism for every n .

An exact functor preserves acyclic complexes in its domain, but K-injectivity tests maps **from every** acyclic complex in the target category. Those tests are not reduced to acyclic module complexes merely by forgetting

actions. On a point with structure ring $\mathbb{F}_2$, Exercise 4 gives an injective and K-injective module whose underlying abelian complex fails the K-injective test. The coefficient-comparison theorem consequently needs the adjunction argument just supplied.

The next lesson constructs internal derived Hom and its tensor adjunction. Later, the projection-formula and base-change lesson uses the unit, counit and coherent composition isomorphisms proved here to define the comparison maps.

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Hom complexes, internal derived Hom and Ext sheaves

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The tensor product combines two inputs, whereas internal Hom records maps from one input to another as an object on the space. Derived internal Hom must detect extensions as well as ordinary maps. Its degree-zero global cohomology recovers maps in the derived category; its other cohomology sheaves record local extensions. These statements involve derived sections, which cannot generally be replaced by ordinary sections of a cohomology sheaf.

The prerequisites are [Derived pullback and pushforward](#), Proposition 3.2 and Theorem 4.1, for restriction and local cohomology; [The derived tensor product and Tor sheaves](#), Lemma 1.1, Lemma 2.1 and Theorem 2.2, for K-flat models and their localization; and [K-injective resolutions in Grothendieck categories](#), Theorems 4.1 and 5.1. We also use maps into K-injectives and their cone closure from Theorem 3.3 and Lemma 3.1 of [Injective modules and bounded-below derived functors](#). The common reading's Theorems 3.1 and 4.2 supply cone triangles and roof cancellation.

Throughout, $\mathcal{O} = \mathcal{O}_X$ is a sheaf of commutative unital rings on an arbitrary topological space. Complexes are cohomologically indexed and may be unbounded in both directions. Tensor totalization uses sums, Hom totalization uses products, and a cone triangle ends in the negative projection. Write $H(A, B)$ for the sheaf Hom complex, reserving Hom for a group of maps.

The construction follows the Stacks project authors' *Cohomology of Sheaves*, “Hom complexes”, “Internal hom”, “Ext sheaves” and “Global derived hom”, in the [AI Integrated Stacks Project edition](#). Composition is natural in its outer variables and dinatural in the repeated middle variable; Section 4 proves the precise identities. Source attribution and the licence for adapted passages appear in the [course notice](#).

1. The sheaf Hom complex

For two module sheaves define the internal Hom by

$$\begin{aligned} \mathcal{H}om(A, B)(U) \\ = \operatorname{Hom}_{\mathcal{O}_U}(A|_U, B|_U). \end{aligned} \tag{1.1}$$

Restriction means restricting a map. Compatible maps on an open cover glue: for each local section of A , their

images agree and glue in B . The glued assignment respects restriction, addition and scalar multiplication because these properties can be checked on the cover. This proves the sheaf condition. Commutativity gives it its $\mathcal{O}$ -module action by multiplying the output.

For complexes set

$$\begin{aligned} H^n(A, B) &= \prod_i \mathcal{H}om(A^i, B^{i+n}), \\ (d\phi)^i &= d_B \phi^i \\ &\quad - (-1)^n \phi^{i+1} d_A. \end{aligned} \tag{1.2}$$

The product is the sheaf product; on every open it is the product of the section groups. Applying the differential twice leaves two mixed terms with coefficients $-(-1)^n$ and $-(-1)^{n+1}$, which cancel. The remaining terms contain d_B^2 or d_A^2 . Thus this is a complex. A degree-zero cycle is a chain map, and a degree-zero boundary is a null-homotopic map. With the usual shifted differential this proves

$$\begin{aligned} &H^n \Gamma(U, H(A, B)) \\ &= \text{Hom}_{K(\mathcal{O}_U)}(A|_U, B|_U[n]). \end{aligned} \tag{1.3}$$

Restriction to an open commutes with (1.1) and (1.2): their values on each smaller open are identical. This observation concerns restriction, not passage to a stalk. An infinite product need not commute with stalks, and a germ of a module map need not be an arbitrary map of stalk modules.

Lemma 1.1 (chain identities). Hom complexes respect homotopies. There is a natural currying isomorphism

$$\begin{aligned} &H(K, H(L, M)) \\ &\cong H(K \otimes L, M). \end{aligned} \tag{1.4}$$

Evaluation and composition are chain maps, given on homogeneous elements by

$$\begin{aligned} \phi \otimes l &\mapsto \phi(l), \\ \phi \otimes \psi &\mapsto \phi\psi. \end{aligned} \tag{1.5}$$

Proof. If $h : B \rightarrow B'[-1]$ is a homotopy between two target maps, the induced Hom homotopy is $\phi \mapsto h\phi$. If $h : A' \rightarrow A[-1]$ is a homotopy between two source maps, it is $\phi \mapsto (-1)^n \phi h$ on degree n . Substitution into (1.2) gives respectively postcomposition and precomposition with $dh + hd$; the mixed terms cancel. Hence both variables descend to homotopy categories.

Currying sends ϕ to the map $k \otimes l \mapsto \phi(k)(l)$, with no additional sign. The tensor relation is exactly the linearity in each variable. In degree n , both sides are the product over pairs of degrees (i, j) of $\mathcal{H}om(K^i \otimes L^j, M^{i+j+n})$. This uses the universal property that Hom out of a sum is a product, followed by reindexing a product. It uses no exactness assertion about products. For homogeneous k , their common differential evaluated at k, l is

$$\begin{aligned} &d_M \phi(k)(l) - (-1)^n \phi(d_K k)(l) \\ &\quad - (-1)^{n+|k|} \phi(k)(d_L l). \end{aligned}$$

Finally $d(\phi(l)) = (d\phi)(l) + (-1)^{|\phi|} \phi(dl)$ and $d(\phi\psi) = (d\phi)\psi + (-1)^{|\phi|} \phi(d\psi)$. These are precisely the tensor differentials, proving (1.5). All constructions are sectionwise and therefore commute with restriction. $\square$

We also need the cone signs. For $g : B \rightarrow B'$, the two finite blocks give $H(A, C(g)) = C(H(A, g))$; the identity identification preserves the differential and negative projection. For $f : K \rightarrow L$, put $h = H(f, I)$. In

degree n , an element of $H(C(f), I)$ is a pair (b, a) , where b has Hom degree n on L and a has degree $n - 1$ on K . The isomorphism to $C(h)[-1]$ is

$$(b, a) \mapsto ((-1)^{n+1}a, b). \quad (1.6)$$

Indeed the source differential is $(db, da - (-1)^n bf)$, and the shifted cone differential on (x, y) is $(-dx - hy, dy)$. Formula (1.6) intertwines them. The source-shift identification $H(K[1], I) \cong H(K, I)[-1]$ multiplies degree n by $(-1)^n$. Together these formulas turn a source cone triangle into the reversed distinguished Hom triangle, with its connecting map determined by the negative original projection. This verifies exactness in the contravariant variable, including its shift sign.

2. Constructing internal derived Hom

Lemma 2.1. Let I be K-injective. A quasi-isomorphism $s : A \rightarrow B$ induces a quasi-isomorphism $H(B, I) \rightarrow H(A, I)$.

Proof. On every open U , the restriction of I is K-injective, by the preceding lesson's Proposition 3.2. Maps into it are unchanged by localization. Thus precomposition by $s|_U$ is a bijection on $\text{Hom}_{K(\mathcal{O}_U)}(-, I|_U[n])$ for every integer n . Formula (1.3) gives a quasi-isomorphism of the section Hom complexes on every open. The cohomology sheaf is the sheafification of section-complex cohomology, as proved in that proposition; sheafifying these isomorphisms proves the claim. Applying the argument to every shift is essential: degree zero alone would not prove a quasi-isomorphism. $\square$

Theorem 2.2. For any L, M , choose a K-injective resolution $M \rightarrow I$. Then

$$\begin{aligned} [L, M] &:= R\mathcal{H}om(L, M) \\ &= H(L, I). \end{aligned} \quad (2.1)$$

defines an object of $D(\mathcal{O})$ and a canonical bifunctor, contravariant in L , covariant in M , exact in each variable. It commutes with restriction to opens.

Proof. Resolutions exist for every unbounded complex by the earlier K-injective theorem. The map from M to another resolution determines a unique homotopy class $I \rightarrow I'$ under the maps from M : maps into a K-injective are the same in the homotopy and derived categories. Such a comparison is a quasi-isomorphism between K-injectives, hence a homotopy equivalence. Lemma 1.1 carries its two homotopy inverses to inverse Hom maps. Comparisons are unique in the homotopy category, so their composites obey the required identity and cocycle laws.

In the source variable Lemma 2.1 inverts every quasi-isomorphism. A roof acts by its numerator and the inverse of its denominator; common refinements give the same action because Hom respects homotopies. In the target variable represent a derived map by the unique homotopy class between K-injective models. Composition of these classes, and commutation with source precomposition, prove bifunctoriality and natural independence of resolutions.

Source cone triangles give reversed Hom triangles by (1.6). A target map between K-injectives has K-injective cone by the earlier cone-closure lemma, so it supplies a model for the entire target triangle. The target cone identity proves exactness there. Shift identifications are those checked after (1.6). Finally restricting $M \rightarrow I$ gives a K-injective resolution on each open, and the sectionwise identity for Hom gives the asserted restriction isomorphism, natural in all roofs. $\square$

Lemma 2.3. If P is K-flat and I K-injective, then $H(P, I)$ is K-injective.

Proof. For an acyclic complex A , Lemma 1.1 and degree-zero homotopy classes give

$$\text{Hom}_K(A, H(P, I)) = \text{Hom}_K(A \otimes P, I) = 0.$$

The tensor is acyclic by K-flatness, and the last group vanishes by K-injectivity. This is the defining test for K-injectivity. No injectivity of the individual terms of $H(P, I)$ is required. $\square$

3. Adjunction, local maps and Ext

Theorem 3.1. There are natural isomorphisms

$$\begin{aligned} [K, [L, M]] &\cong [K \otimes^L L, M], \\ \text{Hom}_D(K, [L, M]) & \\ \cong \text{Hom}_D(K \otimes^L L, M). \end{aligned} \tag{3.1}$$

For every open U and integer n ,

$$\begin{aligned} H^n(U, [L, M]) & \\ \cong \text{Hom}_{D(\mathcal{O}_U)}(L|_U, M|_U[n]). \end{aligned} \tag{3.2}$$

Proof. Resolve L by a K-flat P and M by a K-injective I . Lemma 2.1 permits the model $H(P, I)$ for $[L, M]$; Lemma 2.3 makes this model K-injective. Therefore (1.4) is an actual complex isomorphism computing both sides of the first line of (3.1). The complex $K \otimes P$ computes the derived tensor even when K is not K-flat, by Lemma 1.1 of the tensor lesson. Taking homotopy classes and using maps into K-injectives gives the second line.

Naturality holds first for chain maps by currying. For source maps represented by roofs, Lemma 2.1 inverts the source denominators, and tensor invariance inverts the denominators on the other side. Comparison maps between K-flat models can be represented in their localized homotopy category, by the tensor lesson's Lemma 2.1. Target comparisons are homotopy classes between K-injectives. Thus the same commuting chain squares and their inverse denominator squares prove naturality for every derived morphism, not only actual chain maps on fixed resolutions.

For (3.2), restrict these models to U . Restriction preserves K-flatness and K-injectivity. The K-injective model $H(P, I)|_U$ computes derived sections, so (1.3) identifies its cohomology with the stated group of derived maps. $\square$

Define the Ext sheaves and global derived Hom by

$$\begin{aligned} \mathcal{E}xt^n(L, M) &= \mathcal{H}^n([L, M]), \\ R\text{Hom}_X(L, M) & \\ = R\Gamma(X, [L, M]). \end{aligned} \tag{3.3}$$

The global object is a complex over $\Gamma(X, \mathcal{O})$. Its underlying abelian cohomology can equally be computed by the preceding lesson's coefficient comparison. Formula (3.2) identifies its n -th cohomology with $\text{Hom}_D(L, M[n])$.

In particular, a derived morphism is represented locally by a degree-zero cycle in a Hom complex into a K-injective target, modulo its degree-zero boundaries. Composition in Proposition 4.1 below induces the usual composition of these classes. For a triangle in either input, Theorem 2.2 gives a triangle of internal Hom objects, and derived sections give the resulting long exact sequence of groups of maps. The source-variable connecting map includes the shift sign in (1.6); the target-variable map includes the negative cone projection. Thus the local extension groups and their boundary operations come from the same proved triangle conventions as the global ones. No choice of an unrelated connecting-homomorphism sign is needed.

The Ext sheaf is the sheafification of the presheaf sending U to the right side of (3.2). To see this directly, use the model $H(P, I)$; sheafifying its section-complex cohomology gives its cohomology sheaf. This does not identify ordinary sections of $\mathcal{E}xt^n$ with that derived-map group on an arbitrary open. Sheafification may change sections.

There is also a convenient formula with an arbitrary representative L : the complex $\Gamma(X, \mathbf{H}(L, I))$ computes global derived Hom. Although $\mathbf{H}(L, I)$ need not be K-injective, its comparison to $\mathbf{H}(P, I)$ is a quasi-isomorphism on sections by (1.3) and the K-injective map test. The latter is a K-injective sheaf complex, so its sections compute the derived sections. For module sheaves F, G in degree zero choose a bounded-below injective resolution $G \rightarrow I^{\geq 0}$. Its Hom complex from F has no negative degrees and degree-zero kernel $\mathcal{H}om(F, G)$. Hence $\mathcal{E}xt^n(F, G) = 0$ for $n < 0$, and $\mathcal{E}xt^0(F, G) = \mathcal{H}om(F, G)$.

4. Evaluation, composition and their variances

The adjunction gives a safe way to construct canonical maps without asserting that an arbitrary ordinary tensor of Hom complexes computes a derived tensor. Its counit is evaluation

$$e_{L,M} : [L, M] \otimes^L L \longrightarrow M. \quad (4.1)$$

It is the transpose of the identity of $[L, M]$. On models it is (1.5), after any K-flat replacement needed for its tensor factors. Currying and its naturality show that this construction is independent of those replacements.

Proposition 4.1 (composition). There is a canonical associative, unital composition map

$$\begin{aligned} c_{K,L,M} : [L, M] \otimes^L [K, L] \\ \longrightarrow [K, M]. \end{aligned} \quad (4.2)$$

It is natural contravariantly in K , covariantly in M , and dinatural in L : for $u : L \rightarrow L'$, the two maps from $[L', M] \otimes^L [K, L]$ to $[K, M]$ satisfy

$$\begin{aligned} c_{K,L,M} \circ ([u, M] \otimes 1) \\ = c_{K,L',M} \circ (1 \otimes [K, u]). \end{aligned} \quad (4.3)$$

Proof. Put $A = [L, M]$, $B = [K, L]$. Define (4.2) to be the transpose of successive evaluations

$$\begin{aligned} (A \otimes^L B) \otimes^L K &\longrightarrow A \otimes^L L \\ &\longrightarrow M. \end{aligned} \quad (4.4)$$

Here and below association maps are the coherent tensor isomorphisms proved in the tensor lesson. Evaluation is natural in its target. For a source map u , its defining adjunction gives the identity that evaluating $[u, M](a)$ on l equals evaluating a on $u(l)$. This is an identity of morphisms in the derived category: transpose either side by (3.1), and both are precomposition of the same identity map by u . It therefore holds for roofs as well as chain maps.

Postcomposing (4.4) by a map $M \rightarrow M'$ proves target naturality. Precomposing its last tensor factor by a map $K' \rightarrow K$, and applying the source evaluation identity, proves source naturality. For (4.3), transpose both sides and apply that identity once. Both become successive evaluation with u inserted between the two evaluations. Symbolically both give $a(u(b(k)))$. The adjunction is a bijection of map groups, so equality of the transposes proves equality of the maps.

For four objects, either order of composition transposes to the same three successive evaluations. The tensor associator satisfies the pentagon, so this proves associativity. The identity of L corresponds by (3.2) to a morphism $\mathbf{1} = \mathcal{O} \rightarrow [L, L]$. Evaluation with this morphism is the identity on L , by its transpose; inserting it on either side of (4.4) proves both unit laws. These arguments prove all assertions without changing resolutions in the middle of an unverified diagram. $\square$

The word *dinatural* records the opposing variances of the two occurrences of L . A map $L \rightarrow L'$ induces $[L', M] \rightarrow [L, M]$ but $[K, L] \rightarrow [K, L']$. Thus the source of composition is not a covariant or contravariant

one-variable functor of L . Formula (4.3) is the well-typed compatibility statement. It does not require u to be invertible.

Proposition 4.2 (other canonical maps). Put $L^\vee = [L, \mathbf{1}]$. There are canonical maps

$$\begin{aligned} K &\longrightarrow [L, K \otimes^L L], \\ K \otimes^L [M, L] &\longrightarrow [M, K \otimes^L L], \\ M \otimes^L L^\vee &\longrightarrow [L, M], \\ [L, M] \otimes^L K &\longrightarrow [[K, L], M]. \end{aligned} \tag{4.5}$$

The first is the adjunction unit, natural in K and dinatural in L . The second is natural covariantly in K, L and contravariantly in M . The third is natural covariantly in M , contravariantly in L . The last is natural covariantly in K, M , contravariantly in L .

Proof. Transpose the identity of $K \otimes^L L$ to obtain the first. If $\eta_{K,L}$ denotes it and $u : L \rightarrow L'$, both sides of

$$\begin{aligned} &[u, K \otimes^L L'] \eta_{K,L'} \\ &= [L, K \otimes^L u] \eta_{K,L}. \end{aligned} \tag{4.6}$$

transpose to $1_K \otimes u$. This proves its middle-variable dinaturality; ordinary target naturality proves its K naturality.

For the second map, transpose $1_K \otimes e_{M,L}$. For the third, transpose $1_M \otimes e_{L,\mathbf{1}}$, followed by the unit isomorphism. For the last, its transpose has domain $([L, M] \otimes^L K) \otimes^L [K, L]$. Move K past $[K, L]$ using the Koszul symmetry, evaluate $[K, L]$ on K , and then evaluate $[L, M]$ on the result. Each stated naturality follows from the evaluation identities already proved, followed by the naturality of transposition. On homogeneous chain representatives of degrees p, q, r , this last construction is

$$(\phi \otimes k)(h) = (-1)^{qr} \phi(h(k)). \tag{4.7}$$

The sign comes from moving k past h , not past ϕ . Lemma 1.1 and the tensor symmetry prove it is a chain map whenever the chosen models compute these derived constructions. $\square$

These are maps, with no general claim of invertibility. The perfect-complex lesson proves that the third is an isomorphism when L is perfect. The final exercise below shows a concrete failure without this hypothesis. The unit in (4.5), just like composition, has a repeated variable with opposite variances; writing its compatibility as (4.6) prevents a second misleading blanket functoriality claim.

5. Worked examples

First, $[\mathcal{O}, M] \cong M$. Evaluation at 1 identifies $H(\mathcal{O}, I)$ with I , including its differential. The K-injective resolution identifies this with M . The corresponding global statement is $R\mathrm{Hom}_X(\mathcal{O}, M) = R\Gamma(X, M)$, which may have higher cohomology even when M is a sheaf.

Let $j : U \hookrightarrow X$ be open. For any K-injective resolution $F \rightarrow I$, the extension-by-zero adjunction on every open V gives

$$H(j_! \mathcal{O}_U, I)(V) = \Gamma(V \cap U, I).$$

These identities respect differentials and restrictions. The right side is $j_*(I|_U)(V)$, and $I|_U$ is K-injective. Consequently

$$[j_! \mathcal{O}_U, F] \cong Rj_*(F|_U). \tag{5.1}$$

The ordinary Hom sheaf in degree zero is $j_*F|_U$, not extension by zero. It can have a nonzero boundary stalk. Higher Ext sheaves describe the higher local cohomology of intersections with U , through the local pushforward formula of the preceding lesson.

For a second example take $X = \mathbb{C}$ with holomorphic functions and let k_a be the skyscraper residue module at a . The tensor lesson's parameter resolution is

$$P_a = (\mathcal{O} \xrightarrow{z-a} \mathcal{O}). \quad (5.2)$$

It lies in degrees minus one and zero. This finite free complex can be used in the source of derived Hom. Indeed $H(\mathcal{O}, -)$ is the identity, so a finite filtration by its two terms shows $H(P_a, -)$ preserves quasi-isomorphisms. Thus comparison with any K-injective resolution of k_b proves that $H(P_a, k_b)$ computes $[k_a, k_b]$. It has k_b in degrees zero and one, with differential multiplication by $-(b-a)$. If $b \neq a$, this is invertible and the Hom object is zero. If $b = a$, the differential is zero and

$$[k_a, k_a] \cong k_a \oplus k_a[-1]. \quad (5.3)$$

In particular, the sheaf Ext in degree one is k_a ; all other positive Ext sheaves vanish. These objects describe the derived Hom of modules over the holomorphic structure sheaf, not Hom of arbitrary skyscraper abelian sheaves. Changing the coefficient ring changes the calculation.

Here is an explicit reason for the stalk warning after (1.3). Let $X = \{0\} \cup \{1, 1/2, 1/3, \dots\}$ with its subspace topology, and use the constant ring sheaf k_X for a field. Set $A = \bigoplus_{j \geq 1} k_X$ and $B = k_X$. The sum's universal property gives $\mathcal{H}om(A, B) = \prod_{j \geq 1} k_X$: specifying a map on an open means specifying the image of the unit in each summand. For each j , let f_j be the locally constant function which is one at $1/j$ and zero at every other point. It is locally constant at zero too, since a sufficiently small neighborhood excludes that particular point. Hence $(f_j)_j$ is a section of this product on all of X .

Every individual f_j has zero germ at zero. The induced map of stalk modules $A_0 = \bigoplus_j k \rightarrow B_0 = k$ is therefore zero, since an element of the direct sum uses only finitely many coordinates. But the germ of the section $(f_j)_j$ of the Hom sheaf is nonzero: every neighborhood of zero contains some $1/j$, where its j -th coordinate is one. Thus the canonical comparison $\mathcal{H}om(A, B)_0 \rightarrow \text{Hom}_k(A_0, B_0)$ is not even injective. The issue already appears for ordinary Hom, before any derived construction. Restriction to a fixed open is valid because it preserves the whole sectionwise product; moving to a germ cannot uniformly discard the infinitely many shrinking exceptional points.

6. Exercises with checked solutions

Exercise 1 (easy: the unit). Prove $[\mathcal{O}, M] = M$, and determine the unit $K \rightarrow [\mathcal{O}, K \otimes^L \mathcal{O}]$ under the tensor and Hom unit identifications.

Solution. A map $\mathcal{O}|_V \rightarrow I^n|_V$ is determined by the image of 1, with every section a sent to a times that image. These identifications commute with restriction and the differential, so $H(\mathcal{O}, I) = I$. For the unit, the adjunction transpose of the identity of $K \otimes^L \mathcal{O}$ is the map in question. Evaluation at 1 and the tensor unit both send that identity back to 1_K . Hence the unit becomes the identity of K , in every degree and for every unbounded complex.

Exercise 2 (medium: currying without boundedness). Explain why (3.1) holds without a finiteness condition on any complex. In particular, identify the step where products are needed and distinguish it from an assertion that products of exact sequences are exact.

Solution. Choose a K-flat $P \rightarrow L$ and K-injective $M \rightarrow I$. In Hom degree n , currying identifies the product indexed by i, j of maps $K^i \otimes P^j \rightarrow I^{i+j+n}$ on both sides. On the tensor side the source in each degree is a direct sum, so specifying a map means independently specifying a map on every summand. Rearranging the resulting products gives exactly the Hom product on the iterated side. Formula (1.4)'s differential calculation proves the identification is a complex isomorphism.

K-flatness makes $K \otimes P$ a model for the derived tensor; Lemma 2.3 makes $H(P, I)$ an admissible K-injective target. Consequently both Hom complexes compute the stated derived Hom objects. Quasi-isomorphisms in the first variable are handled on every open by Lemma 2.1, and target comparisons by homotopy equivalences. No step applies a product to an arbitrary exact sequence of sheaves. The product's universal property and the two resolution properties are sufficient.

Exercise 3 (medium: a middle-variable example). Over a field, take $K = M = k$, $L = k^2$, $L' = k$, and $u(x, y) = x$, all in degree zero. Verify (4.3) and explain why it is the appropriate replacement for one-variable naturality.

Solution. Write $a : L' \rightarrow M$ as multiplication by α , and $b : K \rightarrow L$ as $t \mapsto (\beta t, \gamma t)$. Precomposing a by u gives the row $(\alpha, 0)$. Its product with the column (β, γ) is multiplication by $\alpha\beta$. Postcomposing b by u gives multiplication by β , so composing with a gives the same answer. Bilinearity proves equality on the tensor product. All these vector spaces are flat and injective, so the calculation is also the derived one.

The map on the first factor goes from $\text{Hom}(L', M)$ to $\text{Hom}(L, M)$, whereas the map on the second goes from $\text{Hom}(K, L)$ to $\text{Hom}(K, L')$. Their directions oppose each other. Since u is a noninvertible projection, reversing one direction cannot be done by an inverse. Dinaturality compares the two composites on their common mixed source and does exactly what the calculation verifies.

Exercise 4 (hard: the K-injective target test). Let P be K-flat and I K-injective. Prove that $H(P, I)$ is K-injective using an explicit correspondence between maps and homotopies. Explain why termwise flatness alone does not justify this argument.

Solution. A chain map $A \rightarrow H(P, I)$ carries to a chain map $A \otimes P \rightarrow I$ by (1.4). A homotopy is a degree-minus-one element whose Hom differential is the difference of the two maps. Since (1.4) is an isomorphism of complexes, it carries exactly such homotopies in either direction. Thus it gives a bijection on homotopy classes. If A is acyclic, K-flatness makes $A \otimes P$ acyclic, and K-injectivity of I says every resulting chain map is null-homotopic. Uncurrying that null-homotopy proves the defining condition for $H(P, I)$.

The flat-resolution lesson, Section 5, gives an unbounded termwise-free complex over the dual numbers whose tensor with an acyclic complex is not acyclic. Termwise flatness therefore cannot supply the crucial implication for an unbounded P . Bounded-above termwise-flat complexes do supply it, by that lesson's Lemma 2.3; the distinction is between that proved bounded case and the general K-flat condition.

Exercise 5 (hard: a dual map that fails). Let $R = k[\varepsilon]/(\varepsilon^2)$ on a one-point space and $N = R/(\varepsilon)$. Compute $N^\vee$, compare $N \otimes_R^L N^\vee$ with $[N, N]$, and show the canonical degree-zero map is zero.

Solution. The only ideals of R are 0 , (ε) , R . A map from (ε) to R sends ε to an element killed by ε , necessarily $c\varepsilon$, and extends by multiplication by c . Maps from the other two ideals extend immediately. Baer's criterion, proved in the bounded-derived lesson's Lemma 2.1, makes R injective. It is therefore K-injective concentrated in degree zero. Consequently

$$N^\vee = \text{Hom}_R(N, R) = (\varepsilon) \cong N,$$

concentrated in degree zero. A free resolution of N has one copy of R in every nonpositive degree, all negative differentials multiplication by ε , and the quotient augmentation in degree zero. It is exact in negative degrees because the image and kernel of multiplication by ε both equal (ε) .

Tensoring this bounded-above free resolution with $N^\vee \cong N$ makes all differentials zero. Thus the left object has cohomology N in every nonpositive degree and zero in positive degrees. Hom from that resolution to N likewise has zero differential but lies in nonnegative degrees. It computes $[N, N]$: a map $a : P \rightarrow C$ to any acyclic complex has a null-homotopy constructed downwards. In degree zero, a^0 lands in the cycles of C^0 , and projectivity lifts it through $d : C^{-1} \rightarrow Z^0 C$ to h^0 . Having chosen h^{n+1} , the map $a^n - h^{n+1}d_P$ lands in $Z^n C$ by the chain-map equation and the preceding homotopy equation. Lift it to $h^n : P^n \rightarrow C^{n-1}$. This constructs $a = dh + hd$ in every nonpositive degree, with no infinite summation. Applying the same argument to

every shift proves $\mathrm{Hom}^\bullet(P, C)$ acyclic. Cone compatibility then proves invariance under every target quasi-isomorphism, including $N \rightarrow I$. The right object therefore has cohomology N in every nonnegative degree, and zero in negative degrees.

Finally degree-zero evaluation sends $m \otimes \phi$ to the map $n \mapsto \phi(n)m$. Every $\phi(n)$ lies in (ε) , which acts by zero on N . Hence this induced map on degree-zero cohomology is zero. The canonical dual map is far from an isomorphism in this example. Finite duality in the perfect-complex lesson will require a hypothesis this residue module does not satisfy.

Sections with support and the localization triangle

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A section can vanish away from a closed subset while carrying information on that subset. Derived sections with support also detect classes which cannot be represented by a degree-zero section. Localization compares this information with what remains on the open complement. The point of the triangle is to include the boundary map: it measures the obstruction to extending a class across the missing subset.

The prerequisites are [Sheaves of modules on a ringed space](#), Theorems 2.1 and 4.4; [Injective modules and bounded-below derived functors](#), Lemma 1.1, Proposition 1.3 and Theorem 4.1(4), for injective flasqueness and acyclic resolutions; [K-injective resolutions in Grothendieck categories](#), Theorems 4.1 and 5.1; and [Derived pullback and pushforward](#), Proposition 3.2 and Theorem 4.1. The cone construction and exact-sequence triangle are Theorem 3.1 and Proposition 5.2 of the common reading. For cup products we use the tensor lesson's Theorem 2.2 and the internal-Hom lesson's Theorem 3.1.

Sections 1–3 work for any sheaf of unital rings, including noncommutative rings, and unbounded complexes of left modules. Section 5's tensor and cup products assume commutativity. For a closed inclusion $i : Z \hookrightarrow X$, use the unchanged coefficient sheaf $\mathcal{O}_Z = i^{-1}\mathcal{O}_X$; this is distinct from taking a quotient coefficient ring on a closed geometric subspace. Put $j : U = X \setminus Z \hookrightarrow X$.

The construction follows the Stacks project authors' *Cohomology of Sheaves*, “Cohomology with support in a closed subset, II”, Tags 0G6Z–0G79, and *Sheaves of Modules*, in the [AI Integrated Stacks Project edition](#). We develop the support adjunction, localization triangle and canonical product and pullback maps. The Euclidean example includes the singular-cochain comparison it uses. Source attribution and the licence for adapted passages appear in the [course notice](#).

1. Closed supports and their adjunction

The support of a section $s \in F(V)$ is the set of points where its germ is nonzero. It is closed in V : a zero germ means the section vanishes on a neighborhood. Define the ambient support subsheaf by

$$\mathcal{S}_Z^0 F = \ker(F \rightarrow j_* F|_U). \quad (1.1)$$

Its sections are exactly those supported in $Z \cap V$. It is a module subsheaf, since scalar multiplication cannot create a nonzero germ where the section vanished. Kernels show that this additive functor is left exact. Its stalks vanish off Z .

A sheaf with zero stalks off a closed set is canonically the direct image of its restriction to that set. Indeed $i_* G$ has stalk G_z at $z \in Z$ and zero off Z : neighborhoods intersected with Z are cofinal neighborhoods in Z , and a point outside Z has a neighborhood disjoint from it. Thus the unit $F \rightarrow i_* i^{-1} F$ is an isomorphism for such F . The same stalk description proves that i_* is exact, $i^{-1} i_*$ is the identity, and i_* is fully faithful. Set

$$\begin{aligned} \mathcal{H}_Z F &= i^{-1} \mathcal{S}_Z^0 F, \\ i_* \mathcal{H}_Z F &= \mathcal{S}_Z^0 F. \end{aligned} \quad (1.2)$$

Proposition 1.1. The functor $\mathcal{H}_Z$ is right adjoint to the exact functor i_* . It preserves injective objects and K-injective complexes. There is a derived adjunction $i_* \dashv R\mathcal{H}_Z$, and $R\mathcal{H}_Z(i_*N) = N$.

Proof. A map $i_*G \rightarrow F$ has image with zero stalks off Z , so it factors uniquely through (1.1). Full faithfulness of i_* then identifies it with a unique map $G \rightarrow \mathcal{H}_ZF$. Conversely such a map pushes forward and includes into F ; these operations are inverse and natural. Since i_* is exact, Hom into $\mathcal{H}_ZI$ is exact if Hom into I is exact, proving injectivity preservation. If A is an acyclic complex on Z , chain adjunction identifies maps and homotopies $A \rightarrow \mathcal{H}_ZI$ with those $i_*A \rightarrow I$. The latter source is acyclic, so these maps are null-homotopic for K-injective I .

Choose K-injective models in this same adjunction. The map test into K-injectives gives the derived adjunction. Also i_* preserves K-injectives, because its left adjoint i^{-1} is exact; the identical acyclic-source argument proves this. Resolving N by J therefore computes $R\mathcal{H}_Z(i_*N)$ by $\mathcal{H}_Z(i_*J) = J$, proving the final assertion. All arguments use left module maps only. $\square$

Write $D_Z(\mathcal{O}_X)$ for objects whose cohomology sheaves vanish off Z . Exact restriction shows this condition is equivalent to $K|_U = 0$; cone long exact sequences show it defines a triangulated subcategory. The stalk argument for the unit applies to every cohomology sheaf, so $i_* : D(\mathcal{O}_Z) \rightarrow D_Z(\mathcal{O}_X)$ is an equivalence with inverse i^{-1} . Proposition 1.1 then says

$$\mathcal{S}_Z K := i_* R\mathcal{H}_Z K \quad (1.3)$$

is right adjoint to the inclusion of D_Z . Its counit $\mathcal{S}_Z K \rightarrow K$ is universal among maps from objects supported on Z . In particular it is an isomorphism if $K \in D_Z$.

Define $\Gamma_Z(X, F) = \Gamma(X, \mathcal{S}_Z^0 F)$ and derive it with K-injectives. Because $\mathcal{H}_ZI$ is K-injective, (1.2) gives the full local-to-global identity

$$R\Gamma_Z(X, K) = R\Gamma(Z, R\mathcal{H}_Z K). \quad (1.4)$$

On the right we restrict the scalar action from $\Gamma(Z, \mathcal{O}_Z)$ through the map from $\Gamma(X, \mathcal{O}_X)$. This also proves that taking derived sections of (1.3) gives the left side. These identities hold for all unbounded complexes.

2. Localization and its connecting sign

Theorem 2.1. There are triangles natural in K :

$$\begin{aligned} \mathcal{S}_Z K &\longrightarrow K \longrightarrow Rj_*(K|_U) \\ &\longrightarrow \mathcal{S}_Z K[1], \end{aligned} \quad (2.1)$$

and

$$\begin{aligned} R\Gamma_Z(X, K) &\longrightarrow R\Gamma(X, K) \\ &\longrightarrow R\Gamma(U, K|_U) \\ &\longrightarrow R\Gamma_Z(X, K)[1]. \end{aligned} \quad (2.2)$$

Proof. Choose $K \rightarrow I$ K-injective with injective terms. Every term is flasque, and its restriction to U is K-injective. On every open V , flasqueness makes $I^n(V) \rightarrow I^n(V \cap U)$ surjective. The kernel is precisely supported sections. Thus there are termwise short exact sequences

$$\begin{aligned} 0 &\rightarrow i_* \mathcal{H}_Z I \rightarrow I \\ &\rightarrow j_*(I|_U) \rightarrow 0, \\ 0 &\rightarrow \Gamma_Z(X, I) \rightarrow \Gamma(X, I) \\ &\rightarrow \Gamma(U, I) \rightarrow 0. \end{aligned} \quad (2.3)$$

Each term is the indicated derived-functor model by Proposition 1.1 and open restriction. The common reading's exact-sequence triangle proves (2.1) and (2.2). A derived map is a unique homotopy class between the K-injective models. Applying the functors gives a map of exact sequences; the cone and quotient construction is natural for that map. Homotopic maps give homotopic maps on each term and identical maps of derived triangles. Resolution comparisons are homotopy equivalences, proving natural independence of the models. $\square$

The sign calculation is as follows. For $0 \rightarrow A \xrightarrow{a} B \xrightarrow{b} C \rightarrow 0$, the cone has differential $d(y, x) = (dy + a(x), -dx)$, projection $p(y, x) = x$, and quasi-isomorphism $q(y, x) = b(y)$ to C . A cycle $z \in C^n$ lifts to $y \in B^n$; write $dy = a(x)$. Injectivity of a gives $dx = 0$. The cone cycle $(y, -x)$ maps to z , so

$$H^n(p)H^n(q)^{-1}[z] = -[x]. \quad (2.4)$$

Our triangle's last map is $-pq^{-1}$, hence induces the positive lift boundary $[z] \mapsto [x]$. A convention using the positive cone projection has the opposite boundary. Negating two arrows gives an isomorphic triangle: multiply one of its three objects by minus one. Thus one must track the third-arrow convention when transporting a formula from another source, rather than change one boundary informally.

Proposition 2.2. If an open V is disjoint from Z , then $R\mathcal{H}_Z(Rv_*N) = 0$ for $v : V \hookrightarrow X$. Module and underlying-abelian supported cohomology agree canonically, both globally and as support sheaves.

Proof. A K-injective J on V has K-injective v_*J , since restriction is exact. Its sections over W and $W \setminus Z$ are both $J(W \cap V)$, so its support kernel is zero. This proves the first assertion. For coefficient comparison, resolve a module complex by I , and its underlying abelian complex further by an abelian K-injective J . Their support kernels and restrictions give a map of the exact-sequence triangles (2.3), with injective terms used for both resolutions. The middle and open-complement maps on derived global sections are isomorphisms by Theorem 4.1 of the preceding lesson. The long exact sequences, or the Five Lemma on consecutive five terms, make the support map an isomorphism. Apply this argument on every open and sheafify to obtain the local support comparison; restriction of both coefficient models to opens is K-injective. Finally restrict the ambient comparison to Z to get $R\mathcal{H}_Z$. This supplies the source's omitted local verification. $\square$

3. Locally closed pairs

The construction also applies to a locally closed W . Choose an open V in which W is closed, and define the ambient support functor by direct image from V of its closed-support subsheaf. It is independent of this choice. For another such open $V' \subset V$, a supported section on $T \cap V'$ glues with zero on $T \cap (V \setminus W)$: their overlap is outside its support, and these two opens cover $T \cap V$. This is the inverse to restriction of supported sections. Compare arbitrary choices through their intersection. These inverse maps commute with restrictions, differentials and scalars. For an open W , this functor is $w_*F|_W$, not a subsheaf of sections of F extended by zero.

Proposition 3.1. If W is locally closed and W' is closed in W , there is a natural triangle

$$\begin{aligned} R\Gamma_{W'}^{\text{sh}}K &\longrightarrow R\Gamma_W^{\text{sh}}K \\ \longrightarrow R\Gamma_{W \setminus W'}^{\text{sh}}K &\longrightarrow R\Gamma_{W'}^{\text{sh}}K[1]. \end{aligned} \quad (3.1)$$

Here the superscript distinguishes these ambient sheaf functors from global sections.

Proof. Choose V with W closed in V ; then W' is also closed in V , and $W \setminus W'$ is closed in $V \setminus W'$. For a flasque F , a supported section on $(T \cap V) \setminus W'$ extends to $T \cap V$. Outside W the extension is still zero, since that open subset lies inside its original domain. The extension therefore has support in W . Its kernel under restriction consists exactly of sections supported in W' . This proves a short exact sequence of the three support sheaves on every open T , even though direct image from V need not generally be exact. Apply it termwise to a K-injective resolution with injective, hence flasque, terms. The cone argument of Theorem 2.1 proves the triangle and its functoriality, with the same positive lift boundary. $\square$

4. Integer coefficients near a Euclidean point

We prove the comparison needed for this example, rather than assume homotopy invariance for arbitrary sheaf cohomology. Let V be any open subset of $\mathbb{R}^n$ and A any abelian group. Let $S_q(V)$ be the free abelian group on continuous singular simplices $\Delta^q \rightarrow V$, with boundary the alternating sum of faces. Its cochains are $C^q(V; A) = \text{Hom}_{\mathbb{Z}}(S_q(V), A)$, with differential $c \mapsto c\partial$. Let $\mathcal{C}^q$ be the sheafification of this cochain presheaf.

Lemma 4.1 (cochain comparison). The augmented complex $A_V \rightarrow \mathcal{C}^\bullet$ is a flasque resolution, and

$$H^q(V, A_V) = H^q(\mathcal{C}^\bullet(V; A)). \quad (4.1)$$

The comparison is natural under restrictions to smaller opens.

Proof. We supply the three required steps. First, a cover of an open Euclidean set has a locally finite open refinement (V_a) whose closures lie in members of the original cover. Here is an explicit compactness argument. Exhaust the open set by compact sets $K_m = \{x : |x| \leq m, \text{dist}(x, \mathbb{R}^n \setminus V) \geq 1/m\}$; omit the distance condition when the complement is empty. They satisfy $K_m \subset \text{int } K_{m+1}$ and exhaust V . Cover each compact layer $K_m \setminus \text{int } K_{m-1}$ by finitely many balls with closures in cover members and in $\text{int } K_{m+1} \setminus K_{m-2}$. Set the negative-index compact sets empty. The balls cover V ; every point has a neighborhood in some K_m , which meets only finitely many subsequent layers, so they are locally finite.

A section of $\mathcal{C}^q$ is represented by cochains c_a on cover members. Using this refinement, assign to each singular simplex contained in some V_a the value given by one such c_a ; assign zero if it is contained in none. This is a global cochain representing the section. To verify the assertion, fix a point x . Only finitely many refinement members meet a sufficiently small neighborhood. Remove those whose closures do not contain x . For the remaining finite list, their representing cochains are defined near x , and their germs agree. Shrink the neighborhood so all these cochains agree on every simplex in it, and so it is contained in one refinement member containing x . The global assignment then agrees with that representative there, irrespective of the choices for simplices. This proves surjectivity of $C^q(V; A) \rightarrow \Gamma(V, \mathcal{C}^q)$. The same proof works on every smaller open. Extend a representing cochain by zero on simplices not contained in that smaller open; this proves flasqueness of $\mathcal{C}^q$.

Second, on a convex ball straight-line contraction proves that singular cohomology is A in degree zero and zero in positive degrees. The chain verification is the prism identity: a homotopy H from f to g assigns to a q -simplex the alternating sum, with signs $(-1)^i$, of the $q+1$ simplices with vertices $(v_0, 0), \dots, (v_i, 0), (v_i, 1), \dots, (v_q, 1)$ in $\Delta^q \times [0, 1]$, followed by H . Interior faces cancel in pairs, the two horizontal faces give $g - f$, and the other faces give the negative prism of the boundary. Thus $\partial P + P\partial = g_* - f_*$. A constant map factors through the point's chain complex; its augmented homology contracts (insert the unique vertex), and dually its cohomology is as stated. Taking the exact filtered colimit over convex-ball neighborhoods proves that $A_V \rightarrow \mathcal{C}^\bullet$ is exact on stalks.

Third, the kernel of global cochains mapping to sheaf sections is the complex $N^\bullet$ of cochains locally zero: each point has a neighborhood on all of whose singular simplices the cochain vanishes. We prove that this kernel is acyclic. For an open cover $\mathcal{U}$, let $S_*^{\mathcal{U}}$ be the subcomplex generated by simplices contained in one cover member, and let $N_{\mathcal{U}}^\bullet$ be its annihilator. The small-chain inclusion has a retraction r and homotopy h satisfying

$$\begin{aligned} 1 - jr &= \partial h + h\partial, \\ rj &= 1, \quad hj = 0. \end{aligned} \quad (4.2)$$

Here is the full construction. Barycentric subdivision s replaces a simplex by its oriented barycentric triangulation. Internal faces cancel, and boundary faces are subdivided faces, so it is a chain map. There is a homotopy T with $s - 1 = \partial T + T\partial$, supported inside each simplex image. To construct it without another theorem, work first in the standard simplex. In dimension zero take $T = 0$. In dimension q , subtract the already constructed homotopies on its faces from $s\iota_q - \iota_q$. The result is a cycle by the lower-dimensional identity. Cone that cycle to the barycenter inside the convex simplex, then push it forward by the singular

simplex. The cone boundary formula $\partial(b * c) = c - b * \partial c$ proves the identity. Write T_m for the sum of iterated homotopies, so $s^m - 1 = \partial T_m + T_m \partial$; it has the same support control.

Induct on simplex dimension to construct (4.2). Once r, h are constructed on faces of σ , put $z = \sigma - h\partial\sigma$. The induction identity gives $\partial z = jr\partial\sigma$. Choose m so that every simplex in $s^m z$ is small, and define

$$\begin{aligned} jr\sigma &= s^m z - T_m jr\partial\sigma, \\ h\sigma &= -T_m z. \end{aligned} \tag{4.3}$$

The correction is small: $r\partial\sigma$ is small, and T_m keeps each simplex inside its image. The boundary of the first line is $jr\partial\sigma$; subtracting it from z gives $-\partial T_m z$, proving the homotopy identity. For a small σ , its faces have $h = 0$; choose $m = 0$, obtaining $r\sigma = \sigma$ and $h\sigma = 0$. The required m always exists: barycentric subdivision reduces mesh in dimension $q > 0$ by at most $q/(q+1)$. Indeed two vertices of a subdivided simplex are barycenters of nested faces, and their distance is bounded by that fraction of the original diameter. Uniform continuity on the compact standard simplex and a finite subcover of its inverse-image cover then make repeated subdivisions small. The finitely many simplices in z can use a common m .

Dualizing (4.2) contracts $N_{\mathcal{U}}^\bullet$: $cjr = 0$, and ch still vanishes on small chains because $hj = 0$. Every locally zero cochain belongs to some $N_{\mathcal{U}}$, and common refinement orders these complexes as a filtered union. Exact filtered colimits make $N^\bullet$ acyclic. Our first step identifies the section complex of $\mathcal{C}^\bullet$ with $C^\bullet/N^\bullet$; the quotient is therefore quasi-isomorphic to singular cochains. Finally flasque acyclicity and the bounded-below acyclic-resolution theorem from the bounded-derived lesson compute sheaf cohomology by this section complex. All maps used in the quotient comparison are restriction maps, giving its naturality. $\square$

Proposition 4.2. For $n \geq 0$,

$$R\Gamma_{\{0\}}(\mathbb{R}^n, \mathbb{Z}) \cong \mathbb{Z}[-n]. \tag{4.4}$$

Proof. The comparison proves $H^q(\mathbb{R}^n, \mathbb{Z}) = \mathbb{Z}$ for $q = 0$ and zero otherwise, by straight-line contraction. For $n > 0$, the punctured space deformation retracts onto S^{n-1} by radial rescaling; the prism identity proves invariance of singular cohomology under this homotopy.

We recall and prove the sphere calculation needed here. For a cover by two opens, its small-chain complex is the sum of the two singular chain subcomplexes; their intersection is the chain complex of their intersection. Cochains therefore have the short exact sequence with middle term $C^\bullet(U; \mathbb{Z}) \oplus C^\bullet(V; \mathbb{Z})$, last term $C^\bullet(U \cap V; \mathbb{Z})$, and map the difference of restrictions. Surjectivity follows by extending any cochain on the intersection by zero to either open. The small-chain equivalence just proved gives the usual long exact Mayer-Vietoris sequence, with no sheaf comparison required on the sphere itself. Cover S^m by complements of its two poles. Each is homeomorphic to $\mathbb{R}^m$, and their intersection retracts to S^{m-1} . Starting with S^0 , whose two points have $H^0 = \mathbb{Z}^2$, the sequence proves inductively that the reduced cohomology of S^m is $\mathbb{Z}$ in degree m and zero elsewhere.

For $n = 1$, the restriction from the line to its two complementary rays is the diagonal $\mathbb{Z} \rightarrow \mathbb{Z}^2$. Its cokernel is $\mathbb{Z}$. For $n > 1$ the degree-zero restriction is an isomorphism, and the sole reduced punctured-space class has degree $n - 1$. The long exact sequence of (2.2) puts the sole supported class in degree n . For $n = 0$, the support is the entire one-point space and the result is immediate. A complex with just this cohomology is isomorphic to that cohomology module in its degree: the good-truncation inclusion and quotient give the requisite quasi-isomorphisms, as in the common reading's Proposition 5.2. This proves (4.4). $\square$

The same computation on balls centered at zero gives $\mathcal{S}_{\{0\}}\mathbb{Z} = \mathbb{Z}_{\{0\}}[-n]$. The transition isomorphisms follow from the natural comparison and radial rescaling of punctured balls. This is supported cohomology, rather than ordinary sections of the constant sheaf supported at the point; those ordinary sections are zero when $n > 0$.

5. Cup products and support pullback

Assume commutative coefficients in this section. Tensoring an object supported on Z with any derived object keeps that support: a K-flat model makes its restriction to the complement acyclic. The universal counit in (1.3) therefore gives a unique factorization of $K \otimes^{\mathbf{L}} \mathcal{S}_Z M \rightarrow K \otimes^{\mathbf{L}} M$ through $\mathcal{S}_Z(K \otimes^{\mathbf{L}} M)$. On closed-set coefficients it is the canonical map

$$\begin{aligned} i^{-1}K \otimes^{\mathbf{L}} R\mathcal{H}_Z M \\ \longrightarrow R\mathcal{H}_Z(K \otimes^{\mathbf{L}} M). \end{aligned} \quad (5.1)$$

Indeed $K \otimes^{\mathbf{L}} i_* N = i_*(i^{-1}K \otimes^{\mathbf{L}} N)$: take a K-flat model for K , and check the ordinary tensor comparison on stalks, identical over $\mathcal{O}_z$ on Z and zero off it. This proves the identity without a projection-formula assumption. The universal factorization proves naturality and coherence under successive tensor products, because the composites have the same image under the counit.

The internal-Hom lesson identifies cohomology classes with maps from $\mathcal{O}$ to shifted objects. Tensoring such maps defines cup products. Apply the factorization above to get

$$\begin{aligned} H^p(X, K) \times H_Z^q(X, M) \\ \longrightarrow H_Z^{p+q}(X, K \otimes^{\mathbf{L}} M). \end{aligned} \quad (5.2)$$

Forgetting supports commutes with cup product, since composing the factorization with its counit is precisely the ordinary tensor map. This proves the omitted compatibility in Tag 0G77. On homogeneous cycles the product is their tensor, whose differential is $da \otimes b + (-1)^p a \otimes db$; changing either cycle by a boundary changes it by a boundary. The coherent shift identification gives exactly these signs.

For a morphism $f : X' \rightarrow X$, set $Z' = f^{-1}Z$. Derived pullback carries supported objects to objects supported on Z' : off that set, a K-flat acyclic stalk remains acyclic after arbitrary scalar extension, by the tensor lesson's Lemma 1.1. Pulling back the support counit therefore factors uniquely through

$$Lf^* \mathcal{S}_Z K \longrightarrow \mathcal{S}_{Z'} Lf^* K. \quad (5.3)$$

For $g : Z' \rightarrow Z$ it equivalently gives $Lg^* R\mathcal{H}_Z K \rightarrow R\mathcal{H}_{Z'} Lf^* K$. To justify the equivalence directly, a K-flat model P on Z has K-flat $i_* P$ on X , by the stalk test. The canonical ordinary closed-set comparison $f^* i_* P \rightarrow i'_* g^* P$ is an isomorphism on each stalk, proving the derived identity. Universal uniqueness proves that (5.3) commutes with forgetting supports and composes for two pullbacks. On cohomology, pullback uses the unit of $Lf^* \dashv Rf_*$ and the global Leray identity from lesson six; its naturality and the counit factorization prove the supported/ordinary pullback square commutes. This is a canonical map, without a general assertion that it is invertible.

6. A two-point calculation and exercises

Let $X = \{c, \eta\}$ have opens $\emptyset, \{\eta\}, X$; its closed point is c . With a constant ring sheaf, a module sheaf is exactly an arrow $r : M \rightarrow N$. Its global sections are M , its sections on the open point are N , and exactness is exactness at both entries. Thus global sections on this space and on its open point are exact functors. Also $j_* N = (N \xrightarrow{1} N)$ is exact. The localization triangle gives

$$\begin{aligned} H_c^0(X, F) &= \ker r, \\ H_c^1(X, F) &= \operatorname{coker} r, \\ H_c^q(X, F) &= 0 \quad (q \neq 0, 1). \end{aligned} \quad (6.1)$$

The corresponding ambient cohomology sheaves are these modules at c , zero at η . Ordinary supported sections only see the kernel, while the degree-one group measures the failed extension of a section from η . The first exercise shows that failure before any resolutions are chosen.

Exercise 1 (easy: left exactness and its limit). Prove left exactness of closed supported sections directly from their definition. On the two-point space give an exact sequence for which they fail to be right exact.

Solution. In an exact sequence $0 \rightarrow F \rightarrow G \rightarrow H$, a supported section of G mapping to zero in H lifts uniquely to F , by left exactness of ordinary sections. Its restriction off the closed set vanishes because its image in G does, and $F \rightarrow G$ is injective. Thus it is supported. This proves exactness at the first two terms, globally and on every open. On the two-point space the exact sequence of arrows $0 \rightarrow (0 \rightarrow k) \rightarrow (k \xrightarrow{1} k) \rightarrow (k \rightarrow 0) \rightarrow 0$ is exact at each entry. Its closed supported sections are respectively $0, 0, k$, so the last map is not surjective. All the maps used are module maps over the same ring, and the counterexample works with noncommutative coefficients too.

Exercise 2 (medium: the localization fibre). For an arrow $r : M \rightarrow N$, give a complex computing the global supported object, including the connecting map $N \rightarrow H_c^1(X, F)$.

Solution. Since the other two global terms are $M[0]$ and $N[0]$, the supported object is the shifted cone $C(r)[-1]$. It has M in degree zero, N in degree one, and differential $-r$. Its map to $M[0]$ is the identity in degree zero. Rotating the cone triangle with last map $-p$ gives the next map r and the final map $N[0] \rightarrow C(r)$ as the positive cone inclusion. Thus its boundary on cohomology is the positive quotient $N \rightarrow N/r(M)$. The degree-minus-one shift is why the fibre differential is $-r$. Replacing it by $+r$ by negating its N term would also negate the coordinate description of the boundary; one cannot make that replacement while leaving the boundary description unchanged. This proves (6.1) and checks the sign convention in a fully explicit sheaf model.

Exercise 3 (medium: the Euclidean group and its sign). Compute all $H_{\{0\}}^q(\mathbb{R}^n, \mathbb{Z})$. For $n = 1$, label the complementary rays in negative, positive order and determine the boundary of their locally constant section (a, b) .

Solution. Proposition 4.2 gives $\mathbb{Z}$ in degree n and zero in all other degrees, including the whole-support case $n = 0$. In dimension one the degree-zero restriction is diagonal, so the boundary factors through $\mathbb{Z}^2/\mathbb{Z}(1, 1)$. Its positive coordinate is $b - a$. To check the sign with actual cochains, extend the function on the two rays arbitrarily to a zero-cochain on the line. Its differential on an oriented interval crossing zero is its value at the positive endpoint minus its value at the negative endpoint, namely $b - a$. This differential vanishes on chains contained in either ray and represents the relative degree-one class.

The flasque resolution in Lemma 4.1 also computes supported cohomology: each of its terms is acyclic for sections on the line and on the complement, and its restriction is onto, so Theorem 2.1 makes it acyclic for supported sections. The bounded-below acyclic-term theorem therefore applies. Naturality of the singular comparison identifies its cone with the relative cochain cone. Equation (2.4) says the triangle's negative projection gives precisely this positive lift differential. Thus the increasing orientation of the real line gives $\delta(a, b) = b - a$, rather than its negative. In higher dimensions the boundary takes a chosen linking-sphere class to the corresponding supported class with this same positive lift convention; changing the orientation negates both chosen generators.

Exercise 4 (hard: cup products with support). Prove that forgetting supports in (5.2) gives the ordinary cup product. Prove also that first multiplying two ordinary classes and then a supported class agrees, after association, with the successive supported cup products.

Solution. Let $S = \mathcal{S}_Z M$. The source $K \otimes^L S$ lies in D_Z , so the right adjunction identifies maps from it to $\mathcal{S}_Z(K \otimes^L M)$ with maps from it to $K \otimes^L M$. The defining transpose is 1_K tensor the support counit. Composing the supported map with the new counit therefore gives exactly that tensor map. A class is a map from the tensor unit to a shifted object; tensoring the two class maps and composing these equal morphisms proves compatibility with forgetting supports on cohomology, in every pair of degrees.

For objects K, L, M , both successive support products are maps from $K \otimes^L L \otimes^L \mathcal{S}_Z M$, an object of D_Z , to $\mathcal{S}_Z(K \otimes^L L \otimes^L M)$. Under the adjunction both are $1_K \otimes 1_L$ tensor the counit. Its uniqueness proves equality; the tensor associator identifies their sources and targets. Applying this to shifted class maps proves the assertion for classes. No exactness of global sections, flatness of the classes' sheaves, or finiteness of their degrees is used. K-flat models and the proved tensor coherence justify the entire construction.

Exercise 5 (hard: pullback is a map, not always an isomorphism). Explain why (5.3) composes for two ringed-space morphisms and give a case where it is not an isomorphism.

Solution. Pulling back the original counit through two maps gives the same morphism as pulling it back through their composite, by lesson six's coherent pullback composition. Every source involved is supported on the inverse-image closed subset. The universal factorization into its support right adjoint is unique, so the two resulting maps agree after the canonical composition identification.

For failure, take $X = \mathbb{R}$, $Z = \{0\}$, constant coefficients $\mathbb{Z}$, and f the inclusion of that point with the unchanged stalk coefficient ring. Pullback is the exact stalk functor. Proposition 4.2 and its ball calculation give $Lf^*S_Z\mathbb{Z} = \mathbb{Z}[-1]$. The inverse-image support is the entire point, so $S_Z Lf^*\mathbb{Z} = \mathbb{Z}[0]$. Their cohomology groups occur in different degrees, and the canonical map cannot be an isomorphism. All coefficient maps are identity maps; this failure concerns the support operation, not nonflat change of coefficient rings.

Perfect complexes and duals on a ringed space

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A perfect complex is locally a finite calculation with finite projective coefficients. This finiteness has two complementary descriptions: arbitrarily accurate finite approximations, together with a bound on the degrees that tensor products can create; and a dual that permits evaluation and coevaluation. We prove the equivalence on arbitrary ringed spaces. Local rings simplify the models, but are not a hypothesis of the main theorem.

The prerequisites are [Flat modules and K-flat resolutions](#), Lemmas 1.2–1.3, 2.2–2.4, Theorem 3.1 and Lemma 4.1; [The derived tensor product and Tor sheaves](#), Theorem 2.2 and Theorem 3.3; [Hom complexes, internal derived Hom and Ext sheaves](#), Lemma 1.1, Theorems 2.2 and 3.1 and the canonical maps in Section 4; and [Derived pullback and pushforward](#), Theorem 1.1 and Proposition 3.2. [Injective modules and bounded-below derived functors](#), Theorem 3.3, supplies maps into K-injectives. The common reading supplies cones, their long exact cohomology sequences, and localization by roofs. The support lesson's Theorem 2.1 is used once for extension across an open complement.

Throughout, $\mathcal{O} = \mathcal{O}_X$ is commutative and unital, and all derived categories are unbounded. Write $[K, L] = R\mathcal{H}om(K, L)$. A “finite projective sheaf” here means a direct summand of $\mathcal{O}^r$ for finite r ; a “locally finite projective sheaf” has that property on an open cover. Bounds stated locally need not be uniform on the space.

The construction follows the Stacks project authors’ *Cohomology of Sheaves*, “Strictly perfect complexes”, “Pseudo-coherent modules”, “Tor dimension”, “Perfect complexes” and “Duals”, in the [AI Integrated Stacks Project edition](#). We prove the finite-model criterion and duality, including the finite-cell argument needed for the converse. Source attribution and the licence for adapted passages appear in the [course notice](#).

1. Finite models and local maps

A complex is **strictly perfect** if it is bounded and each term is finite projective. An object is **perfect** if, on an open cover, it is isomorphic in the derived category to a strictly perfect complex. Cones, shifts and tensor products of strictly perfect complexes are strictly perfect: in each degree only finitely many summands occur; a tensor product of summands of $\mathcal{O}^r$ and $\mathcal{O}^s$ is a summand of $\mathcal{O}^{rs}$. Pullback preserves these splittings.

Lemma 1.1 (finite lifting and local representatives). A map from a finite projective sheaf to the target of a sheaf epimorphism lifts near every point. If P is strictly perfect with lowest term in degree a , and $H^j(C) = 0$ for $j \geq a$, every chain map $P \rightarrow C$ is locally null-homotopic. Every derived map $P \rightarrow F$ is locally represented by a chain map, and a chain map that is zero in the derived category is locally null-homotopic.

Proof. First lift the finitely many images of the basis of $\mathcal{O}^r$ on a common neighbourhood, then restrict the resulting lift to its summand. For the homotopy, descend from the highest nonzero degree b of P . The degree- b component lands in cycles. Since $H^b(C) = 0$, the epimorphism $C^{b-1} \rightarrow Z^b(C)$ permits a local lift h^b . At degree i , the residual map $f^i - h^{i+1}d_P$ lands in cycles because the chain-map equation and the already

constructed homotopy in degree $i + 1$ give zero after applying d_C . Lift it to C^{i-1} . There are finitely many degrees and shrinkings, and the result is $f = dh + hd$.

Take a K-injective resolution $F \rightarrow I$. Maps into I represent derived maps by the bounded-derived lemma's Theorem 3.3. If C is acyclic, then $H(P, C)$ is an acyclic sheaf complex: Hom from a finite projective term is a summand of a finite power of the exact identity functor, and the finite filtration by terms of P proves the assertion. Consequently $H(P, F) \rightarrow H(P, I)$ is a quasi-isomorphism. A degree-zero cycle on the right lifts locally to a cycle on the left modulo a boundary, since its cohomology-sheaf map is an isomorphism. This gives an actual representative. A cycle on the left whose image is a boundary is locally a boundary on the left, giving the last assertion. The same reasoning shows that a derived map $P \rightarrow C$, under the stated cohomology vanishing, is locally zero: first represent it, then apply the downward homotopy. $\square$

It follows that perfectness is independent of the chosen representative: locally realize a strictly perfect model's derived isomorphism by a chain map to that representative; it is a quasi-isomorphism because its cohomology maps are isomorphisms.

For later use, an acyclic top pair in a strictly perfect complex can be cancelled locally. If the top term is P^b and $H^b(P) = 0$, the epimorphism $P^{b-1} \rightarrow P^b$ splits by Lemma 1.1. Its kernel is finite projective, and the complex decomposes as that shortened complex plus the contractible identity disk on P^b . Repeat finitely often.

Lemma 1.2 (finite presentation and flatness). A flat, finitely presented sheaf is locally finite projective. If all stalk rings are local, locally finite projective sheaves are finite locally free.

Proof. Locally write a presentation $\mathcal{O}^m \xrightarrow{A} \mathcal{O}^n \rightarrow F \rightarrow 0$, and put $L = \text{im } A$. Flatness of F makes $L \subset \mathcal{O}^n$ pure: tensoring this inclusion with any sheaf is injective, by the flat-quotient lemma. Let l_α be the columns of $A = (a_{i\alpha})$. The finite equations

$$\sum_i a_{i\alpha} y_i = l_\alpha \quad (1 \leq \alpha \leq m) \quad (1.1)$$

have the ambient solutions $y_i = e_i$. Set $T = \text{coker}(A^t : \mathcal{O}^n \rightarrow \mathcal{O}^m)$. The tuple (l_α) maps to zero in $T \otimes \mathcal{O}^n$, hence in $T \otimes L$ by purity. Right exactness of tensor says it locally lies in the image of $A^t \otimes L$. Thus (1.1) has solutions in L on a neighbourhood. Sending e_i to y_i gives a retraction $\mathcal{O}^n \rightarrow L$, because it fixes the generators l_α . The quotient F is a finite free summand.

For completeness, a finite projective module over a local ring $(R, \mathfrak{m})$ is free. Nakayama's lemma follows directly: if finite generators satisfy $v = Bv$ with entries of B in $\mathfrak{m}$, then $\det(1 - B)$ is a unit, and the adjugate identity gives $v = 0$. Lift a basis modulo $\mathfrak{m}$ to a finite projective module P . Nakayama applied to the cokernel gives a surjection $R^r \rightarrow P$. It splits; its finite kernel has zero reduction modulo $\mathfrak{m}$, so another application of Nakayama kills that kernel.

A free stalk of a finitely presented sheaf spreads to a free sheaf near that point. Lift its finite basis, and kill the stalk-zero cokernel using finitely many generators. The resulting surjection has finitely generated kernel: from a finite presentation, lift maps between the two finite free presentations; the original finite relations and the columns of the difference between their composite and the identity generate the kernel. Its zero stalk therefore kills finitely many kernel generators on a common neighbourhood. The lifted basis is an isomorphism there. Apply this to the finite projective sheaf. $\square$

For arbitrary stalk rings, “projective” cannot be replaced by “free”. On the one-point space with ring $k \times k$, the summand $(1, 0)R$ is not R^r for any r : its two component dimensions are $(1, 0)$, whereas those of R^r are (r, r) .

Proposition 1.3 (Hom from a finite model). For strictly perfect P and any complex F , $H(P, F)$ represents $[P, F]$. If F is K-flat, this Hom complex is K-flat.

Proof. The finite-filtration argument in Lemma 1.1 makes $H(P, -)$ preserve quasi-isomorphisms, so compare $F \rightarrow I$ with the definition $[P, F] = H(P, I)$. Define

$$\begin{aligned}(P^\vee)^n &= \mathcal{H}om(P^{-n}, \mathcal{O}), \\ d_{P^\vee}^n &= (-1)^{n+1} (d_P^{-n-1})^t.\end{aligned}\tag{1.2}$$

Evaluation identifies $F \otimes P^\vee$ with $H(P, F)$: on homogeneous elements it sends $f \otimes \phi$ to $p \mapsto f\phi(p)$. The identity is clear for finite free terms and passes to summands. The Hom differential and the sign in (1.2) agree with the tensor differential. The dual is bounded with flat terms, hence K-flat; tensoring two K-flats remains K-flat. $\square$

2. Finite approximations

An object K is **m -pseudo-coherent** if locally there is a map $P \rightarrow K$ from a strictly perfect complex, inducing isomorphisms on H^j for $j > m$ and a surjection on H^m . Equivalently, its cone has zero cohomology in degrees $j \geq m$. It is **pseudo-coherent** if this holds for every integer m . The cover may depend on m . Lemma 1.1 makes this definition equivalent to using actual maps to any chosen complex representative.

Smaller thresholds impose stronger conditions. An m -approximation is also an $(m+1)$ -approximation, and shifting by $[r]$ changes the threshold to $m-r$. Such an object is locally bounded above, because its approximation has finitely many nonzero terms. On a quasi-compact space finitely many neighbourhoods give a uniform upper bound.

Derived pullback preserves m -pseudo-coherence. Indeed, a complex with cohomology only in degrees $\leq c$ has a bounded-above flat model with terms only in those degrees, by good truncation and Lemma 4.1 of the flat lesson. Its derived pullback has the same upper bound. Pull back the approximation triangle; the strict model remains strict and the cone remains in degrees $< m$. Similarly, if $K \in D^{\leq a}$ and $L \in D^{\leq b}$, these flat models show $K \otimes^L L \in D^{\leq a+b}$.

Proposition 2.1 (triangles and tensor bounds). In a triangle $K \rightarrow L \rightarrow M \rightarrow K[1]$, the following implications hold:

1. If K is $(m+1)$ -pseudo-coherent and L is m -pseudo-coherent, then M is m -pseudo-coherent.
2. If K, M are m -pseudo-coherent, then L is m -pseudo-coherent.
3. If L is $(m+1)$ -pseudo-coherent and M is m -pseudo-coherent, then K is $(m+1)$ -pseudo-coherent.

If $K \in D^{\leq a}$ is n -pseudo-coherent and $L \in D^{\leq b}$ is m -pseudo-coherent, their tensor product is $\max(n+b, m+a)$ -pseudo-coherent.

Proof. Choose approximations $P \rightarrow K, Q \rightarrow L$ at thresholds $m+1, m$. Brutally cut P below $m+1$, preserving that approximation. For $C = \text{Cone}(Q \rightarrow L)$, the map $P \rightarrow L \rightarrow C$ is locally zero by Lemma 1.1: C has no cohomology in degrees $\geq m$, whereas P starts at $m+1$. Exactness of Hom applied to the triangle $Q \rightarrow L \rightarrow C$ lifts $P \rightarrow L$ to a derived map $P \rightarrow Q$; Lemma 1.1 realizes it by a chain map locally. The triangle axiom gives a map $\text{Cone}(P \rightarrow Q) \rightarrow M$.

Here is the required cohomology check. Cone cohomology in degree j is an extension of the cokernel in degree j by the kernel in degree $j+1$. For $j > m$, the two cokernels are isomorphic: the map on $H^j(Q)$ is an isomorphism and the map from P 's cohomology is onto at $j = m+1$, an isomorphism above it. The kernels in degree $j+1$ are isomorphic. Hence the extension map is an isomorphism. At $j = m$, surjectivity for Q makes the cokernel map onto, and surjectivity for P at $m+1$, with the isomorphism for Q there, makes the kernel map onto. Lifting first the kernel component and then the cokernel component proves surjectivity for the extension. Thus the cone is an approximation. Rotation and the shift rule give the other two implications.

For the tensor claim choose approximations $P \rightarrow K, Q \rightarrow L$ at thresholds n, m . They can have terms only in degrees $\leq a, \leq b$: cancel acyclic top pairs; if the threshold exceeds the relevant bound, the zero complex is already an approximation. Factor the tensor map through $K \otimes^L Q$. Its first cone is $\text{Cone}(P \rightarrow K) \otimes Q$, bounded above by $n+b-1$; its second cone is $K \otimes^L \text{Cone}(Q \rightarrow L)$, bounded above by $a+m-1$. The octahedral cone triangle bounds the composite cone by the larger of these numbers. The source $P \otimes Q$ is strictly perfect, proving the assertion without a Tor spectral sequence. Taking arbitrarily small thresholds proves tensor closure for pseudo-coherent objects. $\square$

Proposition 2.2 (summands and criteria). Summands of m -pseudo-coherent objects are m -pseudo-coherent. A locally bounded-above complex whose term in degree i is $(m-i)$ -pseudo-coherent is m -pseudo-coherent; the same holds if its cohomology sheaf in degree i is $(m-i)$ -pseudo-coherent. A module is 0-pseudo-coherent exactly when it is of finite type, and (-1) -pseudo-coherent exactly when it is finitely presented.

Proof. Suppose $K \oplus L$ is m -pseudo-coherent. Its endomorphism $1_K \oplus 0$ has cone $L \oplus L[1]$. Proposition 2.1 applies since m -pseudo-coherence implies $(m+1)$ -pseudo-coherence. Shift repeatedly; each $L[n] \oplus L[n+1]$ is m -pseudo-coherent. Work near a point where L is bounded above. For large n , $L[n]$ has zero cohomology in degrees $\geq m$, so the zero approximation works. Descend using the split triangles

$$\begin{aligned} L[n] &\longrightarrow L[n] \oplus L[n-1] \\ &\longrightarrow L[n-1] \longrightarrow L[n+1]. \end{aligned} \tag{2.2}$$

The first implication of Proposition 2.1 supplies each preceding step. Only finitely many steps are needed for a fixed threshold at a fixed point. Exchanging K, L finishes the summand assertion.

For the term criterion, remove all degrees below $r \leq m$. The finite upper tail is m -pseudo-coherent by its filtration into terms $F^i[-i]$ and the extension implication. The lower quotient is bounded above by $r-1 < m$, so has the zero approximation. The same extension implication proves the claim. For the cohomology criterion use the finite good-truncation filtration, with quotients $H^i(K)[-i]$, and its lower tail in degrees $< r$.

If $K \in D^{\leq m}$, cancel the top acyclic terms in an m -approximation: $H^m(K)$ is a quotient of a finite projective sheaf, hence finite type. If $K \in D^{\leq m+1}$, the same cancellation leaves $H^{m+1}(K)$ as the cokernel of a map of two finite projective sheaves, hence finitely presented. Apply these facts to a module in degree zero. Conversely, a finite generating surjection in degree zero supplies a 0-approximation; a finite presentation in degrees $-1, 0$ supplies a (-1) -approximation. $\square$

3. Tor amplitude and the perfectness criterion

An object K has **Tor amplitude in** $[a, b]$ if, for every module sheaf F in degree zero, $K \otimes^L F$ has cohomology only in degrees a through b . It has **locally finite Tor amplitude** if such an interval exists on each member of an open cover. Taking $F = \mathcal{O}$ also bounds K itself.

Proposition 3.1 (amplitude models). Amplitude in $[a, b]$ is equivalent to having a representative of flat sheaves with terms only in $[a, b]$. It is also equivalent to every stalk having that amplitude over its stalk ring. Derived pullback preserves the interval; tensoring objects with intervals $[a, b], [c, d]$ gives interval $[a+c, b+d]$. Summands inherit amplitude.

Proof. A bounded complex of flat sheaves is K-flat. Its tensor with a module has terms only in the asserted interval, proving one implication. Conversely, resolve a good upper truncation of K by the bounded-above flat construction, obtaining P with terms only in degrees $\leq b$. Since $H^i(P) = 0$ below a , the flat tail ending in P^a resolves

$$F_a = \text{coker}(P^{a-1} \rightarrow P^a). \tag{3.1}$$

For every test module T , it computes $\text{Tor}_1(F_a, T) = H^{a-1}(P \otimes T) = 0$. The flatness criterion makes F_a flat. Replace the lower tail by F_a in degree a . This is the good lower truncation, quasi-isomorphic to K , and supplies the required model.

Stalks commute with derived tensor and cohomology, by the flat lesson's stalk criterion. For the converse stalk test, any module over $\mathcal{O}_x$ is the stalk at x of its exact skyscraper sheaf. Testing that sheaf proves the stalk interval from the global interval. Conversely, test a sheaf on all stalks to prove its tensor cohomology vanishes. Pullback of the bounded flat model remains bounded and flat; tensor two such models to add the bounds. Cohomology of a finite direct sum splits, proving the summand assertion. Finally, the tensor long exact sequence shows that if K, L have intervals $[a_K, b_K], [a_L, b_L]$, a cone of $K \rightarrow L$ has interval

$$\begin{aligned} a_M &= \min(a_L, a_K - 1), \\ b_M &= \max(b_L, b_K - 1). \end{aligned} \tag{3.2}$$

Rotation gives the other two-out-of-three bounds. $\square$

Theorem 3.2 (perfectness). An object is perfect if and only if it is pseudo-coherent and has locally finite Tor amplitude. More precisely, amplitude $[a, b]$ and $(a - 1)$ -pseudo-coherence already suffice, and give local strictly perfect models with terms in $[a, b]$.

Proof. A strictly perfect model is an approximation at every threshold and is a bounded flat model. This proves necessity. For the stronger converse, choose a strictly perfect approximation $E \rightarrow K$ through degree $a - 1$, and brutally cut E below $a - 1$. The approximation and $H^{a-1}(K) = 0$ show that its cone is

$$\begin{aligned} C &\simeq N[2 - a], \\ N &= \ker(E^{a-1} \rightarrow E^a). \end{aligned} \tag{3.3}$$

Indeed, in degree $a - 2$ the cone cohomology is that kernel, and all other degrees vanish. Tensor the triangle with an arbitrary module T . The lower amplitude bound on K and the tensor long exact sequence give

$$\begin{aligned} &\mathrm{im}(N \otimes T \rightarrow E^{a-1} \otimes T) \\ &= \ker(E^{a-1} \otimes T \rightarrow E^a \otimes T). \end{aligned} \tag{3.4}$$

To identify the Tor group here, take a bounded-above flat resolution of N , and append $E^{a-1} \rightarrow E^a$. This is a flat resolution of $E' = \mathrm{coker}(E^{a-1} \rightarrow E^a)$, whose first tensor homology is exactly the kernel modulo the image in (3.4). Thus $\mathrm{Tor}_1(E', T) = 0$ for all T , and E' is flat. It is finitely presented as the cokernel of a map between finite projective sheaves. Lemma 1.2 makes it locally finite projective.

Replace the bottom pair by E' in degree a . The resulting complex is quasi-isomorphic to K . Cancel its finitely many acyclic top pairs above b , as in Section 1; none of these cancellations crosses degree a . This gives the asserted strictly perfect model. Localize the hypotheses to deduce the general equivalence. $\square$

Corollary 3.3 (permanence). Perfect objects are closed under shifts, cones, derived tensor products, retracts and derived pullback.

Proof. For cones, locally represent the two inputs by strict models and their map by an actual chain map, using Lemma 1.1. Its strict cone is the third object. Rotation handles any two inputs. Tensor and pullback use their strict models directly. Summands use Proposition 2.2, Proposition 3.1 and Theorem 3.2.

A retract is an actual summand here. If $\sigma : K \rightarrow E$ and $r : E \rightarrow K$ satisfy $r\sigma = 1$, complete σ to a triangle $K \rightarrow E \rightarrow C \xrightarrow{h} K[1]$. Since $\sigma[1]h = 0$, composing with $r[1]$ gives $h = 0$. Exactness of Hom lifts 1_C to $t : C \rightarrow E$; replace t by $t - \sigma r t$. Then $rt = 0$, and the maps $(\sigma, t) : K \oplus C \rightarrow E$ and $(r, E \rightarrow C)$ are inverse: their induced cohomology maps split the triangle, so the first is an isomorphism, and the composite on $K \oplus C$ is the identity. The summand result therefore applies to every retract. $\square$

On a quasi-compact space perfect objects have uniform bounds, by a finite subcover. Without quasi-compactness this fails: on the discrete space of nonnegative integers, with stalk ring k , take $K|_{\{n\}} = k[n]$. Each neighbourhood $\{n\}$ has a strict model, but the global nonzero cohomology degrees $-n$ have no lower bound.

Two useful locality consequences complete the picture. If $j : U \hookrightarrow X$, E is perfect on U , and its cohomology is supported on $T \subset U$ with T closed in X , then Rj_*E is perfect. It restricts to E on U and to zero on $X \setminus T$: restriction of a K -injective pushforward computes the pushforward from $U \setminus T$, where the object is zero. These opens cover X .

If the stalk rings are local and E is perfect, the set where all $H^i(E)_x$ are finite free is open. Near such an x choose a strict model in $[a, b]$. Its top cohomology is finitely presented; Lemma 1.2's free-stalk argument

makes it free near x . Split the epimorphism from the top term to that cohomology, then split the preceding term onto its image. Remove the resulting contractible pair and the free cohomology summand. Induct on the finite length. The remaining cohomology stalks still satisfy the hypothesis, so finitely many shrinkings make all cohomology sheaves finite locally free. This also proves maximality of the stated open set.

4. Duals and the two tensor–Hom maps

For any object define $K^\vee = [K, \mathcal{O}]$. The symmetric monoidal structure, with graded symmetry $u \otimes v \mapsto (-1)^{|u||v|} v \otimes u$, was proved in the tensor lesson's Theorem 2.2. No boundedness is required for that structure.

Theorem 4.1 (perfect duality). If K is perfect, so is $K^\vee$, and the canonical maps

$$\begin{aligned} K &\longrightarrow K^{\vee\vee}, \\ M \otimes^L K^\vee &\longrightarrow [K, M] \end{aligned} \tag{4.1}$$

are isomorphisms. In particular, for every integer r ,

$$\begin{aligned} H^r R\Gamma(X, M \otimes^L K^\vee) \\ = \operatorname{Hom}_{D(\mathcal{O})}(K, M[r]). \end{aligned} \tag{4.2}$$

Two further canonical maps are isomorphisms:

$$[K, L] \otimes^L M \longrightarrow [K, L \otimes^L M], \tag{4.3}$$

$$[L, M] \otimes^L K \longrightarrow [[K, L], M]. \tag{4.4}$$

Proof. These are globally defined maps from the Hom lesson's Section 4. Their cones restrict to zero precisely when they are isomorphisms. Thus their invertibility can be checked on strict-model neighbourhoods. Proposition 1.3 proves the second map in (4.1) there, and (1.2) shows that the dual model is strictly perfect. The double-dual differential under the ordinary termwise identification is $-d_P$. Consequently the canonical bidual map in degree i is

$$p \longmapsto (\phi \mapsto (-1)^i \phi(p)). \tag{4.5}$$

This sign makes it a chain map and each term is an isomorphism. Equation (4.2) follows from the internal-Hom lesson's derived-section identification in every shift.

For (4.3), use $[P, L] = L \otimes P^\vee$. The map moves M past $P^\vee$ and associates the three factors, giving $(L \otimes^L M) \otimes P^\vee$. It is therefore an isomorphism. Explicitly, for homogeneous ϕ, m, p , its evaluation is $(-1)^{|m||p|} \phi(p) \otimes m$; this is the same sign as moving the dual factor past m .

For (4.4), the source and target are exact functors of K , and its natural map respects their triangle comparisons, by the chain identities and transpose construction in the Hom lesson. For $K = \mathcal{O}$ it is the identity. For a finite free sheaf it is the componentwise identity on a finite power of $[L, M]$. It passes to finite projective summands and shifts. Filter a bounded strict model by its terms; the two triangle long exact sequences show inductively that the middle map is an isomorphism when the outer two are. This proves (4.4). $\square$

There is a source-locator distinction: in the pinned edition, Tag 0G40 proves (4.4), the evaluation-and-more map. It does not denote (4.3). Both assertions hold for perfect K , as proved above, and both can fail without that hypothesis.

A **left dual** of K is an object N with maps

$$\begin{aligned} \eta : \mathcal{O} &\longrightarrow K \otimes^L N, \\ \epsilon : N \otimes^L K &\longrightarrow \mathcal{O} \end{aligned} \tag{4.6}$$

such that the composites

$$\begin{aligned} K &\xrightarrow{\eta \otimes 1} K \otimes N \otimes K \xrightarrow{1 \otimes \epsilon} K, \\ N &\xrightarrow{1 \otimes \eta} N \otimes K \otimes N \xrightarrow{\epsilon \otimes 1} N \end{aligned} \quad (4.7)$$

are identities; tensors in (4.7) are derived and association constraints are understood. Symmetry gives the corresponding right dual, so we say **dualizable**.

Theorem 4.2 (duals are exactly perfect objects). Every perfect object has dual $K^\vee$, and every dualizable object is perfect. Its dual is canonically isomorphic to $K^\vee$.

Proof: existence and identities. The natural isomorphism $[K, B] = B \otimes^L K^\vee$ identifies the closed tensor–Hom adjunction with

$$- \otimes^L K \dashv - \otimes^L K^\vee. \quad (4.8)$$

Let η be its unit at $\mathcal{O}$, and ϵ its counit there. Equivalently, η corresponds to 1_K under $K \otimes K^\vee \cong [K, K]$, and ϵ is evaluation. The counit at B is $1_B \otimes \epsilon$, since (4.1)'s map is defined by transposing this evaluation. The adjunction's first triangle identity at $\mathcal{O}$ gives the first identity in (4.7). Consequently $1_A \otimes \eta$ transposes to $1_{A \otimes K}$ for every A , and is the unit at A . The second adjunction identity, at $\mathcal{O}$, now gives the second identity in (4.7). These are global identities of derived maps, not an inference that equality of maps can be tested locally.

On a strict free model, $\eta(1) = \sum_{i,\alpha} e_{i,\alpha} \otimes e_{i,\alpha}^*$, with no degree sign. If $de_{i,\alpha} = \sum_{\beta} a_{\beta\alpha} e_{i+1,\beta}$, (1.2) gives the dual differential coefficient $(-1)^i a_{\beta\alpha}$ on $e_{i+1,\beta}^*$. Its tensor differential has the additional sign $(-1)^{i+1}$, so the two contributions cancel. Evaluating either zigzag contracts the appropriate dual basis and gives the identity. Summands inherit this description. The sign in the bidual map (4.5) and the absence of a sign in coevaluation are compatible: they are different maps with different factor orders.

Proof: converse. Fix $x \in X$ and a dual N with (4.6)–(4.7). Resolve K by the actual cycle-attachment construction of Theorem 3.1 of the flat lesson, obtaining $P \rightarrow K$. Its graded terms are sums of $B_U = j_{U!} \mathcal{O}_U$. Stage zero consists of identity disks; each later stage adds generators whose differentials lie in the preceding stage. Resolve N by a K-flat complex Q with flat terms.

By Lemma 1.1, near x the map $\eta : \mathcal{O} \rightarrow P \otimes Q$ is an actual chain map. Its value at one is locally a finite sum of homogeneous tensors $p_t \otimes q_t$. Each germ p_t involves finitely many generators and a finite stage. Include those generators and all generators occurring in their differentials. A later-stage differential only uses earlier stages; its germ has finite support. At stage zero include the other member of every disk encountered. Thus closing this finite set under differentials requires finitely many steps and produces finitely many generators in finitely many degrees.

This germ argument gives an actual local subcomplex. At each step choose representatives of the finitely many differential germs. Their expressions involve finitely many summands after shrinking. A summand B_U with $x \notin U$ has zero stalk there; kill its finitely many section contributions on a smaller neighbourhood and omit it. For the remaining summands shrink inside their finitely many U 's, where they are copies of $\mathcal{O}$. Equality of each chosen expression with the actual differential is a germ equality, so shrink to make all these finitely many equalities hold. The result is a bounded finite free subcomplex

$$E \subset P|_V \quad (4.9)$$

on a neighbourhood V of x , termwise a graded summand, containing all the p_t . Tensoring this inclusion with Q is injective termwise (also by flatness of Q). Since $\eta(1)$ is a cycle, its expression in $E \otimes Q$ is a cycle. Hence η factors through $\tilde{\eta} : \mathcal{O} \rightarrow E \otimes^L N$.

Define $\sigma : K|_V \rightarrow E$ by $\tilde{\eta} \otimes 1_K$, followed by $1_E \otimes \epsilon$, and let $r : E \rightarrow K|_V$ be inclusion followed by the resolution isomorphism. The first identity in (4.7) gives $r\sigma = 1_K$. Corollary 3.3 makes this retract of a strictly

perfect object perfect. Every point has such a neighbourhood, proving the converse. This uses finite free cells, rather than incorrectly assuming that a submodule generated by finitely many sections is projective.

Finally any dual supplies an adjunction $- \otimes K \dashv - \otimes N$: insert η to transpose a map, and insert ϵ to transpose back; the two identities make these operations inverse. Right adjoints are uniquely isomorphic by their natural Hom bijections (evaluate one bijection on the identity, then the other, to obtain inverse maps). Comparing with the closed tensor–Hom adjunction at $\mathcal{O}$ identifies N canonically with $[K, \mathcal{O}]$. $\square$

5. Residues and finiteness examples

A smooth analytic surface residue is perfect. On $\mathbb{C}^2$ let $\mathcal{O}$ be holomorphic functions and S the residue skyscraper at the origin. Its resolution, in degrees $-2, -1, 0$, is

$$\mathcal{O} \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} \mathcal{O}^2 \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} \mathcal{O} \longrightarrow S. \quad (5.1)$$

Taylor expansion shows that a germ vanishing at zero lies in (x, y) : separate the terms divisible by x and those depending only on y , which are divisible by y . Multiplication by x is injective on convergent power series, and multiplication by y is injective modulo x . Thus $xa + yb = 0$ implies $b = xc$ and $a = -yc$, proving middle exactness. The first map is injective. Away from zero one coordinate is a unit and the complex contracts by division by that coordinate. This proves exactness on every stalk, hence perfectness. With any vector-space residue test at zero all differentials vanish, so its Tor groups in degrees $0, 1, 2$ are k, k^2, k . The local Tor amplitude is exactly $[-2, 0]$.

A singular residue need not be perfect. Over $R = k[\epsilon]/(\epsilon^2)$, the residue k has the free resolution

$$\cdots \xrightarrow{\epsilon} R \xrightarrow{\epsilon} R \xrightarrow{\epsilon} R \longrightarrow k \quad (5.2)$$

in nonpositive degrees. Kernel and image of multiplication by ϵ both equal ϵR . Every finite brutal lower truncation gives an approximation: it is an isomorphism in all higher degrees and surjective at the bottom threshold. Thus k is pseudo-coherent. Tensoring (5.2) with k kills every differential, so $\mathrm{Tor}_n^R(k, k) = k$ for every $n \geq 0$. It has no finite Tor amplitude, is not perfect, and is not dualizable. The singularity hypothesis matters; a surface residue alone does not imply nonperfectness.

Perfectness does not force finite cohomology over every ring. Let

$$\begin{aligned} R &= k[x] \oplus Y, & Y &= \bigoplus_{n \geq 1} ky_n, \\ Y^2 &= 0, & xY &= 0. \end{aligned} \quad (5.3)$$

The two-term complex $R \xrightarrow{x} R$, in degrees $-1, 0$, is strictly perfect. But its negative cohomology is Y , which is not finitely generated: the action on Y factors through k , and finitely many elements span only a finite-dimensional subspace. Hence even a perfect, and therefore pseudo-coherent, complex need not have finite-type cohomology over a noncoherent ring. The safe assertions are the approximation definition and Proposition 2.2's explicitly bounded top-cohomology conclusions.

6. Exercises with checked solutions

Exercise 1 (easy: the shifted unit). For $K = \mathcal{O}[1]$, compute its dual, coevaluation and bidual map, and check both identities in (4.7).

Solution. Its generator e has degree -1 , and the dual generator e^* has degree 1 . Thus $K^\vee = \mathcal{O}[-1]$, $\eta(1) = e \otimes e^*$, and $\epsilon(e^* \otimes e) = 1$. The first zigzag sends e to $e \otimes e^* \otimes e$ and back to e ; the second does the analogous contraction for e^* . No factor swap occurs. The bidual map is $e \mapsto -e^{**}$ by (4.5). Multiplying coevaluation

instead by -1 while keeping this evaluation would make both zigzags negative identities; that is not the correct dual data.

Exercise 2 (medium: one approximation suffices). Suppose F is a module sheaf of Tor amplitude $[0, 0]$. What is the weakest threshold in Theorem 3.2 that proves perfectness? Characterize such F directly.

Solution. The threshold is -1 , which is finite presentation by Proposition 2.2. Amplitude $[0, 0]$ means $\mathrm{Tor}_1(F, T) = 0$ for every module T , and therefore flatness. Conversely a flat module in degree zero has that amplitude, since tensor is exact. Lemma 1.2 now says that precisely the flat finitely presented sheaves are locally finite projective, hence perfect in degree zero. The threshold records finite relations as well as finite generators; finite type alone does not supply the finite presentation used in the splitting proof.

Exercise 3 (medium: the coefficient sign). Let P have one finite free term in degree s . For a homogeneous map $\phi : P \rightarrow L$ of degree h and an element $m \in M^t$, calculate (4.3). Explain why it remains an isomorphism for a bounded finite projective complex.

Solution. Evaluation must first move m past $p \in P^s$, giving $(-1)^{ts} \phi(p) \otimes m$. In the model $H(P, L) = L \otimes P^\vee$, the dual term has degree $-s$; swapping it with m gives the same sign $(-1)^{-st} = (-1)^{st}$. This is a tensor symmetry isomorphism. Finite free powers give componentwise copies, finite projective summands inherit an inverse, and bounded complexes are finite extensions of their terms. Applying the compatible triangle maps and long exact sequences proves the general result. K -flat models for L, M compute the derived tensors, so their possible unboundedness causes no additional finite-sum restriction.

Exercise 4 (hard: finite cells rather than finite generators). In the converse proof of Theorem 4.2, why is closing the chosen generators under their differentials a finite process, and why does stage zero need special treatment? Would merely taking the subcomplex generated by the tensor entries and their differentials be sufficient?

Solution. Each chosen germ has finite support in a finite stage. A cell from a positive stage has differential in strictly earlier stages, so following its dependencies decreases a nonnegative integer. Every branch has finite length and every vertex finitely many descendants, giving a finite set. A stage-zero disk has two generators and its lower generator's differential is its upper mate in the same stage; including the mate completes that branch, since its differential is zero. Finite intersection of their defining opens makes the remaining cells free on one neighbourhood. Contributions with zero germ are killed by finitely many shrinkings. The span of this selected subset of graded free cells is a bounded finite free complex and a graded summand. In contrast, finitely many arbitrary sections may generate a nonprojective submodule: the ideal $\epsilon R \subset R$ over the dual numbers is generated by one element but is isomorphic to the nonflat residue k . Such a subcomplex would not supply a perfect object for the retract argument. Selecting actual cells repairs exactly this problem.

Exercise 5 (hard: global duality with no uniform bound). On the discrete space from Section 3, construct the dual of $K|_{\{n\}} = k[n]$ explicitly and check that the global coevaluation is defined although it has infinitely many degree components across the space.

Solution. Take $N|_{\{n\}} = k[-n]$. At point n , send one to the tensor of its degree- $-n$ generator and the degree- n dual generator, and evaluate that pair to one. Sheaves on a discrete space are specified independently at the points; these formulas define sheaf maps. The tensor totalization uses sheaf direct sums, whose sections are locally finite families, not necessarily globally finite families. The singleton neighbourhoods make this family locally finite, so coevaluation is a legitimate section of the tensor sheaf. Both zigzags are identities on every singleton and hence are identities of these actual sheaf maps. This example is perfect and dualizable but has no global finite cohomological bound. It shows why neither definition may silently impose a uniform interval.

The projection formula and the base change map

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The projection formula compares tensoring after pushforward with tensoring before it. Base change compares pullback after pushforward with pushforward after pullback. Each comparison has a canonical direction, supplied by adjunction. Their existence is general; their invertibility needs hypotheses. Keeping these two questions separate prevents a flatness assumption from being mistaken for a universal base-change theorem.

The prerequisites are [Derived pullback and pushforward](#), Theorem 1.1, Proposition 1.2, Theorem 2.2, Propositions 3.1–3.2 and 4.2; [Perfect complexes and duals](#), Lemma 1.1, Corollary 3.3 and Theorems 4.1–4.2; [The derived tensor product and Tor sheaves](#), Theorems 1.2 and 2.2; and the exact pullback and closed-support arguments in the first lesson's Theorem 4.4 and the support lesson's Proposition 1.1. We use maps into K-injectives from the bounded-derived lesson's Theorem 3.3 and unbounded resolutions from the K-injective lesson's Theorem 4.1.

All structure sheaves are commutative and unital. Objects may be unbounded. Let $f : X \rightarrow Y$ be a morphism of ringed spaces. Every tensor in a derived formula below is over the structure sheaf of the indicated space. Ordinary tensors and functors are expressly distinguished when models are used.

The construction follows the Stacks project authors' *Cohomology of Sheaves*, “Projection formula” (Tags 01E7, 01E8, 0B54 and 0B55), “The base change map” (Tag 02N7), and the unbounded base-change and composition results, in the [AI Integrated Stacks Project edition](#). We prove the local identifications and compatibility diagrams, keeping the existence and invertibility hypotheses explicit. Source attribution and the licence for adapted passages appear in the [course notice](#).

1. Finite locally free coefficients

Lemma 1.1. If V is finite locally free and I is an injective module sheaf on a ringed space, then $V \otimes I$ is injective. Tensoring a K-injective complex with V also gives a K-injective complex.

Proof. Put $V^* = \mathcal{H}om(V, \mathcal{O})$. The ordinary dual-basis evaluation maps give a natural bijection

$$\begin{aligned} \operatorname{Hom}(A, V \otimes I) \\ \cong \operatorname{Hom}(A \otimes V^*, I). \end{aligned} \tag{1.1}$$

Indeed, this is the finite free tensor–Hom identity on a trivializing open; the canonical evaluation and coevaluation maps define inverse operations globally, so the local descriptions glue. Both V and V^* are flat by the stalk criterion. For a monomorphism $A \rightarrow B$, tensoring with V^* remains monic. Injectivity of I , followed by (1.1), extends every map $A \rightarrow V \otimes I$ across B . This proves injectivity.

The same degree-zero duality induces a bijection of chain maps modulo homotopy from a complex A to $V \otimes I^\bullet$, and from $A \otimes V^*$ to $I^\bullet$. An acyclic A stays acyclic after this exact tensor. The K-injective acyclic test therefore proves the second assertion. The ranks can vary on the space; global constancy of rank was not used. $\square$

Theorem 1.2 (finite coefficient projection). For a finite locally free sheaf V on Y and any $E \in D(\mathcal{O}_X)$, there is a natural isomorphism

$$Rf_* E \otimes V \cong Rf_*(E \otimes f^*V). \tag{1.2}$$

For a module sheaf F and $q \geq 0$, this induces

$$\begin{aligned} R^q f_* F \otimes V \\ \cong R^q f_*(F \otimes f^*V). \end{aligned} \tag{1.3}$$

Proof. The ordinary projection map is the adjoint of the tensor of the ordinary counit $f^* f_* F \rightarrow F$ with f^*V . On an open $W \subset Y$ trivializing V as $\mathcal{O}_W^r$, put $F_W = F|_{f^{-1}W}$. Its restriction identifies with

$$(f_W)_*(F_W)^r \longrightarrow (f_W)_*(F_W^r). \tag{1.4}$$

It is the identity on each component: the adjoint of each component inclusion is the corresponding counit. Finite sums equal finite products, and pushforward preserves them. Hence (1.4) is an isomorphism. Since the map was globally defined by adjunction, these local isomorphisms prove its global invertibility without choosing compatible trivializations.

Choose a K -injective resolution $E \rightarrow I$. The pullback f^*V is finite locally free, and Lemma 1.1 makes $I \otimes f^*V$ K -injective. Its exact tensor represents $E \otimes f^*V$. Applying the ordinary projection isomorphism termwise gives

$$f_* I \otimes V \cong f_*(I \otimes f^*V), \quad (1.5)$$

which computes both sides of (1.2). All maps are natural in chain maps and homotopies, hence in derived maps through their K -injective representatives. For F in degree zero use its bounded-below injective resolution; Lemma 1.1 makes the tensor another injective resolution. Flatness of V commutes its tensor with kernels, images and cohomology, giving (1.3). This proves the source's bounded-below assertion and the stronger unbounded version. $\square$

2. The general map and perfect coefficients

For any $E \in D(\mathcal{O}_X)$, $K \in D(\mathcal{O}_Y)$, define

$$\begin{aligned} \pi_{f,E,K} : Rf_* E \otimes^L K \\ \longrightarrow Rf_*(E \otimes^L Lf^* K) \end{aligned} \quad (2.1)$$

as the adjoint, under $Lf^* \dashv Rf_*$, of

$$\begin{aligned} Lf^*(Rf_* E \otimes^L K) \\ \cong Lf^* Rf_* E \otimes^L Lf^* K \\ \xrightarrow{\epsilon_f \otimes 1} E \otimes^L Lf^* K. \end{aligned} \quad (2.2)$$

Here ϵ_f is the derived counit. The proven tensor comparison and the adjunction supply every arrow for arbitrary unbounded inputs. This is a definition of a global map, not an attempt to glue local derived morphisms.

Naturality follows by taking adjoints: in either variable the relevant square transposes to the naturality square of ϵ_f , with the same tensor identity on the other factor. The same argument proves compatibility with shifts and triangles. More concretely, the resolution functors, tensor cone comparison and adjunction identify the transposes of all three arrows in the cone triangle; they agree with those in (2.2). Thus (2.1) is a morphism of the two triangle-valued functors in K . The map for $K = \mathcal{O}_Y$ is the identity of $Rf_* E$: its transpose is ϵ_f , exactly the transpose of that identity.

Theorem 2.1 (perfect projection formula). If K is perfect on Y , (2.1) is an isomorphism for every E .

Proof. Restrict to an open $W \subset Y$, replacing X by $f^{-1}W$. Proposition 3.2 of the pullback–pushforward lesson identifies the restricted pushforward, while its Theorem 1.1 and Proposition 1.2 identify the restricted pullbacks and tensors. The counit restricts to the same counit by its defining Hom bijection; hence (2.1) restricts to the projection map for f_W .

Now work on a strict-model neighbourhood for K . For $\mathcal{O}_Y$ the map is the identity. Finite direct sums preserve that assertion, and naturality with the inclusion and retraction of a finite projective summand makes the map on the summand an isomorphism too. Shifts preserve it by the comparison above. Filter the bounded strict complex by its brutal lower tails. Each successive triangle has one term equal to a finite projective sheaf in one degree. The compatible two triangle sequences, and their long exact cohomology sequences, prove the assertion inductively on its finite number of terms. The cone of (2.1) is consequently zero on an open cover, so all its cohomology sheaves vanish globally. Thus the map is an isomorphism. $\square$

This proof does not require f to be proper, flat, separated or of finite cohomological dimension. Finiteness is in the coefficient K , and it is local finiteness. In particular, a perfect coefficient without a uniform bound on a noncompact target is allowed.

3. Closed inclusions with arbitrary coefficients

Theorem 3.1. If the underlying map of $f : X \rightarrow Y$ is a homeomorphism onto a closed subset, the projection map (2.1) is an isomorphism for every E, K . The coefficient map need not be flat or an isomorphism.

Proof. Identify the underlying space X with its image Z . At $y \notin Z$, a neighbourhood disjoint from Z makes $(f_*F)_y = 0$. At $y = f(x)$, intersections of neighbourhoods in Y with Z are cofinal among neighbourhoods in X , so $(f_*F)_y = F_x$, with its action restricted from $\mathcal{O}_{Y,y}$. These stalk formulas prove that f_* is exact and commutes with arbitrary sheaf direct sums: the respective comparison maps are exact or isomorphisms on every stalk. This remains true for arbitrary structure sheaves and their given coefficient homomorphism.

For module sheaves the ordinary projection map is an isomorphism on stalks. Outside Z both sides vanish; at $y = f(x)$, with $R = \mathcal{O}_{Y,y}$, $S = \mathcal{O}_{X,x}$, it is the balanced isomorphism

$$\begin{aligned} & F_x \otimes_R M_y \\ & \longrightarrow F_x \otimes_S (S \otimes_R M_y), \\ & e \otimes m \longmapsto e \otimes (1 \otimes m). \end{aligned} \tag{3.1}$$

Its inverse sends $e \otimes (s \otimes m)$ to $se \otimes m$. Balancing verifies both inverse composites, without flatness.

Choose a K -flat model P for K , and any representative G of E . Pullback f^*P is K -flat. Exactness of f_* makes the right side of (2.1) represented by $f_* \text{Tot}(G \otimes f^*P)$, and the left by $\text{Tot}(f_*G \otimes P)$. The ordinary projection maps in each pair of degrees give a chain isomorphism between them: (3.1) is an isomorphism and f_* commutes with the totalization's sums.

This is the canonical derived map. To check it, take a K -flat resolution $Q \rightarrow f_*G$. The map $Q \otimes P \rightarrow f_*G \otimes P$ is a quasi-isomorphism. Under the ordinary adjunction the chain projection map, precomposed with this resolution, transposes to the map

$$f^*Q \otimes f^*P \longrightarrow G \otimes f^*P \tag{3.2}$$

from the adjoint $f^*Q \rightarrow G$. That adjoint represents the derived counit: comparing $G \rightarrow I$ with a K -flat resolution of f_*I , and refining the two roofs by the common reading's fraction theorem, gives a commuting-up-to-homotopy square; its adjoint is precisely the counit construction in the pullback–pushforward lesson's Theorem 2.2. Thus (3.2) is (2.2). The isomorphism proved on these models is (2.1). $\square$

For a one-point coefficient homomorphism $R \rightarrow S$, this says

$$E \otimes_R^L K \cong E \otimes_S^L (S \otimes_R^L K), \tag{3.3}$$

where on the left E is regarded as an R -complex. Even $R \rightarrow R/I$ is allowed. On more general spaces the closed image is what ensures exact pushforward and its stalk description. An open inclusion has different boundary stalks, as Section 5 will demonstrate.

4. Base change, flat models and compatibility

Consider a commutative square of ringed spaces

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{g} & Y. \end{array} \tag{4.1}$$

Commutative includes the coefficient homomorphisms, not just the continuous maps. The square need not be Cartesian.

Theorem 4.1 (canonical base change). For every $E \in D(\mathcal{O}_X)$ there is a natural map

$$b_E : Lg^* Rf_* E \longrightarrow Rf'_* Lg'^* E. \quad (4.2)$$

It respects identity squares, horizontal and vertical composition of squares, and the projection maps. If both g, g' are flat, its construction uses ordinary exact pullbacks, for unbounded as well as bounded-below complexes. These statements construct and compare maps; they assert no general invertibility.

Proof: definition and naturality. Let $A = Lg^* Rf_* E$. Define b_E to be the adjoint under $Lf'^* \dashv Rf'_*$ of

$$\begin{aligned} Lf'^* A &\cong Lg'^* Lf^* Rf_* E \\ &\xrightarrow{Lg'^* \epsilon_{f, E}} Lg'^* E. \end{aligned} \quad (4.3)$$

The pullback composition comparison supplies the isomorphism. Naturality of the counit and of this comparison proves naturality of b for every derived map, by taking its transpose. In particular, no choice of representatives occurs in the definition.

The equivalent pushforward description first takes the unit

$$E \longrightarrow Rg'_* Lg'^* E,$$

pushes it through Rf_* , identifies $Rf_* Rg'_*$ with $Rg_* Rf'_*$ by composition, and takes its adjoint under $Lg^* \dashv Rg_*$. To verify equivalence, take that map's further transpose under Lf'^* . Successive Hom bijections identify the composite adjunctions for $fg' = gf'$; cancel the unit-counit pairs using their triangle identities. What remains is $Lg'^* \epsilon_{f, E}$, which is (4.3). This also identifies exactly which units are meant in model calculations.

Proof: flat construction. Flatness of a ringed-space morphism means that the target-to-source map on each relevant stalk makes the source stalk flat. The stalk formula for pullback then makes g^* and g'^* exact, so they compute Lg^*, Lg'^* on any representatives. Choose $E \rightarrow I$ K-injective and $g'^* E \rightarrow J$ K-injective. Pushforward $g'_* J$ is K-injective because its left adjoint g'^* is exact, as proved in Proposition 4.2 of the pullback-pushforward lesson. The ordinary unit followed by the resolution gives a chain map $E \rightarrow g'_* J$. Maps into K-injectives give a unique homotopy class

$$\beta : I \longrightarrow g'_* J \quad (4.4)$$

extending it. Push (4.4) through f_* and use $f_* g'_* = g_* f'_*$. Ordinary adjunction gives the chain map

$$g^* f_* I \longrightarrow f'_* J. \quad (4.5)$$

The complexes in (4.5) represent the two sides of (4.2); a homotopy of β induces a homotopy of (4.5). Comparison between K-injective resolutions therefore makes this a well-defined natural derived map. A pullback of an injective was never asserted injective.

Here is a direct check that (4.5) agrees with (4.3). Resolve $f_* I$ by a K-flat Q . Since g is flat, $g^* Q \rightarrow g^* f_* I$ is a quasi-isomorphism, and $g^* Q$ is K-flat. The transpose of (4.5), on that model, is

$$\begin{aligned} f'^* g^* Q &\cong g'^* f^* Q \\ &\longrightarrow g'^* I \longrightarrow J. \end{aligned} \quad (4.6)$$

The first arrow after the isomorphism is g'^* of the adjoint to $Q \rightarrow f_* I$, hence of the derived counit. The last arrow is adjoint to β . Its composite with $g'^* E \rightarrow g'^* I$ is the chosen resolution map to J , up to homotopy, by (4.4). Flatness makes that first map a quasi-isomorphism, so the last arrow represents the required identification with $Lg'^* E$. Thus (4.6) is exactly (4.3).

For bounded-below E , injective resolutions may be bounded below with injective terms. This specializes to Tag 02N7: **both horizontal maps are flat and the conclusion is a canonical map**. Its proof does not turn a Cartesian square, flat horizontal arrows, or any combination of those conditions into an isomorphism claim.

Proof: pasting squares. For any adjunction $L \dashv R$, the transpose of $u : A \rightarrow RB$ is $\epsilon_B \circ L(u)$. We use this formula and (4.3). In an identity square (4.3) is a counit, so (4.2) is the identity.

For horizontal pasting, add a square with $h : Y'' \rightarrow Y'$, $h' : X'' \rightarrow X'$ and $f'' : X'' \rightarrow Y''$. The proposed composite is

$$\begin{aligned} & Lh^* Lg^* Rf_* E \\ & \longrightarrow Lh^* Rf'_* Lg'^* E \\ & \longrightarrow Rf''_* Lh'^* Lg'^* E. \end{aligned} \tag{4.7}$$

Transpose under $Lf''^* \dashv Rf''_*$. The defining equation for the second base-change map replaces its pullback followed by the counit with $Lh'^* \epsilon_{f'}$. The defining equation for the first then replaces $\epsilon_{f'} \circ Lf'^*(b_E)$ with $Lg'^* \epsilon_f$. The result is $Lh'^* Lg'^* \epsilon_f$, with the coherent pullback-composition identification. This is the transpose for the outer rectangle, proving equality of the maps.

For vertical pasting take $p : Y \rightarrow Z$, $p' : Y' \rightarrow Z'$, and $h : Z' \rightarrow Z$, with $pg = hp'$. The composite first applies base change for p to $Rf_* E$, then Rp'_* to the base change for f . The counit of the composite adjunction $L(pf)^* \dashv R(pf)_*$, under the composition isomorphisms, is

$$\begin{aligned} & Lf^* Lp^* Rp_* Rf_* E \\ & \xrightarrow{Lf^* \epsilon_p} Lf^* Rf_* E \\ & \xrightarrow{\epsilon_f} E. \end{aligned} \tag{4.8}$$

This formula follows by composing the two Hom adjunction bijections: the identity of the composed right-adjoint value transposes successively through those two counits. Transposing the proposed pasted map uses the same two factors, pulled back through g' . It is therefore the transpose (4.3) for the rectangle. This proves vertical pasting without an omitted diagram verification.

Proof: projection compatibility. Fix $K \in D(\mathcal{O}_Y)$. The two routes from

$$Lg^*(Rf_* E \otimes^L K)$$

to

$$Rf'_*(Lg'^* E \otimes^L Lf'^* Lg^* K) \tag{4.9}$$

are as follows. One pulls back π_f , applies base change to $E \otimes^L Lf^* K$, then uses tensor and pullback composition. The other uses tensor compatibility first, applies $b_E \otimes 1$, then $\pi_{f'}$. Transpose both under Lf'^* . On the common source model the first route is $Lg'^* \epsilon_{f,E} \otimes 1$, with the coefficient factor $Lg'^* Lf^* K$. The second uses

$$\epsilon_{f', Lg'^* E} \circ Lf'^*(b_E) = Lg'^* \epsilon_{f,E}, \tag{4.10}$$

which is precisely (4.3)'s defining transpose equation, and then tensors the same coefficient identity. The two transposes agree globally, proving compatibility. All tensor reassociations here are the coherent ones established earlier. No invertibility of b_E was needed. $\square$

Projection maps also compose for $X \xrightarrow{f} Y \xrightarrow{p} Z$. First apply π_p to Rf_*E and K , then $Rp_*\pi_f$ with coefficient Lp^*K . Their transpose under the composite pullback is the tensor of (4.8) with $L(pf)^*K$, hence is the transpose of π_{pf} . The maps are equal by adjunction. This observation and Theorem 4.1 ensure that iterating these constructions changes neither their direction nor their definition.

5. Two boundary failures and a coefficient calculation

Let $Y = \{0\} \cup \{1/n : n \geq 1\}$ have its subspace topology in $\mathbb{R}$, let $U = Y \setminus \{0\}$, and let $j : U \hookrightarrow Y$. Give both spaces the constant coefficient sheaf k , for a field k . Every point of U is isolated. A sheaf there is a family of vector spaces, and its sections on an open are their product. Products of vector-space short exact sequences are exact: choose a preimage in each component to prove surjectivity; kernels are componentwise. Consequently j_* is exact on these sheaves, and its derived functor is computed on any complex by ordinary pushforward.

Take $F = k_U$. At the boundary point the neighbourhoods contain all sufficiently large n , so

$$\begin{aligned} (j_*F)_0 &= \varinjlim_N \prod_{n \geq N} k \\ &\cong \left(\prod_{n \geq 1} k \right) / \left(\bigoplus_{n \geq 1} k \right). \end{aligned} \tag{5.1}$$

Indeed, extending a tail by zero proves surjectivity from the full product to this colimit, and a sequence dies exactly when a tail is zero, equivalently when its support is finite. Write this quotient as A ; the class of the constant sequence one is nonzero.

Failure with a flat infinite coefficient. Put $K = \bigoplus_{m \geq 1} k_Y$, in degree zero. It is flat, but not perfect, since its stalk at any point is an infinite-dimensional vector space and Proposition 2.2 of the perfect lesson says a perfect module is of finite type. All derived tensors in this example are ordinary. The projection map at zero is

$$\begin{aligned} &A \otimes_k \left(\bigoplus_{m \geq 1} k \right) \\ &\longrightarrow \varinjlim_N \prod_{n \geq N} \left(\bigoplus_{m \geq 1} k \right). \end{aligned} \tag{5.2}$$

An element on the left uses only finitely many coordinate indices m , uniformly over its sequence entries. On the right take the sequence whose n -th value is the n -th basis vector e_n . It is a valid section on the discrete U . No tail of that sequence has its values in a fixed finite coordinate subspace. Hence its germ is not in the image of (5.2), and the projection map is not an isomorphism, even in degree zero. The failure is about finite coefficients, not about flatness: this coefficient is flat, and there is no higher-cohomology complication hiding the failure.

Failure in a Cartesian flat square. Base change the same open inclusion along $g : \{0\} \hookrightarrow Y$, with coefficient k at the point. The Cartesian top-left space is empty, so $g' : \emptyset \rightarrow U$ and $f' : \emptyset \rightarrow \{0\}$. The stalk coefficient map for g is $k \rightarrow k$, hence flat; g' is flat vacuously. For F , (4.2) becomes

$$A[0] \longrightarrow 0. \tag{5.3}$$

This is not an isomorphism because $A \neq 0$. Thus both flat horizontal maps and a Cartesian square still do not imply base change for arbitrary ringed spaces and pushforward. Every hypothesis has been checked in this example; no appeal to a vague exceptional topology is needed.

The contrast with restriction to an open is instructive. If $g : V \hookrightarrow Y$ is open and $X' = f^{-1}V$ has restricted coefficients, the base-change map is an isomorphism. On a K-injective I , restriction to X' remains K-injective, and the two sides are literally the complexes

$$(f_*I)|_V = (f_V)_*(I|_{X'}). \quad (5.4)$$

Their section values on every smaller open coincide. The map (4.3) gives this same equality: its transpose is the restricted counit. This is precisely Proposition 3.2 of the pullback–pushforward lesson. Replacing an open neighbourhood by a boundary point removes the sectionwise equality that proves (5.4), which explains (5.3).

A nonflat closed coefficient map. On one-point spaces take $R = \mathbb{Z}$, $S = \mathbb{Z}/2$, $E = S$, and $K = \mathbb{Z}/2$. The free resolution $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ in degrees $-1, 0$ represents K . Its scalar extension is $S \xrightarrow{0} S$. Thus both sides of (3.3) have one copy of S in each of degrees $-1, 0$, and the projection isomorphism is the identity on those copies. Replacing derived pullback by ordinary pullback would lose the negative cohomology. Closedness guarantees this projection formula even though the coefficient map is nonflat; it does not erase its Tor groups.

6. Exercises with checked solutions

Exercise 1 (easy: pushforward to a point). Let $f : X \rightarrow \{*\}$ have coefficient ring R at the point. Show the projection formula for $K = R^r$, with r finite, directly on a K-injective model. Explain why this calculation does not justify arbitrary infinite free coefficients.

Solution. For $E \rightarrow I$ K-injective, $Lf^*R^r = \mathcal{O}_X^r$, and I^r is K-injective: finite Hom groups split into finite products, so the acyclic test vanishes in each component. Sections commute with finite direct sums, which are finite products. Both sides are therefore represented by $\Gamma(X, I)^r$, and the map is the identity in every degree and on each component. For infinite rank a tensor uses a sheaf direct sum. Its section functor need not commute with that sum, and its pushforward at a boundary may include families with coordinate support varying without a common finite bound. Formula (5.2) supplies a checked counterexample to exactly that inference. The finite proof uses finiteness twice, in the direct-sum identity and in the coefficient dual.

Exercise 2 (medium: more general degree-zero coefficients). Extend (1.3) to a locally finite projective sheaf V that need not be finite locally free. State the needed ring hypothesis, and check independence of the chosen splitting.

Solution. The rings remain commutative, but their stalks need not be local. On an open neighbourhood choose V as a summand of $\mathcal{O}^r$, with inclusion a and retraction b . The ordinary projection maps commute with a, b because they are defined by the counit and tensor functoriality. Their map on the finite power is an isomorphism; restricting it and its inverse to the summand therefore gives an isomorphism on V . The global projection map is already defined by adjunction, so a second splitting gives the same map, rather than a new choice of gluing data. The pullback remains locally finite projective and flat. Its ordinary duality gives (1.1), hence preserves injectives; alternatively this follows locally from finite projective dual bases, with globally canonical maps. Applying these facts to an injective resolution gives (1.3) on every member of the cover, hence globally. This includes the nonfree module $(1, 0)(k \times k)$ on the one-point example in the perfect lesson.

Exercise 3 (medium: a two-term strict model). Prove the projection formula for $K = (P^{-1} \xrightarrow{d} P^0)$, where both terms are finite projective. Specify the triangle and the comparison that is used; a reference to “devissage” alone is insufficient.

Solution. Inclusion of the top term gives the triangle

$$\begin{aligned} P^0[0] &\longrightarrow K \longrightarrow P^{-1}[1] \\ &\longrightarrow P^0[1]. \end{aligned} \tag{6.1}$$

It comes from the degreewise split short exact sequence of the brutal tails, using the common reading's cone convention. Apply the exact functors $Rf_*E \otimes^L -$ and $Rf_*(E \otimes^L Lf^*(-))$. Naturality and triangle compatibility of (2.1) give a map between these triangles, including their last arrows. The maps on P^0 and $P^{-1}[1]$ are isomorphisms by the finite-power, summand and shift cases. The long exact cohomology sequences then give an isomorphism on the middle object: injectivity follows by lifting a zero image to the preceding term and cancelling its isomorphic counterpart; surjectivity follows by lifting the following-term image and correcting by the preceding-term image. Thus the middle cone is acyclic. This proves the two-term case for any differential, rather than only for a split complex.

Exercise 4 (hard: a global duality proof). Give a second proof of Theorem 2.1 using perfect duality and the derived adjunction, without choosing a global strict model for K .

Solution. Pullback preserves the evaluation, coevaluation and their identities, because it is a monoidal functor with coherent comparisons. Hence $Lf^*(K^\vee)$ is the dual of Lf^*K ; uniqueness of the dual identifies it with $(Lf^*K)^\vee$. Write $A = Rf_*E$, $P = Lf^*K$, and $T_X = Lf^*T$. For every $T \in D(\mathcal{O}_Y)$, the dual and pullback adjunctions give successive natural bijections

$$\begin{aligned} &\mathrm{Hom}(T, A \otimes K) \\ &\cong \mathrm{Hom}(T \otimes K^\vee, A) \\ &\cong \mathrm{Hom}(T_X \otimes P^\vee, E) \\ &\cong \mathrm{Hom}(T_X, E \otimes P) \\ &\cong \mathrm{Hom}(T, Rf_*(E \otimes P)). \end{aligned} \tag{6.2}$$

The tensor symbols in this calculation are derived. The first bijection evaluates the coefficient dual, and the third inserts its coevaluation; the inverse operations use the other two maps and their triangle identities. The middle bijection uses the counit of f , and pullback identifies the dual data just described. Composing the operations, the dual evaluation–coevaluation pairs cancel. The resulting map on Hom groups is composition with the map whose pullback transpose is $\epsilon_f \otimes 1$. By (2.2), that map is exactly $\pi_{f,E,K}$. Thus composition with π is a bijection for all T . With T equal to its target, surjectivity supplies a right inverse; with T equal to its source, injectivity makes that right inverse also a left inverse. This proves the canonical map is an isomorphism globally. It also explains why no uniform bound or global free presentation was necessary.

Exercise 5 (medium: Cartesian flat change on one-point spaces). Let $R \rightarrow B$ and a flat homomorphism $R \rightarrow R'$ be given, and let $B' = B \otimes_R R'$. Form the square of one-point ringed spaces with these coefficient rings. Prove that its base-change map is an isomorphism for every unbounded B -complex, and explain why this does not contradict (5.3).

Solution. Both pushforwards are restriction of scalars, hence exact. The map $B \rightarrow B'$ is flat: for a short exact sequence of B -modules, tensoring with B' over B identifies with tensoring the same underlying R -sequence with the flat R' . Thus both horizontal pullbacks are exact. For any complex E , the two base-change values are represented by $R' \otimes_R E$ and $B' \otimes_B E$. Their canonical balanced isomorphism sends $r' \otimes e$ to $(1 \otimes r') \otimes e$, with inverse $(b \otimes r') \otimes e \mapsto r' \otimes be$. It commutes with differentials. The counit for restriction and extension of scalars is multiplication, so its transpose is precisely this balanced map. Theorem 4.1 therefore identifies it with base change. These particular pushforwards have no boundary-section phenomenon. Flatness is sufficient in this coefficient-only situation, while the topological example (5.3) has nonzero boundary germs over an empty fibre. Neither example licenses replacing the other's hypotheses.

Exercise 6 (hard: a nonflat algebraic square). In the preceding square take $R = \mathbb{Z}$, $B = R' = B' = \mathbb{Z}/2$, and $E = B[0]$. Compute the general derived base-change map and compare it with ordinary pullback.

Solution. The lower horizontal coefficient homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}/2$ is not flat: tensoring the injection $2 : \mathbb{Z} \rightarrow \mathbb{Z}$ makes it zero. The upper horizontal homomorphism $B \rightarrow B'$ is the identity, hence flat. The source of

(4.2) is $(\mathbb{Z}/2) \otimes_{\mathbb{Z}}^L (\mathbb{Z}/2)$, represented by $B \xrightarrow{0} B$ in degrees $-1, 0$; the target is $B[0]$. The transpose description is scalar-extension multiplication. On this model it is the identity on degree zero and zero on degree -1 , so it is the usual augmentation. It induces an isomorphism on H^0 but kills the nonzero $H^{-1} = B$. Thus it is not a quasi-isomorphism. Ordinary pullback would keep only the degree-zero tensor and obscure the failure. The square is Cartesian with the ordinary tensor coefficient ring, but that ring records no derived Tor correction. Theorem 4.1 supplies the comparison, while the calculation states exactly why it is not invertible.

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- **Sheaves of modules.** *Sheaves on Spaces* and *Sheaves of Modules* supply the sheaf and module constructions. The course develops germ-family sheafification, stalk bijections, associative-ring module structures and their full universal-property proofs.
- **Complexes and localization.** *Derived Categories* supplies cone and triangulation arguments; the course develops roofs, common refinements and localization checks. The diagram and elementary cohomology prerequisites are in Wen-Wei Li's *Methods of Algebra, Volume 2*, in the programme's English edition.
- **Injective and K-injective resolutions.** *Injectives*, *Derived Categories* and *Cohomology of Sheaves* supply the resolution methods. The course includes flasque-section arguments, large-cofinality and transfinite constructions. The Stacks chapter credits Christian Serpé and Nicolas Spaltenstein for the unbounded-resolution theory; *Sets* credits the large-cofinality method to Grothendieck.
- **Flat and K-flat resolutions.** *Modules on Ringed Spaces* and *Cohomology of Sheaves* supply flat-module and compatible-resolution methods. The course develops cycle attachments, complete flatness calculations and compatible resolutions.
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